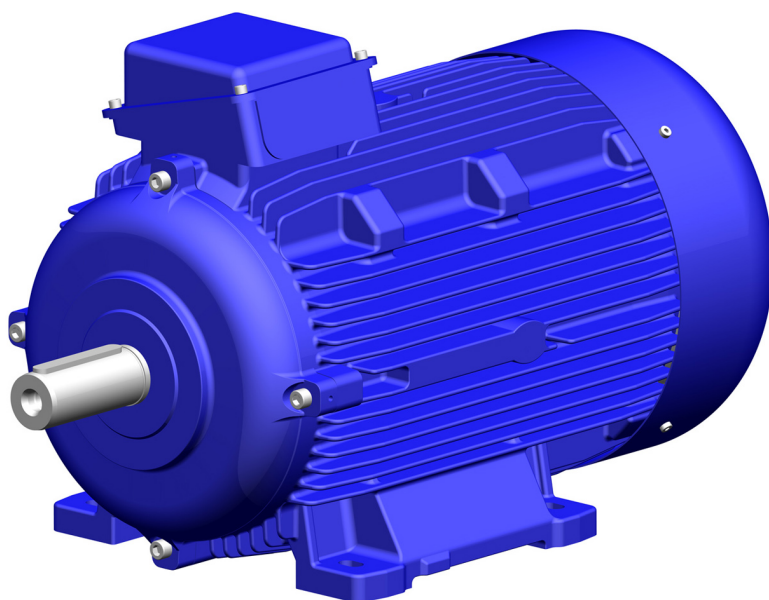




KSB IE4-Motor

Motor Catalogue



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Motor Catalogue KSB IE4-Motor

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Overviews

Overview B3

Table 1: Overview of available motors, type of construction B3, 2-pole

Power [kW]		Motor size	Rated voltage				Frequency	Mat. No.	Page
50 Hz	60 Hz		50 Hz		60 Hz				
			Δ	Y	Δ	Y			
22	22	180M	400	690	460		50/60	05407377	(⇒ Page 9)
30	30	200L	400	690	460		50/60	05407378	(⇒ Page 10)
37	37	200L	400	690	460		50/60	05407379	(⇒ Page 11)
45	45	225M	400	690	460		50/60	05407850	(⇒ Page 12)
55	55	250M	400	690	460		50/60	05407851	(⇒ Page 13)
75	75	280S	400	690	460		50/60	05214325	(⇒ Page 14)
90	90	280M	400	690	460		50/60	05214326	(⇒ Page 15)
110	110	315S	400	690	460		50/60	05214327	(⇒ Page 16)
132	132	315M	400	690	460		50/60	05214321	(⇒ Page 17)

Table 2: Overview of available motors, type of construction B3, 4-pole

Power [kW]		Motor size	Rated voltage				Frequency	Mat. No.	Page
50 Hz	60 Hz		50 Hz		60 Hz				
			Δ	Y	Δ	Y			
22	22	180L	400	690	460		50/60	05407852	(⇒ Page 18)
30	30	200L	400	690	460		50/60	05407853	(⇒ Page 19)
37	37	225S	400	690	460		50/60	05407854	(⇒ Page 20)
45	45	225M	400	690	460		50/60	05407855	(⇒ Page 21)
55	55	250M	400	690	460		50/60	05407856	(⇒ Page 22)
75	75	280S	400	690	460		50/60	05214322	(⇒ Page 23)
90	90	280M	400	690	460		50/60	05214323	(⇒ Page 24)
110	110	315M	400	690	460		50/60	05214324	(⇒ Page 25)
132	132	315M	400	690	460		50/60	05214276	(⇒ Page 26)

Overview V1

Table 3: Overview of available motors, type of construction V1, 2-pole

Power [kW]		Motor size	Rated voltage				Frequency	Mat. No.	Page
50 Hz	60 Hz		50 Hz		60 Hz				
			Δ	Y	Δ	Y			
22	22	180M	400	690	460		50/60	05407841	(⇒ Page 27)
30	30	200L	400	690	460		50/60	05407842	(⇒ Page 28)
37	37	200L	400	690	460		50/60	05407843	(⇒ Page 29)
45	45	225M	400	690	460		50/60	05407844	(⇒ Page 30)
55	55	250M	400	690	460		50/60	05407845	(⇒ Page 31)
75	75	280S	400	690	460		50/60	05214272	(⇒ Page 32)
90	90	280M	400	690	460		50/60	05214273	(⇒ Page 33)
110	110	315S	400	690	460		50/60	05214274	(⇒ Page 34)
132	132	315M	400	690	460		50/60	05214275	(⇒ Page 35)

Table 4: Overview of available motors, type of construction V1, 4-pole

Power [kW]		Motor size	Rated voltage				Frequency	Mat. No.	Page
50 Hz	60 Hz		50 Hz		60 Hz				
			Δ	Y	Δ	Y			
22	22	180L	400	690	460		50/60	05407846	(⇒ Page 36)
30	30	200L	400	690	460		50/60	05407847	(⇒ Page 37)
37	37	225S	400	690	460		50/60	05407848	(⇒ Page 38)

Power [kW]		Motor size	Rated voltage				Frequency	Mat. No.	Page
50 Hz	60 Hz		50 Hz		60 Hz				
			Δ	Y	Δ	Y			
45	45	225M	400	690	460		50/60	05407849	(⇒ Page 39)
55	55	250M	400	690	460		50/60	05408280	(⇒ Page 40)
75	75	280S	400	690	460		50/60	05214277	(⇒ Page 41)
90	90	280M	400	690	460		50/60	05214278	(⇒ Page 42)
110	110	315M	400	690	460		50/60	05214279	(⇒ Page 43)
132	132	315M	400	690	460		50/60	05214320	(⇒ Page 44)

Overview V15

Table 5: Overview of available motors, type of construction V15, 2-pole

Power [kW]		Motor size	Rated voltage				Frequency	Mat. No.	Page
50 Hz	60 Hz		50 Hz		60 Hz				
			Δ	Y	Δ	Y			
22	22	180M	400	690	460		50/60	05407857	(⇒ Page 45)
30	30	200L	400	690	460		50/60	05407858	(⇒ Page 46)
37	37	200L	400	690	460		50/60	05407859	(⇒ Page 47)
45	45	225M	400	690	460		50/60	05407860	(⇒ Page 48)
55	55	250M	400	690	460		50/60	05407861	(⇒ Page 49)
75	75	280S	400	690	460		50/60	05214352	(⇒ Page 50)
90	90	280M	400	690	460		50/60	05214353	(⇒ Page 51)
110	110	315S	400	690	460		50/60	05214354	(⇒ Page 52)
132	132	315M	400	690	460		50/60	05214355	(⇒ Page 53)

Table 6: Overview of available motors, type of construction V15, 4-pole

Power [kW]		Motor size	Rated voltage				Frequency	Mat. No.	Page
50 Hz	60 Hz		50 Hz		60 Hz				
			Δ	Y	Δ	Y			
22	22	180L	400	690	460		50/60	05407862	(⇒ Page 54)
30	30	200L	400	690	460		50/60	05407863	(⇒ Page 55)
37	37	225S	400	690	460		50/60	05407864	(⇒ Page 56)
45	45	225M	400	690	460		50/60	05407865	(⇒ Page 57)
55	55	250M	400	690	460		50/60	05407866	(⇒ Page 58)
75	75	280S	400	690	460		50/60	05214356	(⇒ Page 59)
90	90	280M	400	690	460		50/60	05214357	(⇒ Page 60)
110	110	315M	400	690	460		50/60	05214358	(⇒ Page 61)
132	132	315M	400	690	460		50/60	05214359	(⇒ Page 62)

B3 type of construction, 2-pole, 180M/L

Material: 2-22,0-180M-BWE7F3NGFDWWUWWE¹⁾

Material number: 5443310 (05407377)

Table 7: General data [50 Hz]

General data	+/-	Values
Motor size	Max.	180M
Type of construction	Max.	B3
Starting method	-	Star a. Delta
Motor flange	-	Without
Weight without attachments [kg]	Max.	180
Housing material ²⁾	-	1800
Enclosure	-	IP55
Nominal frequency	[Hz]	50/60
Efficiency class	-	IE4 (Super Premium)
Thermal class F to B	-	Yes
Temperature sensor	-	03
Operation on frequency inverter permitted	-	Yes (acc to motor manufact)
Duty type	-	Continuous running duty (S1)
Direction of rotation	-	Bi-directional
Motor bearings insulated	-	No
Position of fixed motor bearing	-	Drive end
Fixed motor bearing reinforced	-	Without
Radial motor bearing reinforced	-	No
Condensate drain	-	No
Bolted feet	-	Yes

Table 8: Efficiency

Electrical data	+/-	50 Hz	60 Hz
		4/4	3/4
Efficiency [%]	Min.	94	93.6
Power factor	Min.	0.86	0.82

Table 9: Electrical data

Electrical data	+/-	50 Hz	60 Hz
		Δ	Y
Rated voltage [V]	-	400	690
Rated current [A]	-10 %	39.3	22.7
Rated power [kW]	-	22	22
Rated speed [rpm]	Min.	2960	3565
Rated torque [Nm]	Min.	71	59
Starting current [IA/IN]	+20 %	9.7	11

Table 10: Mechanical data

Mechanical data	+/-	Values
		50 Hz
Sound pressure level [dB]	Max.	70
Moment of inertia [kgm ²]	Max.	0.09
Motor canopy	-	No

Table 11: Explosion protection

Explosion protection	+/-	Values
Type of protection	-	Without

Table 12: Environment

Environment	+/-	Values
Ambient temperature [°C]	Max.	40

Table 13: Terminal box

Explosion protection	+/-	Values
Enclosure of terminal box	-	IP55 (TEFC)
Cable entry (Ø)	-	2XM40X1,5

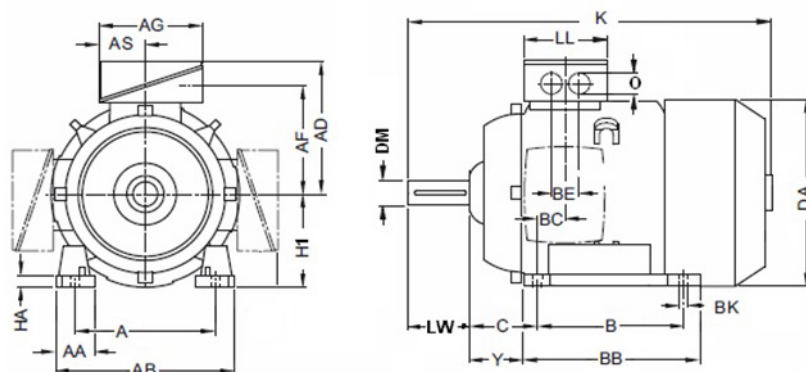

Fig. 1: Envelope dimensions

Table 14: Envelope dimensions A-DLF

A	AA	AB	AD	AF	AG	AS	B	BB	BC	BE	BK	C	DA	DLF
279	51	329	294	239.7	191.5	102.5	241	288	120.5	84	14.5	121	371	

Table 15: Envelope dimensions DM-Z

DM	H1	HA	HH	K	LA	LB	LL	LW	M	N	S	T	Y	Z
48	180	19		684.5			198.5	110					91.1	

¹ The name plate data may vary within the tolerances.

² 1800=grey cast iron, 4400=aluminium

B3 type of construction, 2-pole, 200M/L

Material: 2-30,0-200L-BWE7F3NGFDWWUWWE³⁾

Material number: 5443311 (05407378)

Table 16: General data [50 Hz]

General data	+/-	Values
Motor size	Max.	200L
Type of construction	Max.	B3
Starting method	-	Star a. Delta
Motor flange	-	Without
Weight without attachments [kg]	Max.	210
Housing material ⁴⁾	-	1800
Enclosure	-	IP55
Nominal frequency	[Hz]	50/60
Efficiency class	-	IE4 (Super Premium)
Thermal class F to B	-	Yes
Temperature sensor	-	03
Operation on frequency inverter permitted	-	Yes (acc to motor manufact)
Duty type	-	Continuous running duty (S1)
Direction of rotation	-	Bi-directional
Motor bearings insulated	-	No
Position of fixed motor bearing	-	No
Fixed motor bearing reinforced	-	Without
Radial motor bearing reinforced	-	No
Condensate drain	-	No
Bolted feet	-	Yes

Table 17: Efficiency

Electrical data		+/-	50 Hz			60 Hz		
			4/4	3/4	2/4	4/4	3/4	2/4
Efficiency	[%]	Min.	94.5	94	92.8	93.6	93.4	91.7
Power factor		Min.	0.84	0.8	0.71	0.83	0.78	0.68

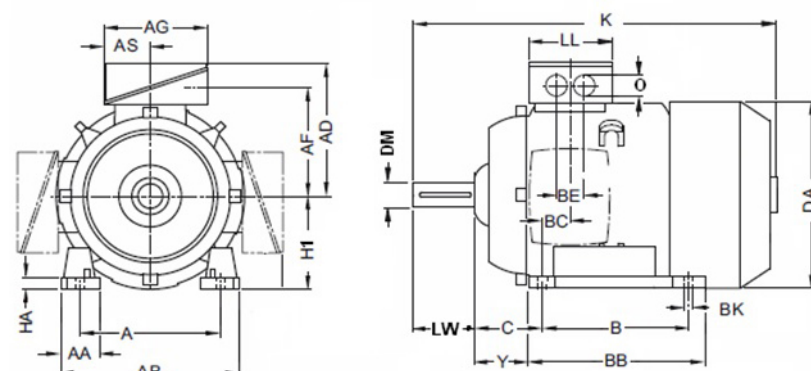


Fig. 2: Envelope dimensions

Table 23: Envelope dimensions A-DLF

A	AA	AB	AD	AF	AG	AS	B	BB	BC	BE	BK	C	DA	DFL
318	65	385	334	272.7	218	119.1	305	369	76	94	18.5	133	414	

Table 24: Envelope dimensions DM-Z

DM	H1	HA	HH	K	LA	LB	LL	LW	M	N	S	T	Y	Z
55	200	30		767.5			228.5	110					89.7	

Table 18: Electrical data

Electrical data		+/-	50 Hz		60 Hz	
			Δ	Y	Δ	Y
Rated voltage	[V]	-	460	690	460	-
Rated current	[A]	-10 %	54.5	31.5	48.5	-
Rated power	[kW]	-	30		30	
Rated speed	[rpm]	Min.	2965		3570	
Rated torque	[Nm]	Min.	96.6		80.2	
Starting current	[I _A /I _N]	+20 %	8.9		10	

Table 19: Mechanical data

Mechanical data		+/-	Values	
			50 Hz	60 Hz
Sound pressure level	[dB]	Max.	75	80
Moment of inertia	[kgm ²]	Max.	0.16	0.16
Motor canopy		-	No	

Table 20: Explosion protection

Explosion protection	+/-	Values
Type of protection	-	Without

Table 21: Environment

Environment	+/-	Values
Ambient temperature [°C]	Max.	40

Table 22: Terminal box

Explosion protection	+/-	Values
Enclosure of terminal box	-	IP55 (TEFC)
Cable entry (Ø)	-	2XM50X1,5

³ The name plate data may vary within the tolerances.

⁴ 1800=grey cast iron, 4400=aluminium

B3 type of construction, 2-pole, 200M/L

Material: 2-37,0-200L-BWE7F3NGFDWWUWWE1⁵⁾

Material number: 5443312 (05407379)

Table 25: General data [50 Hz]

General data	+/-	Values
Motor size	Max.	200L
Type of construction	Max.	B3
Starting method	-	Star a. Delta
Motor flange	-	Without
Weight without attachments [kg]	Max.	260
Housing material ⁶⁾	-	1800
Enclosure	-	IP55
Nominal frequency	[Hz]	50/60
Efficiency class	-	IE4 (Super Premium)
Thermal class F to B	-	Yes
Temperature sensor	-	03
Operation on frequency inverter permitted	-	Yes (acc to motor manufact)
Duty type	-	Continuous running duty (S1)
Direction of rotation	-	Bi-directional
Motor bearings insulated	-	No
Position of fixed motor bearing	-	Drive end
Fixed motor bearing reinforced	-	Without
Radial motor bearing reinforced	-	No
Condensate drain	-	No
Bolted feet	-	Yes

Table 26: Efficiency

Electrical data		+/-	50 Hz			60 Hz		
			4/4	3/4	2/4	4/4	3/4	2/4
Efficiency	[%]	Min.	94.8	94.3	93.3	94.1	93.8	92.4
Power factor		Min.	0.85	0.85	0.72	0.84	0.79	0.7

Table 27: Electrical data

Electrical data		+/-	50 Hz		60 Hz	
			Δ	Y	Δ	Y
Rated voltage	[V]	-	400	690	460	-
Rated current	[A]	-10 %	66.3	38.3	58.8	-
Rated power	[kW]	-	37		37	
Rated speed	[rpm]	Min.	2965		3570	
Rated torque	[Nm]	Min.	119.7		99.1	
Starting current	[I _A /I _N]	+20 %	9.2		10.2	

Table 28: Mechanical data

Mechanical data		+/-	Values	
			50 Hz	60 Hz
Sound pressure level	[dB]	Max.	75	80
Moment of inertia	[kgm²]	Max.	0.19	0.19
Motor canopy		-	No	

Table 29: Explosion protection

Explosion protection	+/-	Values
Type of protection	-	Without

Table 30: Environment

Environment	+/-	Values
Ambient temperature [°C]	Max.	40

Table 31: Terminal box

Explosion protection	+/-	Values
Enclosure of terminal box	-	IP55 (TEFC)
Cable entry (Ø)	-	2XM50X1,5

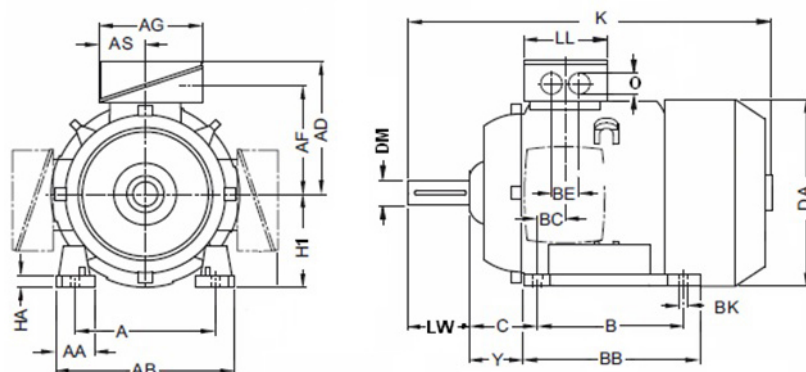

Fig. 3: Envelope dimensions

Table 32: Envelope dimensions A-DLF

A	AA	AB	AD	AF	AG	AS	B	BB	BC	BE	BK	C	DA	DFL
318	65	385	334	272.7	218	119.1	305	369	76	94	18.5	133	414	

Table 33: Envelope dimensions DM-Z

DM	H1	HA	HH	K	LA	LB	LL	LW	M	N	S	T	Y	Z
55	200	30		767.5			228.5	110					89.7	

⁵ The name plate data may vary within the tolerances.

⁶ 1800=grey cast iron, 4400=aluminium

B3 type of construction, 2-pole, 225S/M

Material: 2-45,0-225M-BWE7F3NGHDWWUWWE⁷⁾

Material number: 5443322 (05407850)

Table 34: General data [50 Hz]

General data	+/-	Values
Motor size	Max.	225M
Type of construction	Max.	B3
Starting method	-	Star a. Delta
Motor flange	-	Without
Weight without attachments [kg]	Max.	415
Housing material ⁸⁾	-	1800
Enclosure	-	IP55
Nominal frequency	[Hz]	50/60
Efficiency class	-	IE4 (Super Premium)
Thermal class F to B	-	Yes
Temperature sensor	-	03
Operation on frequency inverter permitted	-	Yes (acc to motor manufact)
Duty type	-	Continuous running duty (S1)
Direction of rotation	-	Bi-directional
Motor bearings insulated	-	No
Position of fixed motor bearing	-	Drive end
Fixed motor bearing reinforced	-	Without
Radial motor bearing reinforced	-	No
Condensate drain	-	No
Bolted feet	-	No

Table 35: Efficiency

Electrical data	+/-	50 Hz	60 Hz
		4/4	3/4
Efficiency [%]	Min.	95	94.3
Power factor	Min.	0.87	0.83

Table 36: Electrical data

Electrical data	+/-	50 Hz	60 Hz
		Δ	Y
Rated voltage [V]	-	400	690
Rated current [A]	-10 %	78.6	45.4
Rated power [kW]	-	45	45
Rated speed [rpm]	Min.	2970	3575
Rated torque [Nm]	Min.	145.19	120.66
Starting current [IA/IN]	+20 %	9.6	10.5

Table 37: Mechanical data

Mechanical data	+/-	Values
		50 Hz
Sound pressure level [dB]	Max.	77
Moment of inertia [kgm ²]	Max.	0.29
Motor canopy	-	No

Table 38: Explosion protection

Explosion protection	+/-	Values
Type of protection	-	Without

Table 39: Environment

Environment	+/-	Values
Ambient temperature [°C]	Max.	40

Table 40: Terminal box

Explosion protection	+/-	Values
Enclosure of terminal box	-	IP55 (TEFC)
Cable entry (O)	-	2XM50X1,5

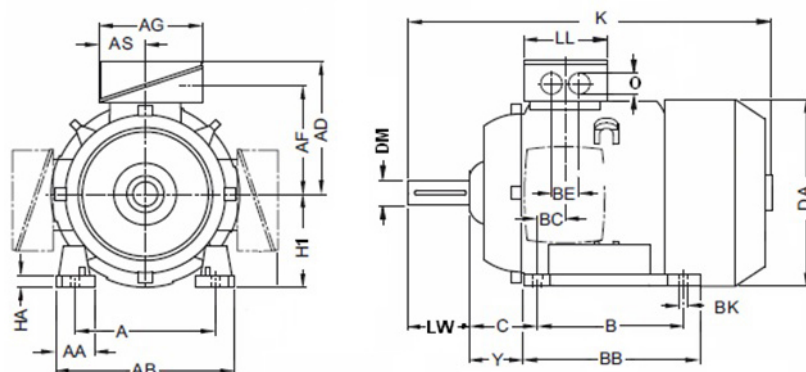


Fig. 4: Envelope dimensions

Table 41: Envelope dimensions A-DLF

A	AA	AB	AD	AF	AG	AS	B	BB	BC	BE	BK	C	DA	DLF
356	80	436	397	314.8	286	165.6	311	439.3	54.6	110	18.5	149	469	

Table 42: Envelope dimensions DM-Z

DM	H1	HA	HH	K	LA	LB	LL	LW	M	N	S	T	Y	Z
55	225	28		856.5			269	110					93.6	

⁷⁾ The name plate data may vary within the tolerances.

⁸⁾ 1800=grey cast iron, 4400=aluminium

B3 type of construction, 2-pole, 250S/M

Material: 2-55,0-250M-BWE7F3NGHDWWUWWE⁹⁾

Material number: 5443323 (05407851)

Table 43: General data [50 Hz]

General data	+/-	Values
Motor size	Max.	250M
Type of construction	Max.	B3
Starting method	-	Star a. Delta
Motor flange	-	Without
Weight without attachments [kg]	Max.	450
Housing material ¹⁰⁾	-	1800
Enclosure	-	IP55
Nominal frequency	[Hz]	50/60
Efficiency class	-	IE4 (Super Premium)
Thermal class F to B	-	Yes
Temperature sensor	-	03
Operation on frequency inverter permitted	-	Yes (acc to motor manufact)
Duty type	-	Continuous running duty (S1)
Direction of rotation	-	Bi-directional
Motor bearings insulated	-	No
Position of fixed motor bearing	-	Drive end
Fixed motor bearing reinforced	-	Without
Radial motor bearing reinforced	-	No
Condensate drain	-	No
Bolted feet	-	No

Table 44: Efficiency

Electrical data		+/-	50 Hz			60 Hz		
			4/4	3/4	2/4	4/4	3/4	2/4
Efficiency	[%]	Min.	95.3	94.5	93.5	94.5	94.2	92.7
Power factor		Min.	0.87	0.83	0.75	0.86	0.82	0.73

Table 45: Electrical data

Electrical data		+/-	50 Hz		60 Hz	
			Δ	Y	Δ	Y
Rated voltage	[V]	-	400	690	460	-
Rated current	[A]	-10 %	95.7	55.3	84.9	-
Rated power	[kW]	-	55		55	
Rated speed	[rpm]	Min.	2970		3570	
Rated torque	[Nm]	Min.	176.58		147.15	
Starting current	[I _A /I _N]	+20 %	9.3		10.5	

Table 46: Mechanical data

Mechanical data		+/-	Values	
			50 Hz	60 Hz
Sound pressure level	[dB]	Max.	77	82
Moment of inertia	[kgm ²]	Max.	0.36	0.36
Motor canopy		-	No	

Table 47: Explosion protection

Explosion protection	+/-	Values
Type of protection	-	Without

Table 48: Environment

Environment	+/-	Values
Ambient temperature [°C]	Max.	40

Table 49: Terminal box

Explosion protection	+/-	Values
Enclosure of terminal box	-	IP55 (TEFC)
Cable entry (Ø)	-	2XM63X1,5

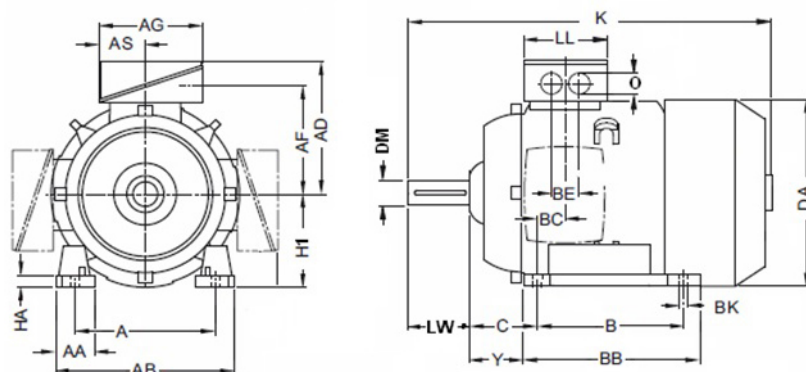


Fig. 5: Envelope dimensions

Table 50: Envelope dimensions A-DFL

A	AA	AB	AD	AF	AG	AS	B	BB	BC	BE	BK	C	DA	DFL
406	100	506	417	334.8	286	165.6	349	500	49	110	24	168	514	

Table 51: Envelope dimensions DM-Z

DM	H1	HA	HH	K	LA	LB	LL	LW	M	N	S	T	Y	Z
60	250	30		965.5			269	140					107	

⁹ The name plate data may vary within the tolerances.

¹⁰ 1800=grey cast iron, 4400=aluminium

B3 type of construction, 2-pole, 280S

Material: 2-75,0-280S-BWE7F3NGFNWWNWKWC¹¹⁾

Material number: 05214325 (5214341)

Table 52: General data [50 Hz]

General data	+/-	Values
Motor size	Max.	280S
Type of construction	Max.	B3
Starting method	-	Star a. Delta
Motor flange	-	Without
Weight without attachments [kg]	Max.	712
Housing material ¹²⁾	-	1800
Enclosure	-	IP55
Nominal frequency	[Hz]	50/60
Efficiency class	-	IE4 (Super Premium)
Thermal class F to B	-	Yes
Temperature sensor	-	03
Operation on frequency inverter permitted	-	No
Duty type	-	Continuous running duty (S1)
Direction of rotation	-	Bi-directional
Motor bearings insulated	-	No
Position of fixed motor bearing	-	Non drive side
Fixed motor bearing reinforced	-	Without
Radial motor bearing reinforced	-	No
Condensate drain	-	Yes
Bolted feet	-	Yes

Table 53: Efficiency

Electrical data		+/-	50 Hz			60 Hz		
			4/4	3/4	2/4	4/4	3/4	2/4
Efficiency	[%]	min.	95.6	94.8	93.6	95	94.8	93
Power factor		Min.	0.87	0.84	0.75	0.86	0.83	0.73

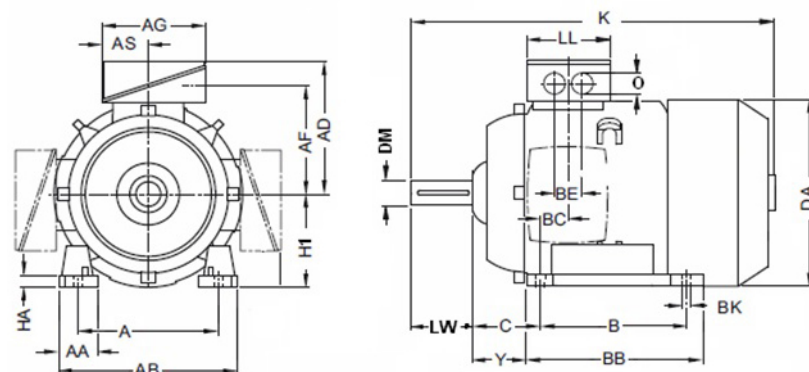


Fig. 6: Envelope dimensions

Table 59: Envelope dimensions A-DLF

A	AA	AB	AD	AF	AG	AS	B	BB	BC	BE	BK	C	DA	DFL
457	100	557	472	391	312	140.4	368	517	76	126	24	190	580	-

Table 60: Envelope dimensions DM-Z

DM	H1	HA	HH	K	LA	LB	LL	LW	M	N	S	T	Y	Z
65	280	42	-	1071	-	-	314	140	-	-	-	-	146	-

Table 54: Electrical data

Electrical data		+/-	50 Hz		60 Hz	
			Δ	Y	Δ	Y
Rated voltage	[V]	-	400	690	460	-
Rated current	[A]	-10 %	130	75.5	115	-
Rated power	[kW]	-	75		75	
Rated speed	[rpm]	Min.	2977		3579	
Rated torque	[Nm]	Min.	241		200	
Starting current	[I _A /I _N]	+20 %	8.3		9	

Table 55: Mechanical data

Mechanical data		+/-	Values	
			50 Hz	60 Hz
Sound pressure level	[dB]	Max.	77	81
Moment of inertia	[kgm²]	Max.	0.94	0.94
Motor canopy		-	No	

Table 56: Explosion protection

Explosion protection	+/-	Values
Type of protection	-	Without

Table 57: Environment

Environment	+/-	Values
Ambient temperature [°C]	Max.	40

Table 58: Terminal box

Explosion protection	+/-	Values
Enclosure of terminal box	-	IP55 (TEFC)
Cable entry (O)	-	2XM63X1,5

¹¹ The name plate data may vary within the tolerances.

¹² 1800=grey cast iron, 4400=aluminium

B3 type of construction, 2-pole, 280M

2-90,0-280M-BWE7F3NGFNWWNWKWC¹³⁾

Material number: 05214326 (5214342)

Table 61: General data [50 Hz]

General data	+/-	Values
Motor size	Max.	280M
Type of construction	Max.	B3
Starting method	-	Star a. Delta
Motor flange	-	Without
Weight without attachments [kg]	Max.	752
Housing material ¹⁴⁾	-	1800
Enclosure	-	IP55
Nominal frequency	[Hz]	50/60
Efficiency class	-	IE4 (Super Premium)
Thermal class F to B	-	Yes
Temperature sensor	-	03
Operation on frequency inverter permitted	-	No
Duty type	-	Continuous running duty (S1)
Direction of rotation	-	Bi-directional
Motor bearings insulated	-	No
Position of fixed motor bearing	-	Non drive side
Fixed motor bearing reinforced	-	Without
Radial motor bearing reinforced	-	No
Condensate drain	-	Yes
Bolted feet	-	Yes

Table 62: Efficiency

Electrical data	+/-	50 Hz			60 Hz		
		4/4	3/4	2/4	4/4	3/4	2/4
Efficiency [%]	Min.	95.8	95.2	94.3	95.4	95	93.5
Power factor	min.	0.88	0.85	0.78	0.87	0.84	0.76

Table 63: Electrical data

Electrical data	+/-	50 Hz		60 Hz	
		Δ	Y	Δ	Y
Rated voltage [V]	-	400	690	460	-
Rated current [A]	-10 %	154	89.3	136	-
Rated power [kW]	-	90		90	
Rated speed [rpm]	min.	2975		3578	
Rated torque [Nm]	Min.	289		240	
Starting current [IA/IN]	+20 %	7.9		8.7	

Table 64: Mechanical data

Mechanical data	+/-	Values	
		50 Hz	60 Hz
Sound pressure level [dB]	Max.	77	81
Moment of inertia [kgm ²]	Max.	1	1
Motor canopy	-	No	

Table 65: Explosion protection

Explosion protection	+/-	Values
Type of protection	-	Without

Table 66: Environment

Environment	+/-	Values
Ambient temperature [°C]	Max.	40

Table 67: Terminal box

Explosion protection	+/-	Values
Enclosure of terminal box	-	IP55 (TEFC)
Cable entry (O)	-	2XM63X1,5

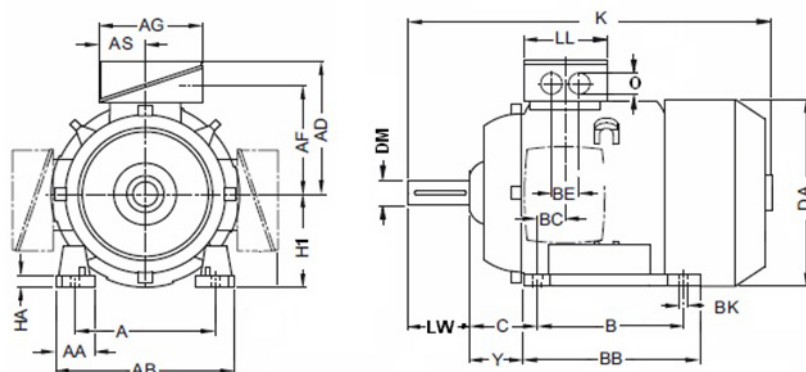


Fig. 7: Envelope dimensions

Table 68: Envelope dimensions A-DLF

A	AA	AB	AD	AF	AG	AS	B	BB	BC	BE	BK	C	DA	DFL
457	100	557	472	391	312	140.4	419	517	76	126	24	190	580	-

Table 69: Envelope dimensions DM-Z

DM	H1	HA	HH	K	LA	LB	LL	LW	M	N	S	T	Y	Z
65	280	42	-	1071	-	-	314	140	-	-	-	-	146	-

¹³ The name plate data may vary within the tolerances.

¹⁴ 1800=grey cast iron, 4400=aluminium

B3 type of construction, 2-pole, 315S

Material: 2-110 -315S-BWE7F3NGHNWWNWKWC¹⁵⁾

Material number: 05214327 (5214343)

Table 70: General data [50 Hz]

General data	+/-	Values
Motor size	Max.	315S
Type of construction	Max.	B3
Starting method	-	Star a. Delta
Motor flange	-	Without
Weight without attachments [kg]	Max.	956
Housing material ¹⁶⁾	-	1800
Enclosure	-	IP55
Nominal frequency	[Hz]	50/60
Efficiency class	-	IE4 (Super Premium)
Thermal class F to B	-	Yes
Temperature sensor	-	03
Operation on frequency inverter permitted	-	No
Duty type	-	Continuous running duty (S1)
Direction of rotation	-	Bi-directional
Motor bearings insulated	-	No
Position of fixed motor bearing	-	Non drive side
Fixed motor bearing reinforced	-	Without
Radial motor bearing reinforced	-	No
Condensate drain	-	Yes
Bolted feet	-	No

Table 71: Efficiency

Electrical data		+/-	50 Hz			60 Hz		
			4/4	3/4	2/4	4/4	3/4	2/4
Efficiency	[%]	Min.	96	95.5	94	95.4	95.4	94.5
Power factor		Min.	0.87	0.84	0.76	0.86	0.83	0.74

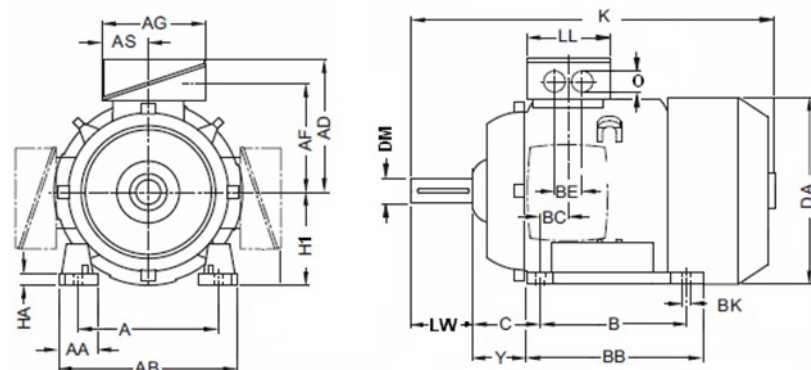

Fig. 8: Envelope dimensions

Table 77: Envelope dimensions A-DLF

A	AA	AB	AD	AF	AG	AS	B	BB	BC	BE	BK	C	DA	DFL
508	120	630	530	432	382	171.9	406	626	48	160	28	216	644	-

Table 78: Envelope dimensions DM-Z

DM	H1	HA	HH	K	LA	LB	LL	LW	M	N	S	T	Y	Z
65	315	48	-	1244	-	-	379	140	-	-	-	-	141	-

Table 72: Electrical data

Electrical data		+/-	50 Hz		60 Hz	
			Δ	Y	Δ	Y
Rated voltage	[V]	-	400	690	460	-
Rated current	[A]	-10 %	190	110	168	-
Rated power	[kW]	-	110		110	
Rated speed	[rpm]	Min.	2976		3577	
Rated torque	[Nm]	Min.	353		294	
Starting current	[IA/IN]	+20 %	8		8.5	

Table 73: Mechanical data

Mechanical data		+/-	Values	
			50 Hz	60 Hz
Sound pressure level	[dB]	Max.	77	81
Moment of inertia	[kgm ²]	Max.	1.4	1.4
Motor canopy		-	No	

Table 74: Explosion protection

Explosion protection	+/-	Values
Type of protection	-	Without

Table 75: Environment

Environment	+/-	Values
Ambient temperature [°C]	Max.	40

Table 76: Terminal box

Explosion protection	+/-	Values
Enclosure of terminal box	-	IP55 (TEFC)
Cable entry (O)	-	2XM63X1,5

¹⁵⁾ The name plate data may vary within the tolerances.

¹⁶⁾ 1800=grey cast iron, 4400=aluminium

B3 type of construction, 2-pole, 315M

Material: 2-132 -315M-BWE7F3NGHNWWNWKWC¹⁷⁾

Material number: 05214321 (5214337)

Table 79: General data [50 Hz]

General data	+/-	Values
Motor size	Max.	315M
Type of construction	Max.	B3
Starting method	-	Star a. Delta
Motor flange	-	Without
Weight without attachments [kg]	Max.	1062
Housing material ¹⁸⁾	-	1800
Enclosure	-	IP55
Nominal frequency	[Hz]	50/60
Efficiency class	-	IE4 (Super Premium)
Thermal class F to B	-	Yes
Temperature sensor	-	03
Operation on frequency inverter permitted	-	No
Duty type	-	Continuous running duty (S1)
Direction of rotation	-	Bi-directional
Motor bearings insulated	-	No
Position of fixed motor bearing	-	Non drive side
Fixed motor bearing reinforced	-	Without
Radial motor bearing reinforced	-	No
Condensate drain	-	Yes
Bolted feet	-	No

Table 80: Efficiency

Electrical data	+/-	50 Hz	60 Hz
		4/4	3/4
Efficiency [%]	Min.	96.2	95.9
Power factor	min.	0.88	0.85

Table 81: Electrical data

Electrical data	+/-	50 Hz	60 Hz
		Δ	Y
Rated voltage [V]	-	400	690
Rated current [A]	-10 %	225	130
Rated power [kW]	-	132	132
Rated speed [rpm]	Min.	2983	3585
Rated torque [Nm]	Min.	423	352
Starting current [IA/IN]	+20 %	8.3	9.2

Table 82: Mechanical data

Mechanical data	+/-	Values
		50 Hz
Sound pressure level [dB]	Max.	77
Moment of inertia [kgm ²]	Max.	1.9
Motor canopy	-	No

Table 83: Explosion protection

Explosion protection	+/-	Values
Type of protection	-	Without

Table 84: Environment

Environment	+/-	Values
Ambient temperature [°C]	Max.	40

Table 85: Terminal box

Explosion protection	+/-	Values
Enclosure of terminal box	-	IP55 (TEFC)
Cable entry (O)	-	2XM63X1,5

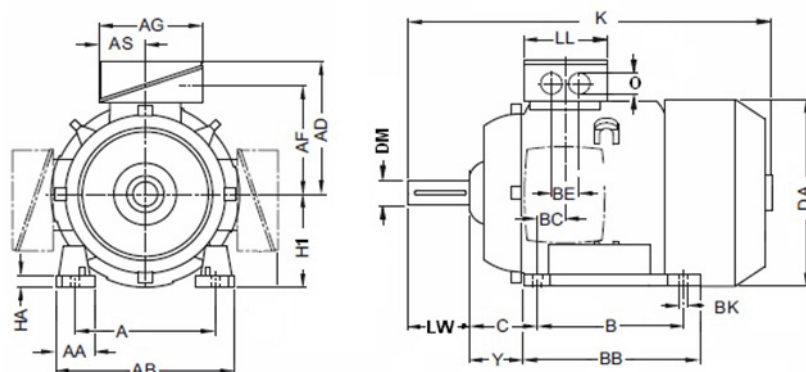


Fig. 9: Envelope dimensions

Table 86: Envelope dimensions A-DLF

A	AA	AB	AD	AF	AG	AS	B	BB	BC	BE	BK	C	DA	DFL
508	120	630	530	432	382	171.9	457	626	48	160	28	216	644	-

Table 87: Envelope dimensions DM-Z

DM	H1	HA	HH	K	LA	LB	LL	LW	M	N	S	T	Y	Z
65	315	48	-	1244	-	-	379	140	-	-	-	-	141	-

¹⁷ The name plate data may vary within the tolerances.

¹⁸ 1800=grey cast iron, 4400=aluminium

B3 type of construction, 4-pole, 180M/L

Material: 4-22,0-180L-BWE7F3NGHDWWUWWE1¹⁹⁾

Material number: 5443324 (05407852)

Table 88: General data [50 Hz]

General data	+/-	Values
Motor size	Max.	180L
Type of construction	Max.	B3
Starting method	-	Star a. Delta
Motor flange	-	Without
Weight without attachments [kg]	Max.	220
Housing material ²⁰⁾	-	1800
Enclosure	-	IP55
Nominal frequency	[Hz]	50/60
Efficiency class	-	IE4 (Super Premium)
Thermal class F to B	-	Yes
Temperature sensor	-	03
Operation on frequency inverter permitted	-	Yes (acc to motor manufact)
Duty type	-	Continuous running duty (S1)
Direction of rotation	-	Bi-directional
Motor bearings insulated	-	No
Position of fixed motor bearing	-	Drive end
Fixed motor bearing reinforced	-	Without
Radial motor bearing reinforced	-	No
Condensate drain	-	No
Bolted feet	-	No

Table 89: Efficiency

Electrical data		+/-	50 Hz			60 Hz		
			4/4	3/4	2/4	4/4	3/4	2/4
Efficiency	[%]	Min.	94.5	94.1	93.2	94.5	94.1	92.9
Power factor		Min.	0.81	0.74	0.62	0.79	0.72	0.6

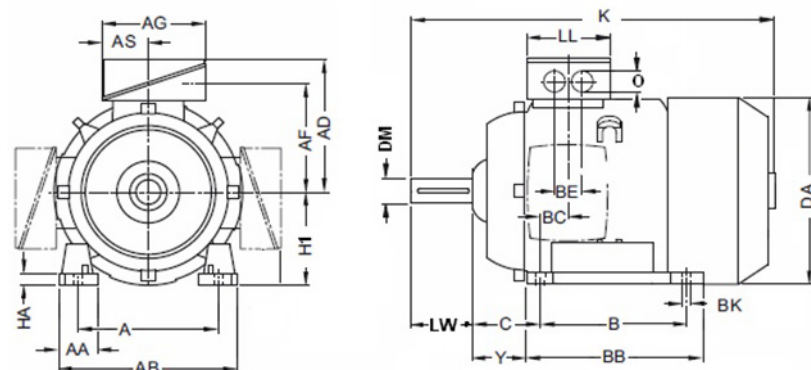

Fig. 10: Envelope dimensions

Table 95: Envelope dimensions A-DLF

A	AA	AB	AD	AF	AG	AS	B	BB	BC	BE	BK	C	DA	DFL
279	51	329	294	245.5	191.5	107.2	279	325	68.4	84.6	14.5	121	371	

Table 96: Envelope dimensions DM-Z

DM	H1	HA	HH	K	LA	LB	LL	LW	M	N	S	T	Y	Z
48	180	19		722.5			198.5	110					91.7	

Table 90: Electrical data

Electrical data		+/-	50 Hz		60 Hz	
			Δ	Y	Δ	Y
Rated voltage	[V]	-	400	690	460	-
Rated current	[A]	-10 %	41.5	24	37	-
Rated power	[kW]	-	22		22	
Rated speed	[rpm]	Min.	1476		1778	
Rated torque	[Nm]	Min.	142.25		118.7	
Starting current	[I _A /I _N]	+20 %	9.1		10.4	

Table 91: Mechanical data

Mechanical data		+/-	Values	
			50 Hz	60 Hz
Sound pressure level	[dB]	Max.	65	68
Moment of inertia	[kgm ²]	Max.	0.21	0.21
Motor canopy		-	No	

Table 92: Explosion protection

Explosion protection	+/-	Values
Type of protection	-	Without

Table 93: Environment

Environment	+/-	Values
Ambient temperature [°C]	Max.	40

Table 94: Terminal box

Explosion protection	+/-	Values
Enclosure of terminal box	-	IP55 (TEFC)
Cable entry (O)	-	2XM40X1,5

¹⁹ The name plate data may vary within the tolerances.

²⁰ 1800=grey cast iron, 4400=aluminium

B3 type of construction, 4-pole, 200M/L

Material: 4-30,0-200L-BWE7F3NGHDWWUWWE²¹⁾

Material number: 5443325 (05407853)

Table 97: General data [50 Hz]

General data	+/-	Values
Motor size	Max.	200L
Type of construction	Max.	B3
Starting method	-	Star a. Delta
Motor flange	-	Without
Weight without attachments [kg]	Max.	255
Housing material ²²⁾	-	1800
Enclosure	-	IP55
Nominal frequency	[Hz]	50/60
Efficiency class	-	IE4 (Super Premium)
Thermal class F to B	-	Yes
Temperature sensor	-	03
Operation on frequency inverter permitted	-	Yes (acc to motor manufact)
Duty type	-	Continuous running duty (S1)
Direction of rotation	-	Bi-directional
Motor bearings insulated	-	No
Position of fixed motor bearing	-	Drive end
Fixed motor bearing reinforced	-	Without
Radial motor bearing reinforced	-	No
Condensate drain	-	No
Bolted feet	-	No

Table 98: Efficiency

Electrical data		+/-	50 Hz			60 Hz		
			4/4	3/4	2/4	4/4	3/4	2/4
Efficiency	[%]	Min.	94.9	94.7	94.2	95	94.8	93.7
Power factor		Min.	0.81	0.75	0.64	0.8	0.74	0.62

Table 99: Electrical data

Electrical data		+/-	50 Hz		60 Hz	
			Δ	Y	Δ	Y
Rated voltage	[V]	-	400	690	460	-
Rated current	[A]	-10 %	56.3	32.6	49.5	-
Rated power	[kW]	-	30		30	
Rated speed	[rpm]	Min.	1480		1783	
Rated torque	[Nm]	Min.	193.26		160.88	
Starting current	[I _A /I _N]	+20 %	8.8		9.8	

Table 100: Mechanical data

Mechanical data		+/-	Values	
			50 Hz	60 Hz
Sound pressure level	[dB]	Max.	66	68
Moment of inertia	[kgm ²]	Max.	0.28	0.28
Motor canopy		-	No	

Table 101: Explosion protection

Explosion protection	+/-	Values
Type of protection	-	Without

Table 102: Environment

Environment	+/-	Values
Ambient temperature [°C]	Max.	40

Table 103: Terminal box

Explosion protection	+/-	Values
Enclosure of terminal box	-	IP55 (TEFC)
Cable entry (Ø)	-	2XM50X1,5

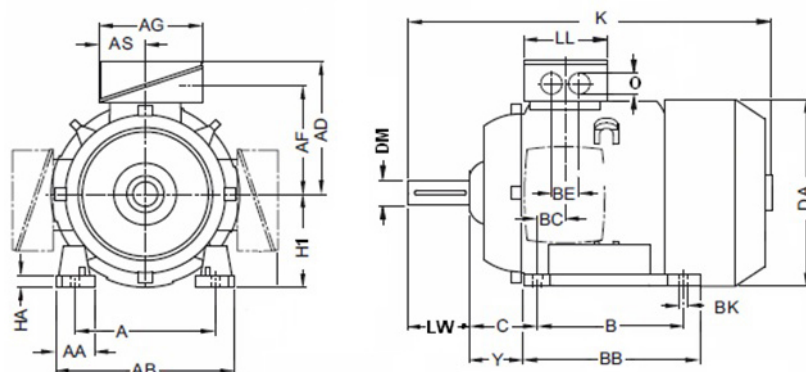

Fig. 11: Envelope dimensions

Table 104: Envelope dimensions A-DFL

A	AA	AB	AD	AF	AG	AS	B	BB	BC	BE	BK	C	DA	DFL
318	65	385	334	286.1	218	150.6	305	369	49.6	100.1	18.5	133	414	

Table 105: Envelope dimensions DM-Z

DM	H1	HA	HH	K	LA	LB	LL	LW	M	N	S	T	Y	Z
55	200	30		767.5			228.5	110					89.7	

²¹ The name plate data may vary within the tolerances.

²² 1800=grey cast iron, 4400=aluminium

B3 type of construction, 4-pole, 225S/M

Material: 4-37,0-225S-BWE7F3NGHDWWUWWEI²³⁾

Material number: 5443326 (05407854)

Table 106: General data [50 Hz]

General data	+/-	Values
Motor size	Max.	225S
Type of construction	Max.	B3
Starting method	-	Star a. Delta
Motor flange	-	Without
Weight without attachments [kg]	Max.	345
Housing material ²⁴⁾	-	1800
Enclosure	-	IP55
Nominal frequency	[Hz]	50/60
Efficiency class	-	IE4 (Super Premium)
Thermal class F to B	-	Yes
Temperature sensor	-	03
Operation on frequency inverter permitted	-	Yes (acc to motor manufact)
Duty type	-	Continuous running duty (S1)
Direction of rotation	-	Bi-directional
Motor bearings insulated	-	No
Position of fixed motor bearing	-	Drive end
Fixed motor bearing reinforced	-	Without
Radial motor bearing reinforced	-	No
Condensate drain	-	No
Bolted feet	-	No

Table 107: Efficiency

Electrical data		+/-	50 Hz			60 Hz		
			4/4	3/4	2/4	4/4	3/4	2/4
Efficiency	[%]	Min.	95.2	94.8	94	95.4	95.1	94
Power factor		Min.	0.82	0.77	0.66	0.81	0.75	0.63

Table 108: Electrical data

Electrical data		+/-	50 Hz		60 Hz	
			Δ	Y	Δ	Y
Rated voltage	[V]	-	400	690	460	-
Rated current	[A]	-10 %	68.4	39.5	60.1	-
Rated power	[kW]	-	37		37	
Rated speed	[rpm]	Min.	1483		1785	
Rated torque	[Nm]	Min.	238.38		198.16	
Starting current	[I _A /I _N]	+20 %	8.5		10	

Table 109: Mechanical data

Mechanical data		+/-	Values	
			50 Hz	60 Hz
Sound pressure level	[dB]	Max.	67	70
Moment of inertia	[kgm ²]	Max.	0.48	0.48
Motor canopy		-	No	

Table 110: Explosion protection

Explosion protection	+/-	Values
Type of protection	-	Without

Table 111: Environment

Environment	+/-	Values
Ambient temperature [°C]	Max.	40

Table 112: Terminal box

Explosion protection	+/-	Values
Enclosure of terminal box	-	IP55 (TEFC)
Cable entry (Ø)	-	2XM50X1,5

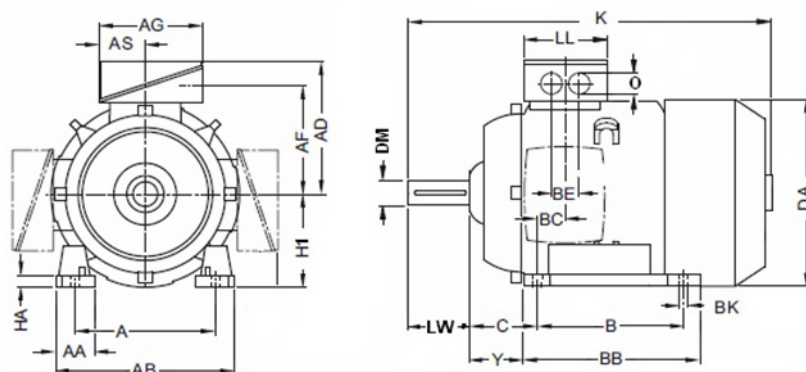

Fig. 12: Envelope dimensions

Table 113: Envelope dimensions A-DLF

A	AA	AB	AD	AF	AG	AS	B	BB	BC	BE	BK	C	DA	DFL
356	80	436	397	314.8	286	165.6	286	439.3	54.6	110	18.5	149	469	

Table 114: Envelope dimensions DM-Z

DM	H1	HA	HH	K	LA	LB	LL	LW	M	N	S	T	Y	Z
60	225	28		886.6			269	140					93.6	

²³ The name plate data may vary within the tolerances.

²⁴ 1800=grey cast iron, 4400=aluminium

B3 type of construction, 4-pole, 225S/M

Material: 4-45,0-225M-BWE7F3NGHDWWUWWE²⁵⁾

Material number: 5443327 (05407855)

Table 115: General data [50 Hz]

General data	+/-	Values
Motor size	Max.	225M
Type of construction	Max.	B3
Starting method	-	Star a. Delta
Motor flange	-	Without
Weight without attachments [kg]	Max.	415
Housing material ²⁶⁾	-	1800
Enclosure	-	IP55
Nominal frequency	[Hz]	50/60
Efficiency class	-	IE4 (Super Premium)
Thermal class F to B	-	Yes
Temperature sensor	-	03
Operation on frequency inverter permitted	-	Yes (acc to motor manufact)
Duty type	-	Continuous running duty (S1)
Direction of rotation	-	Bi-directional
Motor bearings insulated	-	No
Position of fixed motor bearing	-	Drive end
Fixed motor bearing reinforced	-	Without
Radial motor bearing reinforced	-	No
Condensate drain	-	No
Bolted feet	-	No

Table 116: Efficiency

Electrical data		+/-	50 Hz			60 Hz		
			4/4	3/4	2/4	4/4	3/4	2/4
Efficiency	[%]	Min.	95.4	95.3	94.7	95.4	95.4	94.6
Power factor		Min.	0.84	0.8	0.7	0.83	0.78	0.67

Table 117: Electrical data

Electrical data		+/-	50 Hz		60 Hz	
			Δ	Y	Δ	Y
Rated voltage	[V]	-	400	690	460	-
Rated current	[A]	-10 %	81.1	46.9	71.3	-
Rated power	[kW]	-	45		45	
Rated speed	[rpm]	Min.	1483		1785	
Rated torque	[Nm]	Min.	290.38		241.33	
Starting current	[IA/IN]	+20 %	8.7		10.2	

Table 118: Mechanical data

Mechanical data		+/-	Values	
			50 Hz	60 Hz
Sound pressure level	[dB]	Max.	67	70
Moment of inertia	[kgm ²]	Max.	0.64	0.64
Motor canopy		-	No	

Table 119: Explosion protection

Explosion protection	+/-	Values
Type of protection	-	Without

Table 120: Environment

Environment	+/-	Values
Ambient temperature [°C]	Max.	40

Table 121: Terminal box

Explosion protection	+/-	Values
Enclosure of terminal box	-	IP55 (TEFC)
Cable entry (Ø)	-	2XM50X1,5

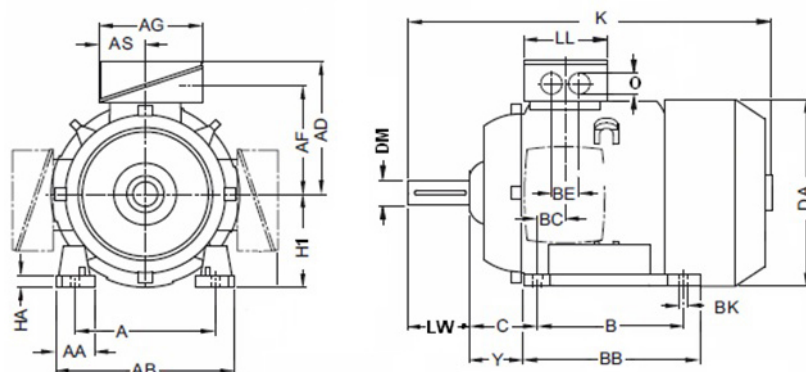

Fig. 13: Envelope dimensions

Table 122: Envelope dimensions A-DFL

A	AA	AB	AD	AF	AG	AS	B	BB	BC	BE	BK	C	DA	DFL
356	80	436	397	314.8	286	165.6	311	439.3	54.6	110	18.5	149	469	

Table 123: Envelope dimensions DM-Z

DM	H1	HA	HH	K	LA	LB	LL	LW	M	N	S	T	Y	Z
60	225	28		886.6			269	140					93.6	

²⁵ The name plate data may vary within the tolerances.

²⁶ 1800=grey cast iron, 4400=aluminium

B3 type of construction, 4-pole, 250S/M

Material: 4-55,0-250M-BWE7F3NGHDWWUWWE²⁷⁾

Material number: 5443328 (05407856)

Table 124: General data [50 Hz]

General data	+/-	Values
Motor size	Max.	250M
Type of construction	Max.	B3
Starting method	-	Star a. Delta
Motor flange	-	Without
Weight without attachments [kg]	Max.	470
Housing material ²⁸⁾	-	1800
Enclosure	-	IP55
Nominal frequency	[Hz]	50/60
Efficiency class	-	IE4 (Super Premium)
Thermal class F to B	-	Yes
Temperature sensor	-	03
Operation on frequency inverter permitted	-	Yes (acc to motor manufact)
Duty type	-	Continuous running duty (S1)
Direction of rotation	-	Bi-directional
Motor bearings insulated	-	No
Position of fixed motor bearing	-	Drive end
Fixed motor bearing reinforced	-	Without
Radial motor bearing reinforced	-	No
Condensate drain	-	No
Bolted feet	-	No

Table 125: Efficiency

Electrical data		+/-	50 Hz			60 Hz		
			4/4	3/4	2/4	4/4	3/4	2/4
Efficiency	[%]	Min.	95.7	95.4	94.8	95.8	95.6	94.7
Power factor		Min.	0.83	0.78	0.67	0.82	0.76	0.65

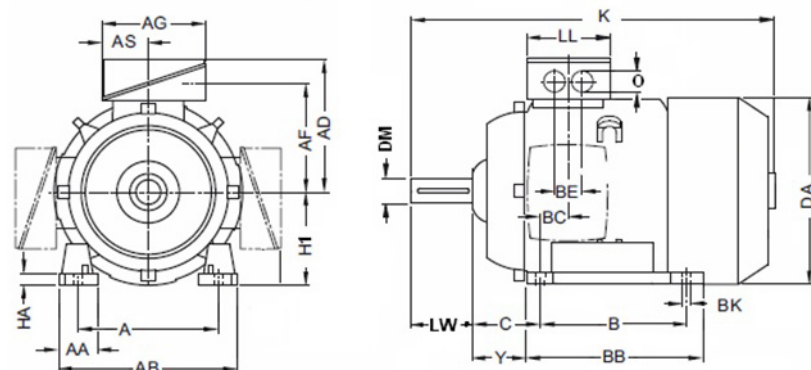


Fig. 14: Envelope dimensions

Table 131: Envelope dimensions A-DFL

A	AA	AB	AD	AF	AG	AS	B	BB	BC	BE	BK	C	DA	DFL
406	100	506	417	334.8	286	165.6	349	500	49	110	24	168	514	

Table 132: Envelope dimensions DM-Z

DM	H1	HA	HH	K	LA	LB	LL	LW	M	N	S	T	Y	Z
65	250	30		965.5			269	140					107	

Table 126: Electrical data

Electrical data		+/-	50 Hz		60 Hz	
			Δ	Y	Δ	Y
Rated voltage	[V]	-	400	690	460	-
Rated current	[A]	-10 %	100	57.8	87.9	-
Rated power	[kW]	-	55		55	
Rated speed	[rpm]	Min.	1485		1787	
Rated torque	[Nm]	Min.	354.14		294.3	
Starting current	[I _A /I _N]	+20 %	8.9		10.2	

Table 127: Mechanical data

Mechanical data		+/-	Values	
			50 Hz	60 Hz
Sound pressure level	[dB]	Max.	67	71
Moment of inertia	[kgm ²]	Max.	0.78	0.78
Motor canopy		-	No	

Table 128: Explosion protection

Explosion protection	+/-	Values
Type of protection	-	Without

Table 129: Environment

Environment	+/-	Values
Ambient temperature [°C]	Max.	40

Table 130: Terminal box

Explosion protection	+/-	Values
Enclosure of terminal box	-	IP55 (TEFC)
Cable entry (O)	-	2XM63X1,5

²⁷ The name plate data may vary within the tolerances.

²⁸ 1800=grey cast iron, 4400=aluminium

B3 type of construction, 4-pole, 280S

Material: 4-75,0-280S-BWE7F3NGFNWWNWKWC²⁹⁾

Material number: 05214322 (5214338)

Table 133: General data [50 Hz]

General data	+/-	Values
Motor size	Max.	280S
Type of construction	Max.	B3
Starting method	-	Star a. Delta
Motor flange	-	Without
Weight without attachments [kg]	Max.	779
Housing material ³⁰⁾	-	1800
Enclosure	-	IP55
Nominal frequency	[Hz]	50/60
Efficiency class	-	IE4 (Super Premium)
Thermal class F to B	-	Yes
Temperature sensor	-	03
Operation on frequency inverter permitted	-	No
Duty type	-	Continuous running duty (S1)
Direction of rotation	-	Bi-directional
Motor bearings insulated	-	No
Position of fixed motor bearing	-	Non drive side
Fixed motor bearing reinforced	-	Without
Radial motor bearing reinforced	-	No
Condensate drain	-	Yes
Bolted feet	-	Yes

Table 134: Efficiency

Electrical data		+/-	50 Hz			60 Hz		
			4/4	3/4	2/4	4/4	3/4	2/4
Efficiency	[%]	Min.	96	95.8	95.3	96.2	95.7	94.8
Power factor		Min.	0.83	0.8	0.7	0.82	0.77	0.66

Table 135: Electrical data

Electrical data		+/-	50 Hz		60 Hz	
			Δ	Y	Δ	Y
Rated voltage	[V]	-	400	690	460	-
Rated current	[A]	-10 %	136	78.8	119	-
Rated power	[kW]	-	75		75	
Rated speed	[rpm]	Min.	1486		1788	
Rated torque	[Nm]	Min.	482		401	
Starting current	[IA/IN]	+20 %	7.7		8.7	

Table 136: Mechanical data

Mechanical data		+/-	Values	
			50 Hz	60 Hz
Sound pressure level	[dB]	Max.	69	73
Moment of inertia	[kgm ²]	Max.	1.7	1.7
Motor canopy		-	No	

Table 137: Explosion protection

Explosion protection	+/-	Values
Type of protection	-	Without

Table 138: Environment

Environment	+/-	Values
Ambient temperature [°C]	Max.	40

Table 139: Terminal box

Explosion protection	+/-	Values
Enclosure of terminal box	-	IP55 (TEFC)
Cable entry (O)	-	2XM63X1,5

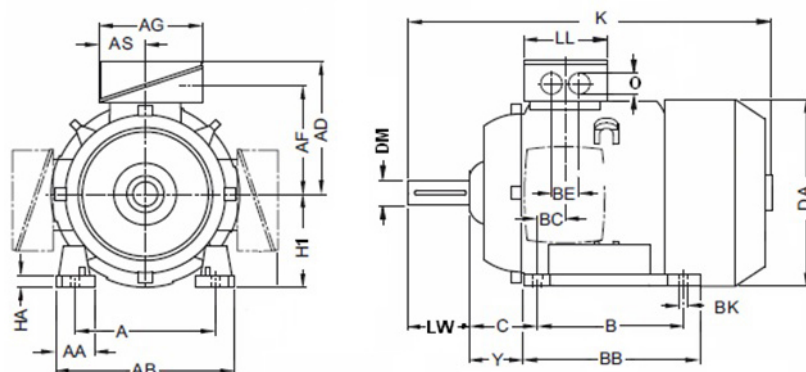


Fig. 15: Envelope dimensions

Table 140: Envelope dimensions A-DFL

A	AA	AB	AD	AF	AG	AS	B	BB	BC	BE	BK	C	DA	DFL
457	100	557	472	391	312	140.4	368	517	76	126	24	190	580	-

Table 141: Envelope dimensions DM-Z

DM	H1	HA	HH	K	LA	LB	LL	LW	M	N	S	T	Y	Z
75	280	42	-	1071	-	-	314	140	-	-	-	-	141	-

²⁹⁾ The name plate data may vary within the tolerances.

³⁰⁾ 1800=grey cast iron, 4400=aluminium

B3 type of construction, 4-pole, 280M

Material: 4-90,0-280M-BWE7F3NGFNWWNWKWC³¹⁾

Material number: 05214323 (5214339)

Table 142: General data [50 Hz]

General data	+/-	Values
Motor size	Max.	280M
Type of construction	Max.	B3
Starting method	-	Star a. Delta
Motor flange	-	Without
Weight without attachments [kg]	Max.	840
Housing material ³²⁾	-	1800
Enclosure	-	IP55
Nominal frequency	[Hz]	50/60
Efficiency class	-	IE4 (Super Premium)
Thermal class F to B	-	Yes
Temperature sensor	-	03
Operation on frequency inverter permitted	-	No
Duty type	-	Continuous running duty (S1)
Direction of rotation	-	Bi-directional
Motor bearings insulated	-	No
Position of fixed motor bearing	-	Non drive side
Fixed motor bearing reinforced	-	Without
Radial motor bearing reinforced	-	No
Condensate drain	-	Yes
Bolted feet	-	Yes

Table 143: Efficiency

Electrical data		+/-	50 Hz			60 Hz		
			4/4	3/4	2/4	4/4	3/4	2/4
Efficiency	[%]	Min.	96.1	96	95.5	96.2	96.1	95.5
Power factor		Min.	0.84	0.81	0.72	0.83	0.79	0.69

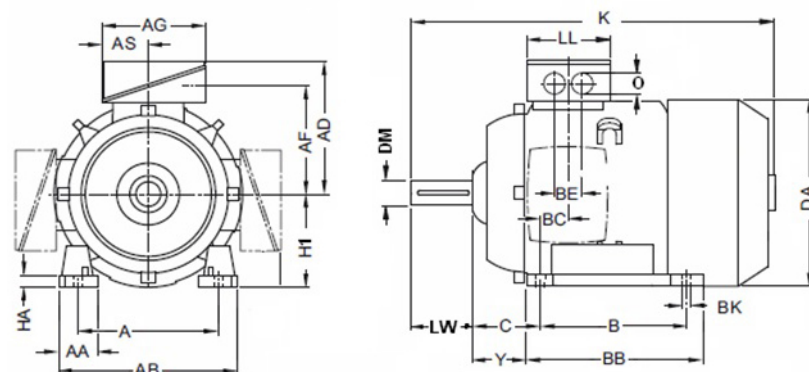


Fig. 16: Envelope dimensions

Table 149: Envelope dimensions A-DFL

A	AA	AB	AD	AF	AG	AS	B	BB	BC	BE	BK	C	DA	DFL
457	100	557	472	391	312	140.4	419	517	76	126	24	190	580	-

Table 150: Envelope dimensions DM-Z

DM	H1	HA	HH	K	LA	LB	LL	LW	M	N	S	T	Y	Z
75	280	42	-	1071	-	-	314	140	-	-	-	-	141	-

Table 144: Electrical data

Electrical data		+/-	50 Hz		60 Hz	
			Δ	Y	Δ	Y
Rated voltage	[V]	-	400	690	460	-
Rated current	[A]	-10 %	161	93.3	141	-
Rated power	[kW]	-	90		90	
Rated speed	[rpm]	Min.	1486		1788	
Rated torque	[Nm]	Min.	579		481	
Starting current	[IA/IN]	+20 %	8		8.9	

Table 145: Mechanical data

Mechanical data		+/-	Values	
			50 Hz	60 Hz
Sound pressure level	[dB]	Max.	69	73
Moment of inertia	[kgm ²]	Max.	2	2
Motor canopy		-	No	

Table 146: Explosion protection

Explosion protection	+/-	Values
Type of protection	-	Without

Table 147: Environment

Environment	+/-	Values
Ambient temperature [°C]	Max.	40

Table 148: Terminal box

Explosion protection	+/-	Values
Enclosure of terminal box	-	IP55 (TEFC)
Cable entry (O)	-	2XM63X1,5

31 The name plate data may vary within the tolerances.

32 1800=grey cast iron, 4400=aluminium

B3 type of construction, 4-pole, 315M

Material: 4-110 -315M-BWE7F3NGHNWWNWKWC³³⁾

Material number: 05214324 (5214340)

Table 151: General data [50 Hz]

General data		+/-	Values
Motor size		Max.	315M
Type of construction		Max.	B3
Starting method		-	Star a. Delta
Motor flange		-	Without
Weight without attachments	[kg]	Max.	987
Housing material ³⁴⁾		-	1800
Enclosure		-	IP55
Nominal frequency	[Hz]		50/60
Efficiency class		-	IE4 (Super Premium)
Thermal class F to B		-	Yes
Temperature sensor		-	03
Operation on frequency inverter permitted		-	No
Duty type		-	Continuous running duty (S1)
Direction of rotation		-	Bi-directional
Motor bearings insulated		-	No
Position of fixed motor bearing		-	Non drive side
Fixed motor bearing reinforced		-	Without
Radial motor bearing reinforced		-	No
Condensate drain		-	Yes
Bolted feet		-	No

Table 152: Efficiency

Electrical data		+/-	50 Hz			60 Hz		
			4/4	3/4	2/4	4/4	3/4	2/4
Efficiency	[%]	Min.	96.3	95.4	96	96.2	96.1	95.3
Power factor		Min.	0.83	0.79	0.69	0.82	0.77	0.67

Table 153: Electrical data

Electrical data		+/-	50 Hz		60 Hz	
			Δ	Y	Δ	Y
Rated voltage	[V]	-	400	690	460	-
Rated current	[A]	-10 %	199	115	175	-
Rated power	[kW]	-	110		110	
Rated speed	[rpm]	Min.	1491		1792	
Rated torque	[Nm]	Min.	706		586	
Starting current	[IA/IN]	+20 %	7		7.9	

Table 154: Mechanical data

Mechanical data		+/-	Values	
			50 Hz	60 Hz
Sound pressure level	[dB]	Max.	71	75
Moment of inertia	[kgm ²]	Max.	2.7	2.7
Motor canopy		-	No	

Table 155: Explosion protection

Explosion protection	+/-	Values
Type of protection	-	Without

Table 156: Environment

Environment	+/-	Values
Ambient temperature	[°C]	Max. 40

Table 157: Terminal box

Explosion protection	+/-	Values
Enclosure of terminal box	-	IP55 (TEFC)
Cable entry (O)	-	2XM63X1,5

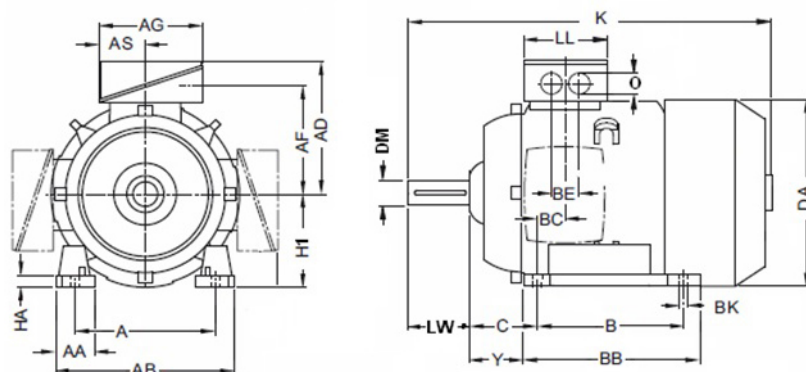

Fig. 17: Envelope dimensions

Table 158: Envelope dimensions A-DFL

A	AA	AB	AD	AF	AG	AS	B	BB	BC	BE	BK	C	DA	DFL
508	120	630	530	432	382	171.9	457	626	48	160	28	216	644	-

Table 159: Envelope dimensions DM-Z

DM	H1	HA	HH	K	LA	LB	LL	LW	M	N	S	T	Y	Z
80	315	48	-	1274	-	-	379	170	-	-	-	-	141	-

³³ The name plate data may vary within the tolerances.

³⁴ 1800=grey cast iron, 4400=aluminium

B3 type of construction, 4-pole, 315M

Material: 4-132 -315M-BWE7F3NGHNWWNWKWC³⁵⁾

Material number: 05214276 (5214332)

Table 160: General data [50 Hz]

General data	+/-	Values
Motor size	Max.	315M
Type of construction	Max.	B3
Starting method	-	Star a. Delta
Motor flange	-	Without
Weight without attachments [kg]	Max.	1035
Housing material ³⁶⁾	-	1800
Enclosure	-	IP55
Nominal frequency	[Hz]	50/60
Efficiency class	-	IE4 (Super Premium)
Thermal class F to B	-	Yes
Temperature sensor	-	03
Operation on frequency inverter permitted	-	No
Duty type	-	Continuous running duty (S1)
Direction of rotation	-	Bi-directional
Motor bearings insulated	-	No
Position of fixed motor bearing	-	Non drive side
Fixed motor bearing reinforced	-	Without
Radial motor bearing reinforced	-	No
Condensate drain	-	Yes
Bolted feet	-	No

Table 161: Efficiency

Electrical data		+/-	50 Hz			60 Hz		
			4/4	3/4	2/4	4/4	3/4	2/4
Efficiency	[%]	Min.	96.4	96.1	95.4	96.5	96.2	95.4
Power factor		Min.	0.83	0.78	0.68	0.82	0.77	0.66

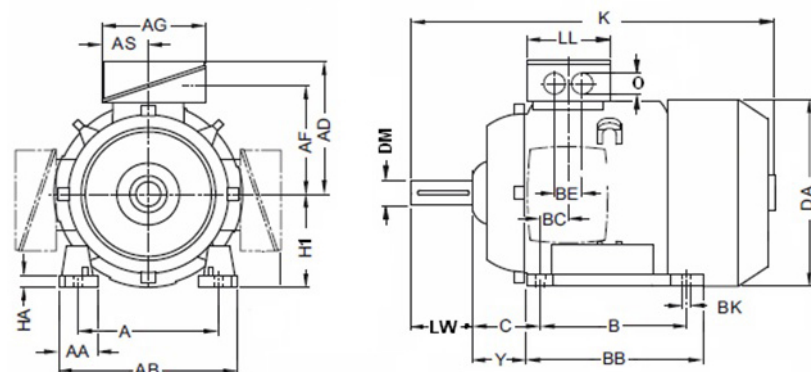

Fig. 18: Envelope dimensions

Table 167: Envelope dimensions A-DFL

A	AA	AB	AD	AF	AG	AS	B	BB	BC	BE	BK	C	DA	DFL
508	120	630	530	432	382	171.9	457	626	48	160	28	216	644	-

Table 168: Envelope dimensions DM-Z

DM	H1	HA	HH	K	LA	LB	LL	LW	M	N	S	T	Y	Z
80	315	48	-	1274	-	-	379	170	-	-	-	-	146	-

Table 162: Electrical data

Electrical data		+/-	50 Hz		60 Hz	
			Δ	Y	Δ	Y
Rated voltage	[V]	-	400	690	460	-
Rated current	[A]	-10 %	238	138	209	-
Rated power	[kW]	-	132		132	
Rated speed	[rpm]	Min.	1491		1792	
Rated torque	[Nm]	Min.	847		704	
Starting current	[I _A /I _N]	+20 %	7.3		8.3	

Table 163: Mechanical data

Mechanical data		+/-	Values	
			50 Hz	60 Hz
Sound pressure level	[dB]	Max.	71	75
Moment of inertia	[kgm²]	Max.	3.1	3.1
Motor canopy		-	No	

Table 164: Explosion protection

Explosion protection	+/-	Values
Type of protection	-	Without

Table 165: Environment

Environment	+/-	Values
Ambient temperature [°C]	Max.	40

Table 166: Terminal box

Explosion protection	+/-	Values
Enclosure of terminal box	-	IP55 (TEFC)
Cable entry (Ø)	-	2XM63X1,5

³⁵⁾ The name plate data may vary within the tolerances.

³⁶⁾ 1800=grey cast iron, 4400=aluminium

V1 type of construction, 2-pole, 180M

Material: 2-22,0-180M-BWE7F3NHWDWFUWWEI³⁷⁾

Material number: 5443313 (05407841)

Table 169: General data [50 Hz]

General data		+/-	Values
Motor size		Max.	180M
Type of construction		Max.	V1
Starting method		-	Star a. Delta
Motor flange		-	IEC 60072 Type FF
Weight without attachments	[kg]	Max.	180
Housing material ³⁸⁾		-	1800
Enclosure		-	IP55
Nominal frequency	[Hz]		50/60
Efficiency class		-	IE4 (Super Premium)
Thermal class F to B		-	Yes
Temperature sensor		-	03
Operation on frequency inverter permitted		-	Yes (acc to motor manufact)
Duty type		-	Continuous running duty (S1)
Direction of rotation		-	Bi-directional
Motor bearings insulated		-	No
Position of fixed motor bearing		-	Drive end
Fixed motor bearing reinforced		-	Without
Radial motor bearing reinforced		-	No
Condensate drain		-	Yes
Bolted feet		-	No

Table 170: Efficiency

Electrical data		+/-	50 Hz			60 Hz		
			4/4	3/4	2/4	4/4	3/4	2/4
Efficiency	[%]	Min.	94	93.6	92.5	93	93	92.2
Power factor		Min.	0.86	0.82	0.73	0.85	0.81	0.7

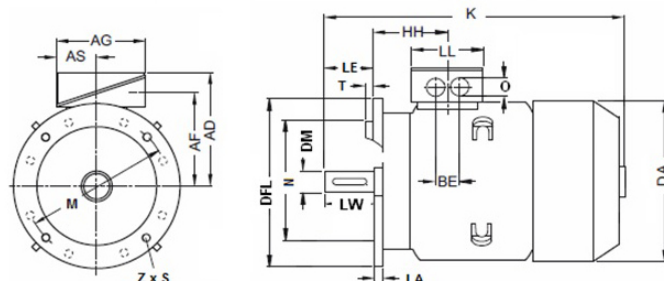


Fig. 19: Envelope dimensions

Table 176: Envelope dimensions A-DFL

A	AA	AB	AD	AF	AG	AS	B	BB	BC	BE	BK	C	DA	DFL
			294	239.7	191.5	102.5				84			389	350

Table 177: Envelope dimensions DM-Z

DM	H1	HA	HH	K	LA	LB	LL	LE	M	N	S	T	Y	Z
48			241.5	684.5	16	574.5	198.5		300	250	18.5	5		4

Table 171: Electrical data

Electrical data		+/-	50 Hz		60 Hz	
			Δ	Y	Δ	Y
Rated voltage	[V]	-	400	690	460	-
Rated current	[A]	-10 %	39.3	22.7	34.9	-
Rated power	[kW]	-	22		22	
Rated speed	[rpm]	Min.	2960		3565	
Rated torque	[Nm]	Min.	71		59	
Starting current	[IA/IN]	+20 %	9.7		11	

Table 172: Mechanical data

Mechanical data		+/-	Values	
			50 Hz	60 Hz
Sound pressure level	[dB]	Max.	70	75
Moment of inertia	[kgm²]	Max.	0.09	0.09
Motor canopy		-	No	

Table 173: Explosion protection

Explosion protection	+/-	Values
Type of protection	-	Without

Table 174: Environment

Environment	+/-	Values
Ambient temperature	[°C]	Max. 40

Table 175: Terminal box

Explosion protection	+/-	Values
Enclosure of terminal box	-	IP55 (TEFC)
Cable entry (Ø)	-	2XM40X1,5

37 The name plate data may vary within the tolerances.

38 1800=grey cast iron, 4400=aluminium

V1 type of construction, 2-pole, 200M/L

Material: 2-30,0-200L-BWE7F3NHWDWFUWWE1³⁹⁾

Material number: 5443314 (05407842)

Table 178: General data [50 Hz]

General data	+/-	Values
Motor size	Max.	200L
Type of construction	Max.	V1
Starting method	-	Star a. Delta
Motor flange	-	IEC 60072 Type FF
Weight without attachments [kg]	Max.	210
Housing material ⁴⁰⁾	-	1800
Enclosure	-	IP55
Nominal frequency	[Hz]	50/60
Efficiency class	-	IE4 (Super Premium)
Thermal class F to B	-	Yes
Temperature sensor	-	03
Operation on frequency inverter permitted	-	Yes (acc to motor manufact)
Duty type	-	Continuous running duty (S1)
Direction of rotation	-	Bi-directional
Motor bearings insulated	-	No
Position of fixed motor bearing	-	Drive end
Fixed motor bearing reinforced	-	Without
Radial motor bearing reinforced	-	No
Condensate drain	-	Yes
Bolted feet	-	No

Table 179: Efficiency

Electrical data		+/-	50 Hz			60 Hz		
			4/4	3/4	2/4	4/4	3/4	2/4
Efficiency	[%]	Min.	94.5	94	92.8	93.6	93.4	91.7
Power factor		Min.	0.84	0.8	0.71	0.83	0.78	0.68

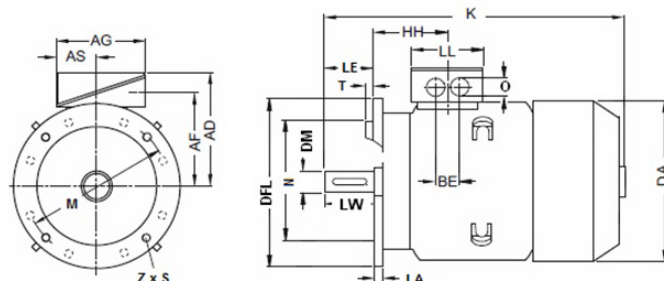

Fig. 20: Envelope dimensions

Table 185: Envelope dimensions A-DFL

A	AA	AB	AD	AF	AG	AS	B	BB	BC	BE	BK	C	DA	DFL
			334	272.7	218	119.1				94			422.5	400

Table 186: Envelope dimensions DM-Z

DM	H1	HA	HH	K	LA	LB	LL	LE	M	N	S	T	Y	Z
55			209	767.5	18	657.5	228.5		350	300	18.5	5		4

Table 180: Electrical data

Electrical data		+/-	50 Hz		60 Hz	
			Δ	Y	Δ	Y
Rated voltage	[V]	-	400	690	460	-
Rated current	[A]	-10 %	54.5	31.5	48.5	-
Rated power	[kW]	-	30		30	
Rated speed	[rpm]	Min.	30		3570	
Rated torque	[Nm]	Min.	96.6		80.2	
Starting current	[IA/IN]	+20 %	8.9		10	

Table 181: Mechanical data

Mechanical data		+/-	Values	
			50 Hz	60 Hz
Sound pressure level	[dB]	Max.	75	80
Moment of inertia	[kgm²]	Max.	0.16	0.16
Motor canopy		-	No	

Table 182: Explosion protection

Explosion protection	+/-	Values
Type of protection	-	Without

Table 183: Environment

Environment	+/-	Values
Ambient temperature [°C]	Max.	40

Table 184: Terminal box

Explosion protection	+/-	Values
Enclosure of terminal box	-	IP55 (TEFC)
Cable entry (Ø)	-	2XM50X1,5

³⁹⁾ The name plate data may vary within the tolerances.

⁴⁰⁾ 1800=grey cast iron, 4400=aluminium

V1 type of construction, 2-pole, 200M/L

Material: 2-37,0-200L-BWE7F3NHWDWFUWWE1⁴¹⁾

Material number: 5443315 (05407843)

Table 187: General data [50 Hz]

General data	+/-	Values
Motor size	Max.	200L
Type of construction	Max.	V1
Starting method	-	Star a. Delta
Motor flange	-	IEC 60072 Type FF
Weight without attachments [kg]	Max.	260
Housing material ⁴²⁾	-	1800
Enclosure	-	IP55
Nominal frequency	[Hz]	50/60
Efficiency class	-	IE4 (Super Premium)
Thermal class F to B	-	Yes
Temperature sensor	-	03
Operation on frequency inverter permitted	-	Yes (acc to motor manufact)
Duty type	-	Continuous running duty (S1)
Direction of rotation	-	Bi-directional
Motor bearings insulated	-	No
Position of fixed motor bearing	-	Drive end
Fixed motor bearing reinforced	-	Without
Radial motor bearing reinforced	-	No
Condensate drain	-	Yes
Bolted feet	-	No

Table 188: Efficiency

Electrical data		+/-	50 Hz			60 Hz		
			4/4	3/4	2/4	4/4	3/4	2/4
Efficiency	[%]	Min.	94.8	94.3	93.3	94.1	93.8	92.4
Power factor		Min.	0.85	0.81	0.72	0.84	0.79	0.7

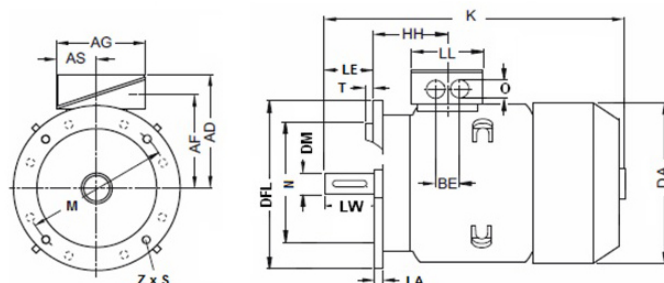


Fig. 21: Envelope dimensions

Table 194: Envelope dimensions A-DFL

A	AA	AB	AD	AF	AG	AS	B	BB	BC	BE	BK	C	DA	DFL
			334	272.7	218	119.1				94			422.5	400

Table 195: Envelope dimensions DM-Z

DM	H1	HA	HH	K	LA	LB	LL	LE	M	N	S	T	Y	Z
55			209	767.5	18	657.5	228.5		350	300	18.5	5		4

Table 189: Electrical data

Electrical data		+/-	50 Hz		60 Hz	
			Δ	Y	Δ	Y
Rated voltage	[V]	-	400	690	460	-
Rated current	[A]	-10 %	66.3	38.3	58.8	-
Rated power	[kW]	-	37		37	
Rated speed	[rpm]	Min.	2965		3570	
Rated torque	[Nm]	Min.	119.7		99.1	
Starting current	[I _A /I _N]	+20 %	9.2		10.2	

Table 190: Mechanical data

Mechanical data		+/-	Values	
			50 Hz	60 Hz
Sound pressure level	[dB]	Max.	75	80
Moment of inertia	[kgm²]	Max.	0.19	0.19
Motor canopy		-	No	

Table 191: Explosion protection

Explosion protection	+/-	Values
Type of protection	-	Without

Table 192: Environment

Environment	+/-	Values
Ambient temperature [°C]	Max.	40

Table 193: Terminal box

Explosion protection	+/-	Values
Enclosure of terminal box	-	IP55 (TEFC)
Cable entry (O)	-	2XM50X1,5

41 The name plate data may vary within the tolerances.

42 1800=grey cast iron, 4400=aluminium

V1 type of construction, 2-pole, 225S/M

Material: 2-45,0-225M-BWE7F3NHWDWFUWWE⁴³⁾

Material number: 5443316 (05407844)

Table 196: General data [50 Hz]

General data	+/-	Values
Motor size	Max.	225M
Type of construction	Max.	V1
Starting method	-	Star a. Delta
Motor flange	-	IEC 60072 Type FF
Weight without attachments [kg]	Max.	415
Housing material ⁴⁴⁾	-	1800
Enclosure	-	IP55
Nominal frequency	[Hz]	50/60
Efficiency class	-	IE4 (Super Premium)
Thermal class F to B	-	Yes
Temperature sensor	-	03
Operation on frequency inverter permitted	-	Yes (acc to motor manufact)
Duty type	-	Continuous running duty (S1)
Direction of rotation	-	Bi-directional
Motor bearings insulated	-	No
Position of fixed motor bearing	-	Drive end
Fixed motor bearing reinforced	-	Without
Radial motor bearing reinforced	-	No
Condensate drain	-	Yes
Bolted feet	-	No

Table 197: Efficiency

Electrical data		+/-	50 Hz			60 Hz		
			4/4	3/4	2/4	4/4	3/4	2/4
Efficiency	[%]	Min.	95	94.3	93.2	94.5	94.1	92.6
Power factor		Min.	0.87	0.83	0.74	0.86	0.81	0.72

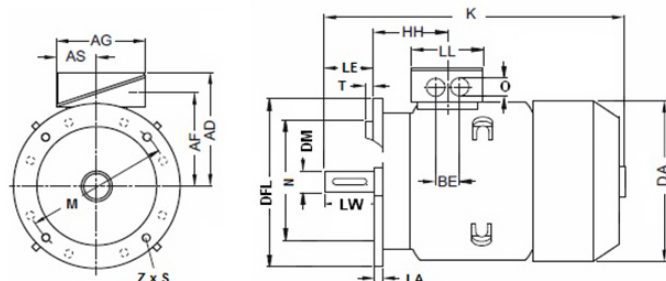


Fig. 22: Envelope dimensions

Table 203: Envelope dimensions A-DFL

A	AA	AB	AD	AF	AG	AS	B	BB	BC	BE	BK	C	DA	DFL
			397	314.8	286	165.6				110			488	450

Table 204: Envelope dimensions DM-Z

DM	H1	HA	HH	K	LA	LB	LL	LE	M	N	S	T	Y	Z
55			203.5	856.5	18	746.5	269		400	350	18.5	5		8

Table 198: Electrical data

Electrical data		+/-	50 Hz		60 Hz	
			Δ	Y	Δ	Y
Rated voltage	[V]	-	400	690	460	-
Rated current	[A]	-10 %	78.6	45.4	69.5	-
Rated power	[kW]	-	45		45	
Rated speed	[rpm]	Min.	2970		3575	
Rated torque	[Nm]	Min.	145.2		120.7	
Starting current	[IA/IN]	+20 %	9.6		10.5	

Table 199: Mechanical data

Mechanical data		+/-	Values	
			50 Hz	60 Hz
Sound pressure level	[dB]	Max.	77	82
Moment of inertia	[kgm²]	Max.	0.29	0.29
Motor canopy		-	No	

Table 200: Explosion protection

Explosion protection	+/-	Values
Type of protection	-	Without

Table 201: Environment

Environment	+/-	Values
Ambient temperature [°C]	Max.	40

Table 202: Terminal box

Explosion protection	+/-	Values
Enclosure of terminal box	-	IP55 (TEFC)
Cable entry (Ø)	-	2XM50X1,5

⁴³ The name plate data may vary within the tolerances.

⁴⁴ 1800=grey cast iron, 4400=aluminium

V1 type of construction, 2-pole, 250S/M

Material: 2-55,0-250M-BWE7F3NHWDWFUWWE1⁴⁵⁾

Material number: 5443317 (05407845)

Table 205: General data [50 Hz]

General data	+/-	Values
Motor size	Max.	250M
Type of construction	Max.	V1
Starting method	-	Star a. Delta
Motor flange	-	IEC 60072 Type FF
Weight without attachments [kg]	Max.	450
Housing material ⁴⁶⁾	-	1800
Enclosure	-	IP55
Nominal frequency	[Hz]	50/60
Efficiency class	-	IE4 (Super Premium)
Thermal class F to B	-	Yes
Temperature sensor	-	03
Operation on frequency inverter permitted	-	Yes (acc to motor manufact)
Duty type	-	Continuous running duty (S1)
Direction of rotation	-	Bi-directional
Motor bearings insulated	-	No
Position of fixed motor bearing	-	Drive end
Fixed motor bearing reinforced	-	Without
Radial motor bearing reinforced	-	No
Condensate drain	-	Yes
Bolted feet	-	No

Table 206: Efficiency

Electrical data		+/-	50 Hz			60 Hz		
			4/4	3/4	2/4	4/4	3/4	2/4
Efficiency	[%]	Min.	95.3	94.5	93.5	94.5	94.2	92.7
Power factor		Min.	0.87	0.83	0.75	0.86	0.82	0.73

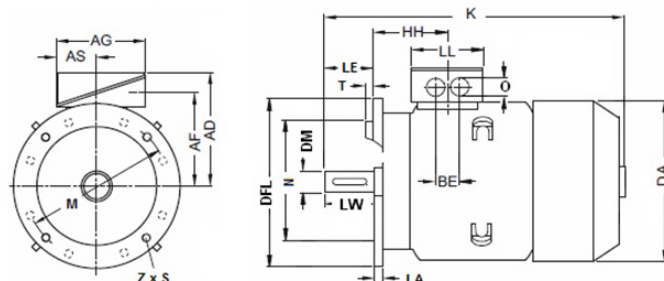


Fig. 23: Envelope dimensions

Table 212: Envelope dimensions A-DFL

A	AA	AB	AD	AF	AG	AS	B	BB	BC	BE	BK	C	DA	DFL
			417	334.8	286	165.6				110			528	550

Table 213: Envelope dimensions DM-Z

DM	H1	HA	HH	K	LA	LB	LL	LE	M	N	S	T	Y	Z
60			217	965.5	20	825.5	269		500	450	18.5	5		8

Table 207: Electrical data

Electrical data		+/-	50 Hz		60 Hz	
			Δ	Y	Δ	Y
Rated voltage	[V]	-	400	690	460	-
Rated current	[A]	-10 %	95.7	55.3	84.9	-
Rated power	[kW]	-	55		55	
Rated speed	[rpm]	Min.	2970		3570	
Rated torque	[Nm]	Min.	176.6		147.2	
Starting current	[I _A /I _N]	+20 %	9.3		10.5	

Table 208: Mechanical data

Mechanical data		+/-	Values	
			50 Hz	60 Hz
Sound pressure level	[dB]	Max.	77	82
Moment of inertia	[kgm²]	Max.	0.36	0.36
Motor canopy		-	No	

Table 209: Explosion protection

Explosion protection	+/-	Values
Type of protection	-	Without

Table 210: Environment

Environment	+/-	Values
Ambient temperature [°C]	Max.	40

Table 211: Terminal box

Explosion protection	+/-	Values
Enclosure of terminal box	-	IP55 (TEFC)
Cable entry (O)	-	2XM63X1,5

⁴⁵ The name plate data may vary within the tolerances.

⁴⁶ 1800=grey cast iron, 4400=aluminium

V1 type of construction, 2-pole, 280S

Material: 2-75,0-280S-BWE7F3NHWDWFWNWKWC⁴⁷⁾

Material number: 05214272 (5214248)

Table 214: General data [50 Hz]

General data	+/-	Values
Motor size	Max.	280S
Type of construction	Max.	V1
Starting method	-	Star a. Delta
Motor flange	-	IEC 60072 Type FF
Weight without attachments [kg]	Max.	724
Housing material ⁴⁸⁾	-	1800
Enclosure	-	IP55
Nominal frequency	[Hz]	50/60
Efficiency class	-	IE4 (Super Premium)
Thermal class F to B	-	Yes
Temperature sensor	-	03
Operation on frequency inverter permitted	-	No
Duty type	-	Continuous running duty (S1)
Direction of rotation	-	Bi-directional
Motor bearings insulated	-	No
Position of fixed motor bearing	-	Drive end
Fixed motor bearing reinforced	-	Without
Radial motor bearing reinforced	-	No
Condensate drain	-	Yes
Bolted feet	-	No

Table 215: Efficiency

Electrical data		+/-	50 Hz			60 Hz		
			4/4	3/4	2/4	4/4	3/4	2/4
Efficiency	[%]	min.	95.6	94.8	93.6	95	94.8	93
Power factor		Min.	0.87	0.84	0.75	0.86	0.83	0.73

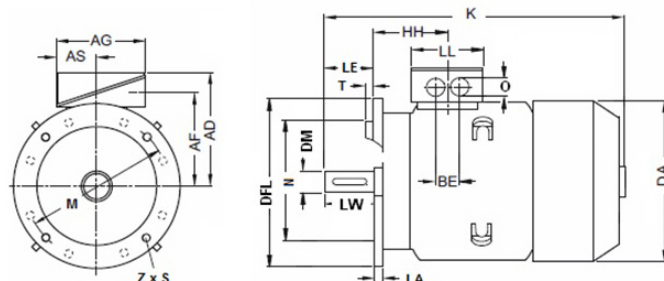

Fig. 24: Envelope dimensions

Table 221: Envelope dimensions A-DFL

A	AA	AB	AD	AF	AG	AS	B	BB	BC	BE	BK	C	DA	DFL
-	-	-	472	391	312	140.4	-	-	-	126	-	-	580	550

Table 222: Envelope dimensions DM-Z

DM	H1	HA	HH	K	LA	LB	LL	LE	M	N	S	T	Y	Z
65	-	-	266	1071	18	931	314	140	500	450	18.5	5	-	8

Table 216: Electrical data

Electrical data		+/-	50 Hz		60 Hz	
			Δ	Y	Δ	Y
Rated voltage	[V]	-	400	690	460	-
Rated current	[A]	-10 %	130	75.5	115	-
Rated power	[kW]	-	75		75	
Rated speed	[rpm]	Min.	2977		3579	
Rated torque	[Nm]	Min.	241		200	
Starting current	[IA/IN]	+20 %	8.3		9	

Table 217: Mechanical data

Mechanical data		+/-	Values	
			50 Hz	60 Hz
Sound pressure level	[dB]	Max.	77	81
Moment of inertia	[kgm²]	Max.	0.94	0.94
Motor canopy		-	No	

Table 218: Explosion protection

Explosion protection	+/-	Values
Type of protection	-	Without

Table 219: Environment

Environment	+/-	Values
Ambient temperature [°C]	Max.	40

Table 220: Terminal box

Explosion protection	+/-	Values
Enclosure of terminal box	-	IP55 (TEFC)
Cable entry (O)	-	2XM63X1,5

⁴⁷⁾ The name plate data may vary within the tolerances.

⁴⁸⁾ 1800=grey cast iron, 4400=aluminium

V1 type of construction, 2-pole, 280M

Material: 2-90,0-280M-BWE7F3NHWDWFNWKWC⁴⁹⁾

Material number: 05214273 (5214249)

Table 223: General data [50 Hz]

General data	+/-	Values
Motor size	Max.	280M
Type of construction	Max.	V1
Starting method	-	Star a. Delta
Motor flange	-	IEC 60072 Type FF
Weight without attachments [kg]	Max.	764
Housing material ⁵⁰⁾	-	1800
Enclosure	-	IP55
Nominal frequency	[Hz]	50/60
Efficiency class	-	IE4 (Super Premium)
Thermal class F to B	-	Yes
Temperature sensor	-	03
Operation on frequency inverter permitted	-	No
Duty type	-	Continuous running duty (S1)
Direction of rotation	-	Bi-directional
Motor bearings insulated	-	No
Position of fixed motor bearing	-	Drive end
Fixed motor bearing reinforced	-	Without
Radial motor bearing reinforced	-	No
Condensate drain	-	Yes
Bolted feet	-	No

Table 224: Efficiency

Electrical data	+/-	50 Hz			60 Hz		
		4/4	3/4	2/4	4/4	3/4	2/4
Efficiency [%]	Min.	95.8	95.2	94.3	95.4	95	93.5
Power factor	min.	0.88	0.85	0.78	0.87	0.84	0.76

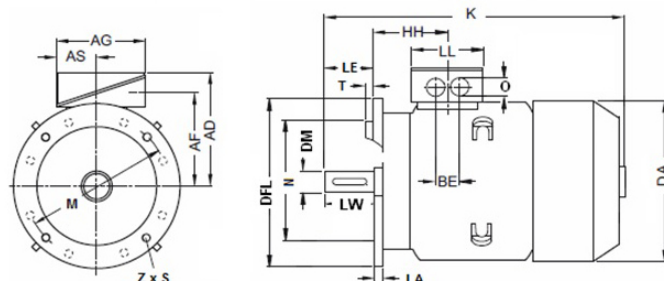


Fig. 25: Envelope dimensions

Table 230: Envelope dimensions A-DFL

A	AA	AB	AD	AF	AG	AS	B	BB	BC	BE	BK	C	DA	DFL
-	-	-	472	391	312	140.4	-	-	-	126	-	-	580	550

Table 231: Envelope dimensions DM-Z

DM	H1	HA	HH	K	LA	LB	LL	LE	M	N	S	T	Y	Z
65	-	-	266	1071	18	931	314	140	500	450	18.5	5	-	8

Table 225: Electrical data

Electrical data	+/-	50 Hz		60 Hz	
		Δ	Y	Δ	Y
Rated voltage [V]	-	400	690	460	-
Rated current [A]	-10 %	154	89.3	136	-
Rated power [kW]	-	90		90	
Rated speed [rpm]	min.	2975		3578	
Rated torque [Nm]	Min.	289		240	
Starting current [IA/IN]	+20 %	7.9		8.7	

Table 226: Mechanical data

Mechanical data	+/-	Values	
		50 Hz	60 Hz
Sound pressure level [dB]	Max.	77	81
Moment of inertia [kgm²]	Max.	1	1
Motor canopy	-	No	

Table 227: Explosion protection

Explosion protection	+/-	Values
Type of protection	-	Without

Table 228: Environment

Environment	+/-	Values
Ambient temperature [°C]	Max.	40

Table 229: Terminal box

Explosion protection	+/-	Values
Enclosure of terminal box	-	IP55 (TEFC)
Cable entry (O)	-	2XM63X1,5

49 The name plate data may vary within the tolerances.

50 1800=grey cast iron, 4400=aluminium

V1 type of construction, 2-pole, 315S

Material: 2-110 -315S-BWE7F3NHWDFNWKWC⁵¹⁾

Material number: 05214274 (5214330)

Table 232: General data [50 Hz]

General data	+/-	Values
Motor size	Max.	315S
Type of construction	Max.	V1
Starting method	-	Star a. Delta
Motor flange	-	IEC 60072 Type FF
Weight without attachments [kg]	Max.	976
Housing material ⁵²⁾	-	1800
Enclosure	-	IP55
Nominal frequency	[Hz]	50/60
Efficiency class	-	IE4 (Super Premium)
Thermal class F to B	-	Yes
Temperature sensor	-	03
Operation on frequency inverter permitted	-	No
Duty type	-	Continuous running duty (S1)
Direction of rotation	-	Bi-directional
Motor bearings insulated	-	No
Position of fixed motor bearing	-	Drive end
Fixed motor bearing reinforced	-	Without
Radial motor bearing reinforced	-	No
Condensate drain	-	Yes
Bolted feet	-	No

Table 233: Efficiency

Electrical data		+/-	50 Hz			60 Hz		
			4/4	3/4	2/4	4/4	3/4	2/4
Efficiency	[%]	Min.	96	95.5	94	95.4	95.4	94.5
Power factor		Min.	0.87	0.84	0.76	0.86	0.83	0.74

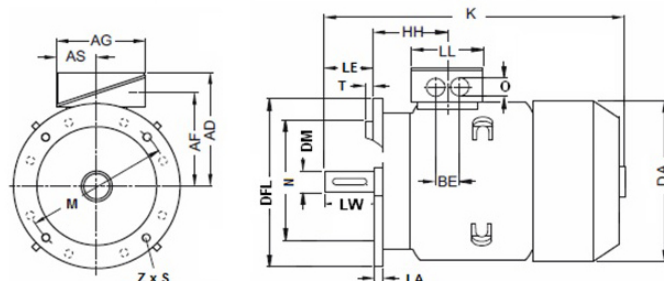


Fig. 26: Envelope dimensions

Table 239: Envelope dimensions A-DFL

A	AA	AB	AD	AF	AG	AS	B	BB	BC	BE	BK	C	DA	DFL
-	-	-	530	432	382	171.9	-	-	-	160	-	-	644	660

Table 240: Envelope dimensions DM-Z

DM	H1	HA	HH	K	LA	LB	LL	LE	M	N	S	T	Y	Z
65	-	-	264	1244	22	1104	379	140	600	550	24	6	-	8

Table 234: Electrical data

Electrical data		+/-	50 Hz		60 Hz	
			Δ	Y	Δ	Y
Rated voltage	[V]	-	400	690	460	-
Rated current	[A]	-10 %	190	110	168	-
Rated power	[kW]	-	110		110	
Rated speed	[rpm]	Min.	2976		3577	
Rated torque	[Nm]	Min.	353		294	
Starting current	[I _A /I _N]	+20 %	8		8.5	

Table 235: Mechanical data

Mechanical data		+/-	Values	
			50 Hz	60 Hz
Sound pressure level	[dB]	Max.	77	81
Moment of inertia	[kgm²]	Max.	1.4	1.4
Motor canopy		-	No	

Table 236: Explosion protection

Explosion protection	+/-	Values
Type of protection	-	Without

Table 237: Environment

Environment	+/-	Values
Ambient temperature [°C]	Max.	40

Table 238: Terminal box

Explosion protection	+/-	Values
Enclosure of terminal box	-	IP55 (TEFC)
Cable entry (O)	-	2XM63X1,5

⁵¹ The name plate data may vary within the tolerances.

⁵² 1800=grey cast iron, 4400=aluminium

V1 type of construction, 2-pole, 315M

Material: 2-132 -315M-BWE7F3NHWDWFWNWKWC⁵³⁾

Material number: 05214275 (5214331)

Table 241: General data [50 Hz]

General data		+/-	Values
Motor size		Max.	315M
Type of construction		Max.	V1
Starting method		-	Star a. Delta
Motor flange		-	IEC 60072 Type FF
Weight without attachments	[kg]	Max.	1080
Housing material ⁵⁴⁾		-	1800
Enclosure		-	IP55
Nominal frequency	[Hz]		50/60
Efficiency class		-	IE4 (Super Premium)
Thermal class F to B		-	Yes
Temperature sensor		-	03
Operation on frequency inverter permitted		-	No
Duty type		-	Continuous running duty (S1)
Direction of rotation		-	Bi-directional
Motor bearings insulated		-	No
Position of fixed motor bearing		-	Drive end
Fixed motor bearing reinforced		-	Without
Radial motor bearing reinforced		-	No
Condensate drain		-	Yes
Bolted feet		-	No

Table 242: Efficiency

Electrical data		+/-	50 Hz			60 Hz		
			4/4	3/4	2/4	4/4	3/4	2/4
Efficiency	[%]	Min.	96.2	95.9	95.1	95.8	95	93
Power factor		min.	0.88	0.85	0.79	0.88	0.85	0.77

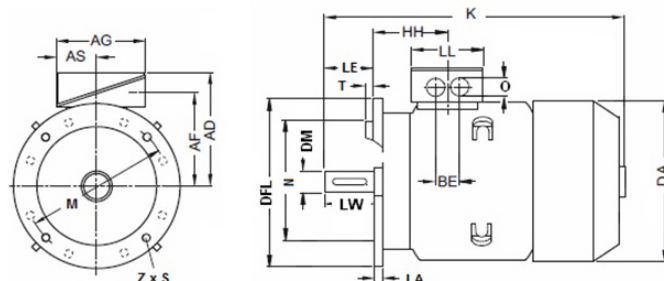

Fig. 27: Envelope dimensions

Table 248: Envelope dimensions A-DFL

A	AA	AB	AD	AF	AG	AS	B	BB	BC	BE	BK	C	DA	DFL
-	-	-	530	432	382	171.9	-	-	-	160	-	-	644	660

Table 249: Envelope dimensions DM-Z

DM	H1	HA	HH	K	LA	LB	LL	LE	M	N	S	T	Y	Z
65	-	-	264	1244	22	1104	379	140	600	550	24	6	-	8

Table 243: Electrical data

Electrical data		+/-	50 Hz		60 Hz	
			Δ	Y	Δ	Y
Rated voltage	[V]	-	400	690	460	-
Rated current	[A]	-10 %	225	130	197	-
Rated power	[kW]	-	132		132	
Rated speed	[rpm]	Min.	2983		3585	
Rated torque	[Nm]	Min.	423		352	
Starting current	[IA/IN]	+20 %	8.3		9.2	

Table 244: Mechanical data

Mechanical data		+/-	Values	
			50 Hz	60 Hz
Sound pressure level	[dB]	Max.	77	81
Moment of inertia	[kgm²]	Max.	1.9	1.9
Motor canopy		-	No	

Table 245: Explosion protection

Explosion protection	+/-	Values
Type of protection	-	Without

Table 246: Environment

Environment	+/-	Values
Ambient temperature	[°C]	Max. 40

Table 247: Terminal box

Explosion protection	+/-	Values
Enclosure of terminal box	-	IP55 (TEFC)
Cable entry (O)	-	2XM63X1,5

53 The name plate data may vary within the tolerances.

54 1800=grey cast iron, 4400=aluminium

V1 type of construction, 4-pole, 180L

Material: 4-22,0-180L-BWE7F3NHWDWFUWWE1⁵⁵⁾

Material number: 5443318 (05407846)

Table 250: General data [50 Hz]

General data	+/-	Values
Motor size	Max.	180L
Type of construction	Max.	V1
Starting method	-	Star a. Delta
Motor flange	-	IEC 60072 Type FF
Weight without attachments [kg]	Max.	220
Housing material ⁵⁶⁾	-	1800
Enclosure	-	IP55
Nominal frequency	[Hz]	50/60
Efficiency class	-	IE4 (Super Premium)
Thermal class F to B	-	Yes
Temperature sensor	-	03
Operation on frequency inverter permitted	-	Yes (acc to motor manufact)
Duty type	-	Continuous running duty (S1)
Direction of rotation	-	Bi-directional
Motor bearings insulated	-	No
Position of fixed motor bearing	-	Drive end
Fixed motor bearing reinforced	-	Without
Radial motor bearing reinforced	-	No
Condensate drain	-	Yes
Bolted feet	-	No

Table 251: Efficiency

Electrical data		+/-	50 Hz			60 Hz		
			4/4	3/4	2/4	4/4	3/4	2/4
Efficiency	[%]	Min.	94.5	94.1	93.2	94.5	0.72	92.9
Power factor		Min.	0.81	0.74	0.62	0.79	0.72	0.6

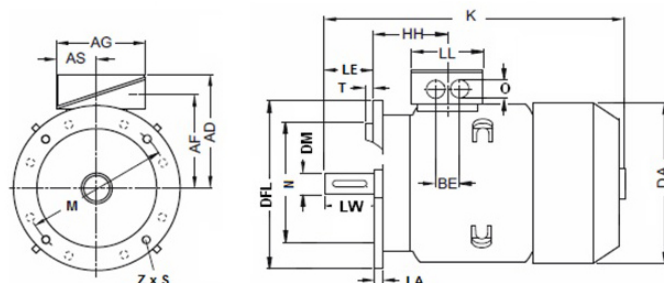

Fig. 28: Envelope dimensions

Table 257: Envelope dimensions A-DFL

A	AA	AB	AD	AF	AG	AS	B	BB	BC	BE	BK	C	DA	DFL
			294	239.7	191.5	102.5				84			389	350

Table 258: Envelope dimensions DM-Z

DM	H1	HA	HH	K	LA	LB	LL	LE	M	N	S	T	Y	Z
48			212.5	722.5	16	612.5	198.5		300	250	18.5	5		4

Table 252: Electrical data

Electrical data		+/-	50 Hz		60 Hz	
			Δ	Y	Δ	Y
Rated voltage	[V]	-	400	690	460	-
Rated current	[A]	-10 %	41.5	24	37	-
Rated power	[kW]	-	22		22	
Rated speed	[rpm]	Min.	1476		1778	
Rated torque	[Nm]	Min.	142.25		118.7	
Starting current	[I _A /I _N]	+20 %	9.1		10.4	

Table 253: Mechanical data

Mechanical data		+/-	Values	
			50 Hz	60 Hz
Sound pressure level	[dB]	Max.	65	68
Moment of inertia	[kgm²]	Max.	0.21	0.21
Motor canopy		-	No	

Table 254: Explosion protection

Explosion protection	+/-	Values
Type of protection	-	Without

Table 255: Environment

Environment	+/-	Values
Ambient temperature [°C]	Max.	40

Table 256: Terminal box

Explosion protection	+/-	Values
Enclosure of terminal box	-	IP55 (TEFC)
Cable entry (Ø)	-	2XM40X1,5

⁵⁵ The name plate data may vary within the tolerances.

⁵⁶ 1800=grey cast iron, 4400=aluminium

V1 type of construction, 4-pole, 200M/L

Material: 4-30,0-200L-BWE7F3NHWDWFUWWE⁵⁷⁾

Material number: 5443319 (05407847)

Table 259: General data [50 Hz]

General data	+/-	Values
Motor size	Max.	200L
Type of construction	Max.	V1
Starting method	-	Star a. Delta
Motor flange	-	IEC 60072 Type FF
Weight without attachments [kg]	Max.	255
Housing material ⁵⁸⁾	-	1800
Enclosure	-	IP55
Nominal frequency	[Hz]	50/60
Efficiency class	-	IE4 (Super Premium)
Thermal class F to B	-	Yes
Temperature sensor	-	03
Operation on frequency inverter permitted	-	Yes (acc to motor manufact)
Duty type	-	Continuous running duty (S1)
Direction of rotation	-	Bi-directional
Motor bearings insulated	-	No
Position of fixed motor bearing	-	Drive end
Fixed motor bearing reinforced	-	Without
Radial motor bearing reinforced	-	No
Condensate drain	-	Yes
Bolted feet	-	No

Table 260: Efficiency

Electrical data		+/-	50 Hz			60 Hz		
			4/4	3/4	2/4	4/4	3/4	2/4
Efficiency	[%]	Min.	94.9	94.7	94.2	95	94.8	93.7
Power factor		Min.	0.81	0.75	0.64	0.8	0.74	0.62

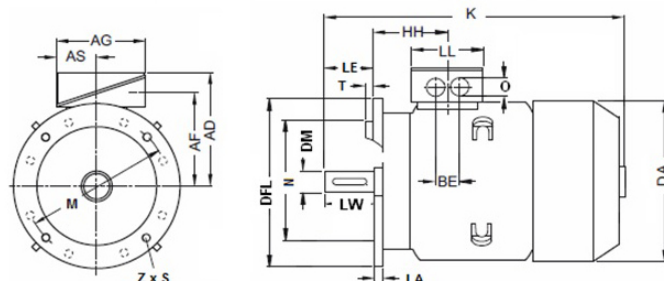


Fig. 29: Envelope dimensions

Table 266: Envelope dimensions A-DFL

A	AA	AB	AD	AF	AG	AS	B	BB	BC	BE	BK	C	DA	DFL
			334	272.7	218	119.1				94			422.5	400

Table 267: Envelope dimensions DM-Z

DM	H1	HA	HH	K	LA	LB	LL	LE	M	N	S	T	Y	Z
55			209	767.5	18	657.5	228.5		350	300	18.5	5		4

Table 261: Electrical data

Electrical data		+/-	50 Hz		60 Hz	
			Δ	Y	Δ	Y
Rated voltage	[V]	-	400	690	460	-
Rated current	[A]	-10 %	56.3	32.6	49.5	-
Rated power	[kW]	-	30		30	
Rated speed	[rpm]	Min.	1480		1783	
Rated torque	[Nm]	Min.	193.26		160.88	
Starting current	[IA/IN]	+20 %	8.8		9.8	

Table 262: Mechanical data

Mechanical data		+/-	Values	
			50 Hz	60 Hz
Sound pressure level	[dB]	Max.	66	68
Moment of inertia	[kgm²]	Max.	0.28	0.28
Motor canopy		-	No	

Table 263: Explosion protection

Explosion protection	+/-	Values
Type of protection	-	Without

Table 264: Environment

Environment	+/-	Values
Ambient temperature [°C]	Max.	40

Table 265: Terminal box

Explosion protection	+/-	Values
Enclosure of terminal box	-	IP55 (TEFC)
Cable entry (O)	-	2XM50X1,5

⁵⁷ The name plate data may vary within the tolerances.

⁵⁸ 1800=grey cast iron, 4400=aluminium

V1 type of construction, 4-pole, 225S/M

Material: 4-37,0-225S-BWE7F3NHWDWFUWWE⁵⁹⁾

Material number: 5443320 (05407848)

Table 268: General data [50 Hz]

General data		+/-	Values
Motor size		Max.	225S
Type of construction		Max.	V1
Starting method		-	Star a. Delta
Motor flange		-	IEC 60072 Type FF
Weight without attachments	[kg]	Max.	345
Housing material ⁶⁰⁾		-	1800
Enclosure		-	IP55
Nominal frequency	[Hz]		50/60
Efficiency class		-	IE4 (Super Premium)
Thermal class F to B		-	Yes
Temperature sensor		-	03
Operation on frequency inverter permitted		-	Yes (acc to motor manufact)
Duty type		-	Continuous running duty (S1)
Direction of rotation		-	Bi-directional
Motor bearings insulated		-	No
Position of fixed motor bearing		-	Drive end
Fixed motor bearing reinforced		-	Without
Radial motor bearing reinforced		-	No
Condensate drain		-	Yes
Bolted feet		-	No

Table 269: Efficiency

Electrical data		+/-	50 Hz			60 Hz		
			4/4	3/4	2/4	4/4	3/4	2/4
Efficiency	[%]	Min.	95.2	94.8	94	95.4	95.1	94
Power factor		Min.	0.82	0.77	0.66	0.81	0.75	0.63

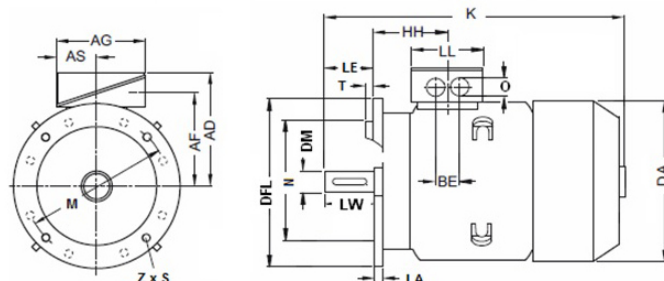

Fig. 30: Envelope dimensions

Table 275: Envelope dimensions A-DFL

A	AA	AB	AD	AF	AG	AS	B	BB	BC	BE	BK	C	DA	DFL
			397	314.8	286	165.6				110			488	450

Table 276: Envelope dimensions DM-Z

DM	H1	HA	HH	K	LA	LB	LL	LE	M	N	S	T	Y	Z
60			203.5	886.6	18	746.6	269		400	350	18.5	5		8

Table 270: Electrical data

Electrical data		+/-	50 Hz		60 Hz	
			Δ	Y	Δ	Y
Rated voltage	[V]	-	400	690	460	-
Rated current	[A]	-10 %	68.4	39.5	60.1	-
Rated power	[kW]	-	37		37	
Rated speed	[rpm]	Min.	1483		1785	
Rated torque	[Nm]	Min.	238.38		198.16	
Starting current	[IA/IN]	+20 %	8.5		10	

Table 271: Mechanical data

Mechanical data		+/-	Values	
			50 Hz	60 Hz
Sound pressure level	[dB]	Max.	67	70
Moment of inertia	[kgm ²]	Max.	0.48	0.48
Motor canopy		-	No	

Table 272: Explosion protection

Explosion protection	+/-	Values
Type of protection	-	Without

Table 273: Environment

Environment	+/-	Values
Ambient temperature	[°C]	Max. 40

Table 274: Terminal box

Explosion protection	+/-	Values
Enclosure of terminal box	-	IP55 (TEFC)
Cable entry (Ø)	-	2XM50X1,5

⁵⁹⁾ The name plate data may vary within the tolerances.

⁶⁰⁾ 1800=grey cast iron, 4400=aluminium

V1 type of construction, 4-pole, 225S/M

Material: 4-45,0-225M-BWE7F3NHWDWFUWWE1⁶¹⁾

Material number: 5443321 (05407849)

Table 277: General data [50 Hz]

General data	+/-	Values
Motor size	Max.	225M
Type of construction	Max.	V1
Starting method	-	Star a. Delta
Motor flange	-	IEC 60072 Type FF
Weight without attachments [kg]	Max.	415
Housing material ⁶²⁾	-	1800
Enclosure	-	IP55
Nominal frequency	[Hz]	50/60
Efficiency class	-	IE4 (Super Premium)
Thermal class F to B	-	Yes
Temperature sensor	-	03
Operation on frequency inverter permitted	-	Yes (acc to motor manufact)
Duty type	-	Continuous running duty (S1)
Direction of rotation	-	Bi-directional
Motor bearings insulated	-	No
Position of fixed motor bearing	-	Drive end
Fixed motor bearing reinforced	-	Without
Radial motor bearing reinforced	-	No
Condensate drain	-	Yes
Bolted feet	-	No

Table 278: Efficiency

Electrical data		+/-	50 Hz			60 Hz		
			4/4	3/4	2/4	4/4	3/4	2/4
Efficiency	[%]	Min.	95.4	95.3	94.7	95.4	95.4	94.6
Power factor		Min.	0.84	0.8	0.7	0.83	0.78	0.67

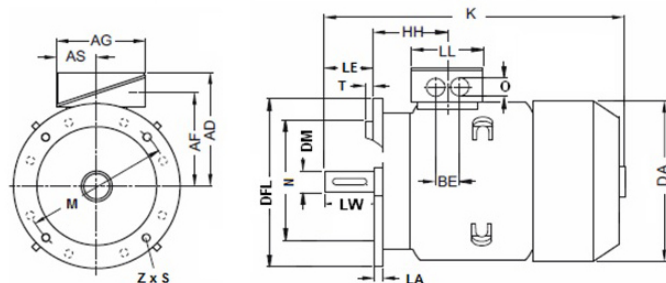


Fig. 31: Envelope dimensions

Table 284: Envelope dimensions A-DFL

A	AA	AB	AD	AF	AG	AS	B	BB	BC	BE	BK	C	DA	DFL
			397	314.8	286	165.6				110			488	450

Table 285: Envelope dimensions DM-Z

DM	H1	HA	HH	K	LA	LB	LL	LE	M	N	S	T	Y	Z
60			203.5	886.6	18	746.6	269		400	350	18.5	5		8

Table 279: Electrical data

Electrical data		+/-	50 Hz		60 Hz	
			Δ	Y	Δ	Y
Rated voltage	[V]	-	400	690	460	-
Rated current	[A]	-10 %	81.1	46.9	71.3	-
Rated power	[kW]	-	45		45	
Rated speed	[rpm]	Min.	1483		1785	
Rated torque	[Nm]	Min.	290.38		241.33	
Starting current	[IA/IN]	+20 %	8.7		10.2	

Table 280: Mechanical data

Mechanical data		+/-	Values	
			50 Hz	60 Hz
Sound pressure level	[dB]	Max.	67	70
Moment of inertia	[kgm²]	Max.	0.64	0.64
Motor canopy		-	No	

Table 281: Explosion protection

Explosion protection	+/-	Values
Type of protection	-	Without

Table 282: Environment

Environment	+/-	Values
Ambient temperature [°C]	Max.	40

Table 283: Terminal box

Explosion protection	+/-	Values
Enclosure of terminal box	-	IP55 (TEFC)
Cable entry (O)	-	2XM50X1,5

61 The name plate data may vary within the tolerances.

62 1800=grey cast iron, 4400=aluminium

V1 type of construction, 4-pole, 250S/M

Material: 4-55,0-250M-BWE7F3NHWDWFUWWE⁶³⁾

Material number: 5443339 (05408280)

Table 286: General data [50 Hz]

General data	+/-	Values
Motor size	Max.	250M
Type of construction	Max.	V1
Starting method	-	Star a. Delta
Motor flange	-	IEC 60072 Type FF
Weight without attachments [kg]	Max.	470
Housing material ⁶⁴⁾	-	1800
Enclosure	-	IP55
Nominal frequency	[Hz]	50/60
Efficiency class	-	IE4 (Super Premium)
Thermal class F to B	-	Yes
Temperature sensor	-	03
Operation on frequency inverter permitted	-	Yes (acc to motor manufact)
Duty type	-	Continuous running duty (S1)
Direction of rotation	-	Bi-directional
Motor bearings insulated	-	No
Position of fixed motor bearing	-	Drive end
Fixed motor bearing reinforced	-	Without
Radial motor bearing reinforced	-	No
Condensate drain	-	Yes
Bolted feet	-	No

Table 287: Efficiency

Electrical data		+/-	50 Hz			60 Hz		
			4/4	3/4	2/4	4/4	3/4	2/4
Efficiency	[%]	Min.	95.7	95.4	94.8	95.8	95.6	94.7
Power factor		Min.	0.83	0.78	0.67	0.82	0.76	0.65

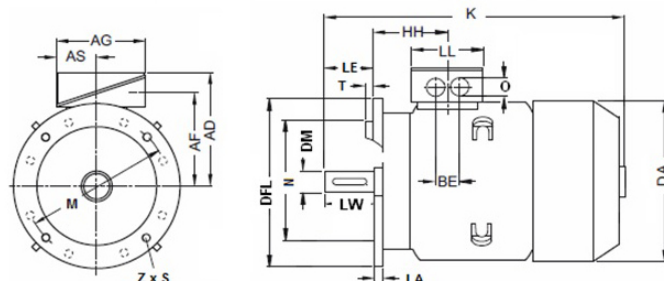


Fig. 32: Envelope dimensions

Table 293: Envelope dimensions A-DFL

A	AA	AB	AD	AF	AG	AS	B	BB	BC	BE	BK	C	DA	DFL
			417	334.8	286	165.6				110			528	550

Table 294: Envelope dimensions DM-Z

DM	H1	HA	HH	K	LA	LB	LL	LE	M	N	S	T	Y	Z
65			217	965.5	20	825.5	269		500	450	18.5	5		8

Table 288: Electrical data

Electrical data		+/-	50 Hz		60 Hz	
			Δ	Y	Δ	Y
Rated voltage	[V]	-	400	690	460	-
Rated current	[A]	-10 %	100	57.8	87.9	-
Rated power	[kW]	-	55		55	
Rated speed	[rpm]	Min.	1485		1787	
Rated torque	[Nm]	Min.	354.14		294.3	
Starting current	[I _A /I _N]	+20 %	8.9		10.2	

Table 289: Mechanical data

Mechanical data		+/-	Values	
			50 Hz	60 Hz
Sound pressure level	[dB]	Max.	67	71
Moment of inertia	[kgm²]	Max.	0.78	0.78
Motor canopy		-	No	

Table 290: Explosion protection

Explosion protection	+/-	Values
Type of protection	-	Without

Table 291: Environment

Environment	+/-	Values
Ambient temperature [°C]	Max.	40

Table 292: Terminal box

Explosion protection	+/-	Values
Enclosure of terminal box	-	IP55 (TEFC)
Cable entry (O)	-	2XM63X1,5

⁶³ The name plate data may vary within the tolerances.

⁶⁴ 1800=grey cast iron, 4400=aluminium

V1 type of construction, 4-pole, 280S

Material: 4-75,0-280S-BWE7F3NHWDWFWNWKWC⁶⁵⁾

Material number: 05214277 (5214333)

Table 295: General data [50 Hz]

General data	+/-	Values
Motor size	Max.	280S
Type of construction	Max.	V1
Starting method	-	Star a. Delta
Motor flange	-	IEC 60072 Type FF
Weight without attachments [kg]	Max.	735
Housing material ⁶⁶⁾	-	1800
Enclosure	-	IP55
Nominal frequency	[Hz]	50/60
Efficiency class	-	IE4 (Super Premium)
Thermal class F to B	-	Yes
Temperature sensor	-	03
Operation on frequency inverter permitted	-	No
Duty type	-	Continuous running duty (S1)
Direction of rotation	-	Bi-directional
Motor bearings insulated	-	No
Position of fixed motor bearing	-	Drive end
Fixed motor bearing reinforced	-	Without
Radial motor bearing reinforced	-	No
Condensate drain	-	Yes
Bolted feet	-	No

Table 296: Efficiency

Electrical data		+/-	50 Hz			60 Hz		
			4/4	3/4	2/4	4/4	3/4	2/4
Efficiency	[%]	Min.	96	95.8	95.3	96.2	95.7	94.8
Power factor		Min.	0.83	0.8	0.7	0.82	0.77	0.66

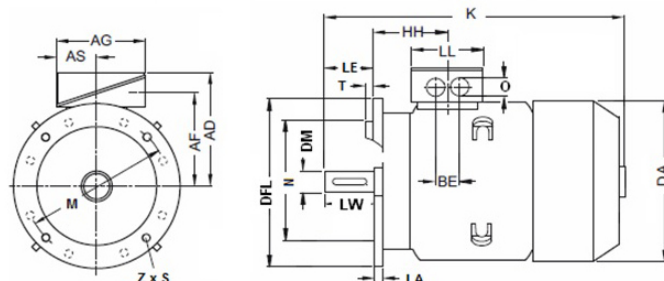


Fig. 33: Envelope dimensions

Table 302: Envelope dimensions A-DFL

A	AA	AB	AD	AF	AG	AS	B	BB	BC	BE	BK	C	DA	DFL
-	-	-	472	391	312	140.4	-	-	-	126	-	-	580	550

Table 303: Envelope dimensions DM-Z

DM	H1	HA	HH	K	LA	LB	LL	LE	M	N	S	T	Y	Z
75	-	-	266	1071	18	931	314	140	500	450	18.5	5	-	8

Table 297: Electrical data

Electrical data		+/-	50 Hz		60 Hz	
			Δ	Y	Δ	Y
Rated voltage	[V]	-	400	690	460	-
Rated current	[A]	-10 %	136	78.8	119	-
Rated power	[kW]	-	75		75	
Rated speed	[rpm]	Min.	1486		1788	
Rated torque	[Nm]	Min.	482		401	
Starting current	[IA/IN]	+20 %	7.7		8.7	

Table 298: Mechanical data

Mechanical data		+/-	Values	
			50 Hz	60 Hz
Sound pressure level	[dB]	Max.	69	73
Moment of inertia	[kgm²]	Max.	1.7	1.7
Motor canopy		-	No	

Table 299: Explosion protection

Explosion protection	+/-	Values
Type of protection	-	Without

Table 300: Environment

Environment	+/-	Values
Ambient temperature [°C]	Max.	40

Table 301: Terminal box

Explosion protection	+/-	Values
Enclosure of terminal box	-	IP55 (TEFC)
Cable entry (O)	-	2XM63X1,5

⁶⁵ The name plate data may vary within the tolerances.

⁶⁶ 1800=grey cast iron, 4400=aluminium

V1 type of construction, 4-pole, 280M

Material: 4-90,0-280M-BWE7F3NHWDWFNWKWC⁶⁷⁾

Material number: 05214278 (5214334)

Table 304: General data [50 Hz]

General data	+/-	Values
Motor size	Max.	280M
Type of construction	Max.	V1
Starting method	-	Star a. Delta
Motor flange	-	IEC 60072 Type FF
Weight without attachments [kg]	Max.	853
Housing material ⁶⁸⁾	-	1800
Enclosure	-	IP55
Nominal frequency	[Hz]	50/60
Efficiency class	-	IE4 (Super Premium)
Thermal class F to B	-	Yes
Temperature sensor	-	03
Operation on frequency inverter permitted	-	No
Duty type	-	Continuous running duty (S1)
Direction of rotation	-	Bi-directional
Motor bearings insulated	-	No
Position of fixed motor bearing	-	Drive end
Fixed motor bearing reinforced	-	No
Radial motor bearing reinforced	-	No
Condensate drain	-	Yes
Bolted feet	-	No

Table 305: Efficiency

Electrical data		+/-	50 Hz			60 Hz		
			4/4	3/4	2/4	4/4	3/4	2/4
Efficiency	[%]	Min.	96.1	96	95.5	96.2	96.1	95.5
Power factor		Min.	0.84	0.81	0.72	0.83	0.79	0.69

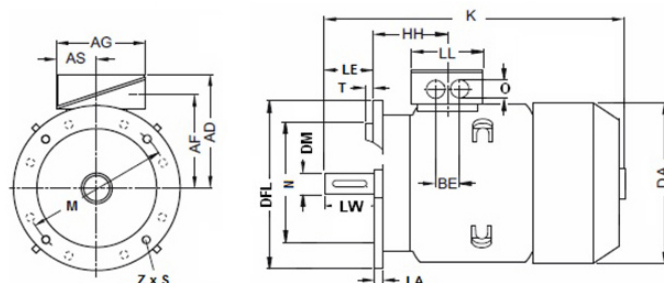


Fig. 34: Envelope dimensions

Table 311: Envelope dimensions A-DFL

A	AA	AB	AD	AF	AG	AS	B	BB	BC	BE	BK	C	DA	DFL
-	-	-	472	391	312	140.4	-	-	-	126	-	-	580	550

Table 312: Envelope dimensions DM-Z

DM	H1	HA	HH	K	LA	LB	LL	LE	M	N	S	T	Y	Z
75	-	-	266	1071	18	931	314	140	500	450	18.5	5	-	8

Table 306: Electrical data

Electrical data		+/-	50 Hz		60 Hz	
			Δ	Y	Δ	Y
Rated voltage	[V]	-	400	690	460	-
Rated current	[A]	-10 %	161	93.3	141	-
Rated power	[kW]	-	90		90	
Rated speed	[rpm]	Min.	1486		1788	
Rated torque	[Nm]	Min.	579		481	
Starting current	[I _A /I _N]	+20 %	8		8.9	

Table 307: Mechanical data

Mechanical data		+/-	Values	
			50 Hz	60 Hz
Sound pressure level	[dB]	Max.	69	73
Moment of inertia	[kgm²]	Max.	2	2
Motor canopy		-	No	

Table 308: Explosion protection

Explosion protection	+/-	Values
Type of protection	-	Without

Table 309: Environment

Environment	+/-	Values
Ambient temperature [°C]	Max.	40

Table 310: Terminal box

Explosion protection	+/-	Values
Enclosure of terminal box	-	IP55 (TEFC)
Cable entry (O)	-	2XM63X1,5

⁶⁷ The name plate data may vary within the tolerances.

⁶⁸ 1800=grey cast iron, 4400=aluminium

V1 type of construction, 4-pole, 315M

Material: 4-110 -315M-BWE7F3NHWDWFWNWKWC⁶⁹⁾

Material number: 05214279 (5214335)

Table 313: General data [50 Hz]

General data	+/-	Values
Motor size	Max.	315M
Type of construction	Max.	V1
Starting method	-	Star a. Delta
Motor flange	-	IEC 60072 Type FF
Weight without attachments [kg]	Max.	1004
Housing material ⁷⁰⁾	-	1800
Enclosure	-	IP55
Nominal frequency	[Hz]	50/60
Efficiency class	-	IE4 (Super Premium)
Thermal class F to B	-	Yes
Temperature sensor	-	03
Operation on frequency inverter permitted	-	No
Duty type	-	Continuous running duty (S1)
Direction of rotation	-	Bi-directional
Motor bearings insulated	-	No
Position of fixed motor bearing	-	Drive end
Fixed motor bearing reinforced	-	Without
Radial motor bearing reinforced	-	No
Condensate drain	-	Yes
Bolted feet	-	No

Table 314: Efficiency

Electrical data	+/-	50 Hz			60 Hz		
		4/4	3/4	2/4	4/4	3/4	2/4
Efficiency [%]	Min.	96.3	95.4	96	96.2	96.1	95.3
Power factor	Min.	0.83	0.79	0.69	0.82	0.77	0.67

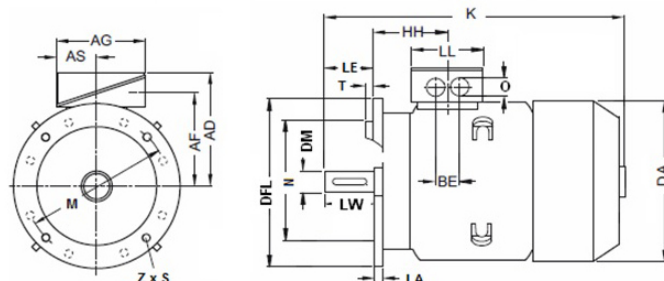


Fig. 35: Envelope dimensions

Table 320: Envelope dimensions A-DFL

A	AA	AB	AD	AF	AG	AS	B	BB	BC	BE	BK	C	DA	DFL
-	-	-	530	432	382	171.9	-	-	-	160	-	-	644	660

Table 321: Envelope dimensions DM-Z

DM	H1	HA	HH	K	LA	LB	LL	LE	M	N	S	T	Y	Z
80	-	-	264	1274	22	1104	379	170	600	550	24	6	-	8

Table 315: Electrical data

Electrical data	+/-	50 Hz		60 Hz	
		Δ	Y	Δ	Y
Rated voltage [V]	-	400	690	460	-
Rated current [A]	-10 %	199	115	175	-
Rated power [kW]	-	110		110	
Rated speed [rpm]	Min.	1491		1792	
Rated torque [Nm]	Min.	706		586	
Starting current [IA/IN]	+20 %	7		7.9	

Table 316: Mechanical data

Mechanical data	+/-	Values	
		50 Hz	60 Hz
Sound pressure level [dB]	Max.	71	75
Moment of inertia [kgm²]	Max.	2.7	2.7
Motor canopy	-	No	

Table 317: Explosion protection

Explosion protection	+/-	Values
Type of protection	-	Without

Table 318: Environment

Environment	+/-	Values
Ambient temperature [°C]	Max.	40

Table 319: Terminal box

Explosion protection	+/-	Values
Enclosure of terminal box	-	IP55 (TEFC)
Cable entry (O)	-	2XM63X1,5

⁶⁹ The name plate data may vary within the tolerances.

⁷⁰ 1800=grey cast iron, 4400=aluminium

V1 type of construction, 4-pole, 315M

Material: 4-132 -315M-BWE7F3NHWDWFWNWKWC⁷¹⁾

Material number: 05214320 (5214336)

Table 322: General data [50 Hz]

General data		+/-	Values
Motor size		Max.	315M
Type of construction		Max.	V1
Starting method		-	Star a. Delta
Motor flange		-	IEC 60072 Type FF
Weight without attachments	[kg]	Max.	1052
Housing material ⁷²⁾		-	1800
Enclosure		-	IP55
Nominal frequency	[Hz]		50/60
Efficiency class		-	IE4 (Super Premium)
Thermal class F to B		-	Yes
Temperature sensor		-	03
Operation on frequency inverter permitted		-	No
Duty type		-	Continuous running duty (S1)
Direction of rotation		-	Bi-directional
Motor bearings insulated		-	No
Position of fixed motor bearing		-	Drive end
Fixed motor bearing reinforced		-	Without
Radial motor bearing reinforced		-	No
Condensate drain		-	Yes
Bolted feet		-	No

Table 323: Efficiency

Electrical data		+/-	50 Hz			60 Hz		
			4/4	3/4	2/4	4/4	3/4	2/4
Efficiency	[%]	Min.	96.4	96.1	95.4	96.5	96.2	95.4
Power factor		Min.	0.83	0.78	0.68	0.82	0.77	0.66

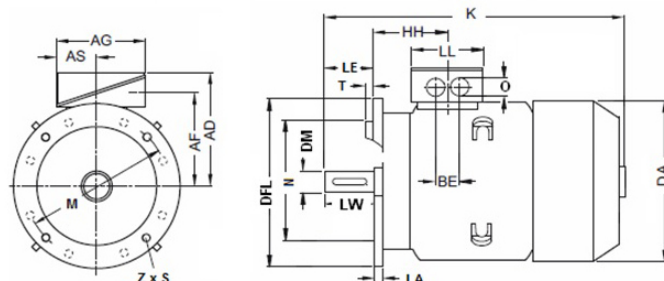

Fig. 36: Envelope dimensions

Table 329: Envelope dimensions A-DFL

A	AA	AB	AD	AF	AG	AS	B	BB	BC	BE	BK	C	DA	DFL
-	-	-	530	432	382	171.9	-	-	-	160	-	-	644	660

Table 330: Envelope dimensions DM-Z

DM	H1	HA	HH	K	LA	LB	LL	LE	M	N	S	T	Y	Z
80	-	-	264	1274	22	1104	379	170	600	550	24	6	-	8

Table 324: Electrical data

Electrical data		+/-	50 Hz		60 Hz	
			Δ	Y	Δ	Y
Rated voltage	[V]	-	400	690	460	-
Rated current	[A]	-10 %	238	138	209	-
Rated power	[kW]	-	132		132	
Rated speed	[rpm]	Min.	1491		1792	
Rated torque	[Nm]	Min.	847		704	
Starting current	[IA/IN]	+20 %	7.3		8.3	

Table 325: Mechanical data

Mechanical data		+/-	Values	
			50 Hz	60 Hz
Sound pressure level	[dB]	Max.	71	75
Moment of inertia	[kgm ²]	Max.	3.1	3.1
Motor canopy		-	No	

Table 326: Explosion protection

Explosion protection	+/-	Values
Type of protection	-	Without

Table 327: Environment

Environment	+/-	Values
Ambient temperature	[°C]	Max. 40

Table 328: Terminal box

Explosion protection	+/-	Values
Enclosure of terminal box	-	IP55 (TEFC)
Cable entry (O)	-	2XM63X1,5

⁷¹ The name plate data may vary within the tolerances.

⁷² 1800=grey cast iron, 4400=aluminium

V15 type of construction, 2-pole, 180M/L

Material: 2-22,0-180M-BWE7F3NGHDFWUWWEI⁷³⁾

Material number: 5443329 (05407857)

Table 331: General data [50 Hz]

General data	+/-	Values
Motor size	Max.	180M
Type of construction	Max.	V15
Starting method	-	Star a. Delta
Motor flange	-	IEC 60072 Type FF
Weight without attachments [kg]	Max.	180
Housing material ⁷⁴⁾	-	1800
Enclosure	-	IP55
Nominal frequency	[Hz]	50/60
Efficiency class	-	IE4 (Super Premium)
Thermal class F to B	-	Yes
Temperature sensor	-	03
Operation on frequency inverter permitted	-	Yes (acc to motor manufact)
Duty type	-	Continuous running duty (S1)
Direction of rotation	-	Bi-directional
Motor bearings insulated	-	No
Position of fixed motor bearing	-	Drive end
Fixed motor bearing reinforced	-	Without
Radial motor bearing reinforced	-	No
Condensate drain	-	Yes
Bolted feet	-	No

Table 332: Efficiency

Electrical data		+/-	50 Hz			60 Hz		
			4/4	3/4	2/4	4/4	3/4	2/4
Efficiency	[%]	Min.	94	93.56	92.52	93	93	92.19
Power factor		Min.	0.86	0.82	0.73	0.85	0.81	0.7

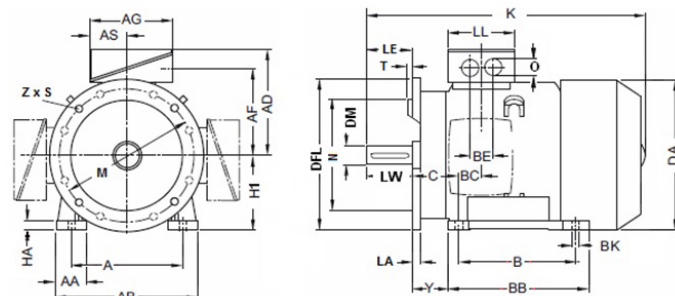

Fig. 37: Envelope dimensions

Table 338: Envelope dimensions A-DFL

A	AA	AB	AD	AF	AG	AS	B	BB	BC	BE	BK	C	DA	DFL
279	51	329	294	239.7	191.5	102.5	241	288	120.5	84	14.5	121	371	350

Table 339: Envelope dimensions DM-Z

DM	H1	HA	HH	K	LA	LB	LL	LE	M	N	S	T	Y	Z
48	180	19	241.5	684.5	16	574.5	198.5	110	300	250	18.5	5	97.5	4

Table 333: Electrical data

Electrical data		+/-	50 Hz		60 Hz	
			Δ	Y	Δ	Y
Rated voltage	[V]	-	400	690	460	-
Rated current	[A]	-10 %	39.3	22.7	34.9	-
Rated power	[kW]	-	22		22	
Rated speed	[rpm]	Min.	2960		3565	
Rated torque	[Nm]	Min.	71.02		58.96	
Starting current	[I _A /I _N]	+20 %	9.7		11	

Table 334: Mechanical data

Mechanical data		+/-	Values	
			50 Hz	60 Hz
Sound pressure level	[dB]	Max.	70	75
Moment of inertia	[kgm²]	Max.	0.09	0.09
Motor canopy		-	No	

Table 335: Explosion protection

Explosion protection	+/-	Values
Type of protection	-	Without

Table 336: Environment

Environment	+/-	Values
Ambient temperature [°C]	Max.	40

Table 337: Terminal box

Explosion protection	+/-	Values
Enclosure of terminal box	-	IP55 (TEFC)
Cable entry (Ø)	-	2XM40X1,5

73 The name plate data may vary within the tolerances.

74 1800=grey cast iron, 4400=aluminium

V15 type of construction, 2-pole, 200M/L

Material: 2-30,0-200L-BWE7F3NGHDFUWWE1⁷⁵⁾

Material number: 5443330 (05407858)

Table 340: General data [50 Hz]

General data	+/-	Values
Motor size	Max.	200L
Type of construction	Max.	V15
Starting method	-	Star a. Delta
Motor flange	-	IEC 60072 Type FF
Weight without attachments [kg]	Max.	210
Housing material ⁷⁶⁾	-	1800
Enclosure	-	IP55
Nominal frequency	[Hz]	50/60
Efficiency class	-	IE4 (Super Premium)
Thermal class F to B	-	Yes
Temperature sensor	-	03
Operation on frequency inverter permitted	-	Yes (acc to motor manufact)
Duty type	-	Continuous running duty (S1)
Direction of rotation	-	Bi-directional
Motor bearings insulated	-	No
Position of fixed motor bearing	-	Drive end
Fixed motor bearing reinforced	-	Without
Radial motor bearing reinforced	-	No
Condensate drain	-	Yes
Bolted feet	-	No

Table 341: Efficiency

Electrical data		+/-	50 Hz			60 Hz		
			4/4	3/4	2/4	4/4	3/4	2/4
Efficiency	[%]	Min.	94.5	94	92.8	93.6	93.4	91.7
Power factor		Min.	0.84	0.8	0.71	0.83	0.78	0.68

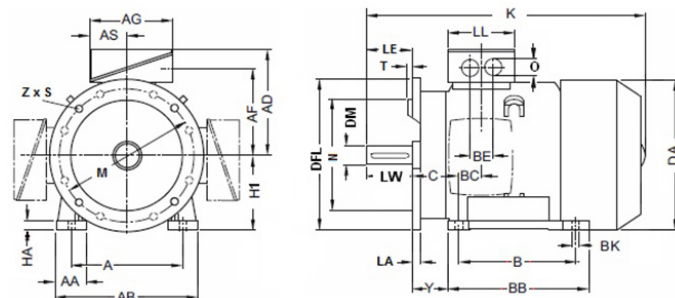

Fig. 38: Envelope dimensions

Table 347: Envelope dimensions A-DFL

A	AA	AB	AD	AF	AG	AS	B	BB	BC	BE	BK	C	DA	DFL
318	65	385	334	272.7	218	119.1	305	369	76	94	18.5	133	414	400

Table 348: Envelope dimensions DM-Z

DM	H1	HA	HH	K	LA	LB	LL	LE	M	N	S	T	Y	Z
55	200	30	209	767.5	18	657.5	228.5	110	350	300	18.5	5	101	4

Table 342: Electrical data

Electrical data		+/-	50 Hz		60 Hz	
			Δ	Y	Δ	Y
Rated voltage	[V]	-	400	690	460	-
Rated current	[A]	-10 %	54.5	31.5	48.5	-
Rated power	[kW]	-	30		30	
Rated speed	[rpm]	Min.	2965		3570	
Rated torque	[Nm]	Min.	96.63		80.25	
Starting current	[I _A /I _N]	+20 %	8.9		10	

Table 343: Mechanical data

Mechanical data		+/-	Values	
			50 Hz	60 Hz
Sound pressure level	[dB]	Max.	75	80
Moment of inertia	[kgm²]	Max.	0.16	0.16
Motor canopy		-	No	

Table 344: Explosion protection

Explosion protection	+/-	Values
Type of protection	-	Without

Table 345: Environment

Environment	+/-	Values
Ambient temperature [°C]	Max.	40

Table 346: Terminal box

Explosion protection	+/-	Values
Enclosure of terminal box	-	IP55 (TEFC)
Cable entry (Ø)	-	2XM50X1,5

⁷⁵ The name plate data may vary within the tolerances.

⁷⁶ 1800=grey cast iron, 4400=aluminium

V15 type of construction, 2-pole, 200M/L

Material: 2-37,0-200L-BWE7F3NGHDFUWWEI⁷⁷⁾

Material number: 5443331 (05407859)

Table 349: General data [50 Hz]

General data	+/-	Values
Motor size	Max.	200L
Type of construction	Max.	V15
Starting method	-	Star a. Delta
Motor flange	-	IEC 60072 Type FF
Weight without attachments [kg]	Max.	260
Housing material ⁷⁸⁾	-	1800
Enclosure	-	IP55
Nominal frequency	[Hz]	50/60
Efficiency class	-	IE4 (Super Premium)
Thermal class F to B	-	Yes
Temperature sensor	-	03
Operation on frequency inverter permitted	-	Yes (acc to motor manufact)
Duty type	-	Continuous running duty (S1)
Direction of rotation	-	Bi-directional
Motor bearings insulated	-	No
Position of fixed motor bearing	-	Drive end
Fixed motor bearing reinforced	-	Without
Radial motor bearing reinforced	-	No
Condensate drain	-	Yes
Bolted feet	-	No

Table 350: Efficiency

Electrical data		+/-	50 Hz			60 Hz		
			4/4	3/4	2/4	4/4	3/4	2/4
Efficiency	[%]	Min.	94.8	94.3	93.3	94.1	93.8	92.4
Power factor		Min.	0.85	0.81	0.72	0.84	0.79	0.7

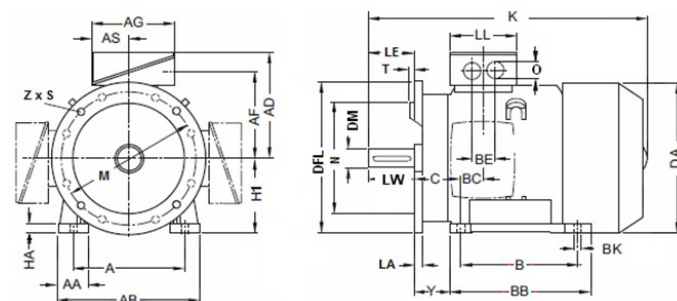


Fig. 39: Envelope dimensions

Table 356: Envelope dimensions A-DFL

A	AA	AB	AD	AF	AG	AS	B	BB	BC	BE	BK	C	DA	DFL
318	65	385	334	272.7	218	119.1	305	369	76	94	18.5	133	414	400

Table 357: Envelope dimensions DM-Z

DM	H1	HA	HH	K	LA	LB	LL	LE	M	N	S	T	Y	Z
55	200	30	209	767.5	18	657.5	228.5	110	350	300	18.5	5	101	4

Table 351: Electrical data

Electrical data		+/-	50 Hz		60 Hz	
			Δ	Y	Δ	Y
Rated voltage	[V]	-	400	690	460	-
Rated current	[A]	-10 %	66.3	38.3	58.8	-
Rated power	[kW]	-	37		37	
Rated speed	[rpm]	Min.	2965		3570	
Rated torque	[Nm]	Min.	119.68		99.08	
Starting current	[IA/IN]	+20 %	9.2		10.2	

Table 352: Mechanical data

Mechanical data		+/-	Values	
			50 Hz	60 Hz
Sound pressure level	[dB]	Max.	75	80
Moment of inertia	[kgm ²]	Max.	0.19	0.19
Motor canopy		-	No	

Table 353: Explosion protection

Explosion protection	+/-	Values
Type of protection	-	Without

Table 354: Environment

Environment	+/-	Values
Ambient temperature [°C]	Max.	40

Table 355: Terminal box

Explosion protection	+/-	Values
Enclosure of terminal box	-	IP55 (TEFC)
Cable entry (Ø)	-	2XM50X1,5

77 The name plate data may vary within the tolerances.

78 1800=grey cast iron, 4400=aluminium

V15 type of construction, 2-pole, 225S/M

Material: 2-45,0-225M-BWE7F3NGHDFUWWE1⁷⁹⁾

Material number: 5443332 (05407860)

Table 358: General data [50 Hz]

General data		+/-	Values
Motor size		Max.	225M
Type of construction		Max.	V15
Starting method		-	Star a. Delta
Motor flange		-	IEC 60072 Type FF
Weight without attachments	[kg]	Max.	415
Housing material ⁸⁰⁾		-	1800
Enclosure		-	IP55
Nominal frequency	[Hz]		50/60
Efficiency class		-	IE4 (Super Premium)
Thermal class F to B		-	Yes
Temperature sensor		-	03
Operation on frequency inverter permitted		-	Yes (acc to motor manufact)
Duty type		-	Continuous running duty (S1)
Direction of rotation		-	Bi-directional
Motor bearings insulated		-	No
Position of fixed motor bearing		-	Drive end
Fixed motor bearing reinforced		-	Without
Radial motor bearing reinforced		-	No
Condensate drain		-	Yes
Bolted feet		-	No

Table 359: Efficiency

Electrical data		+/-	50 Hz			60 Hz		
			4/4	3/4	2/4	4/4	3/4	2/4
Efficiency	[%]	Min.	95	94.3	93.2	94.5	94.1	92.6
Power factor		Min.	0.87	0.83	0.74	0.86	0.81	0.72

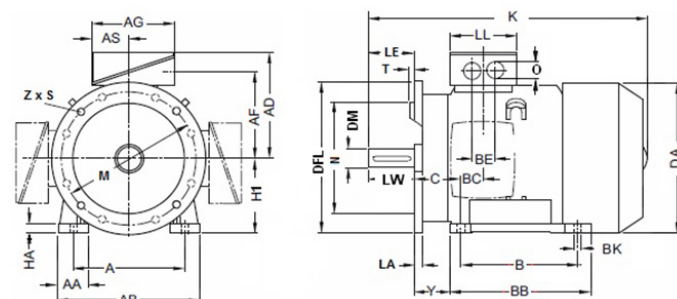

Fig. 40: Envelope dimensions

Table 365: Envelope dimensions A-DFL

A	AA	AB	AD	AF	AG	AS	B	BB	BC	BE	BK	C	DA	DFL
356	80	436	397	314.8	286	165.6	311	439.3	54.6	110	18.5	149	469	450

Table 366: Envelope dimensions DM-Z

DM	H1	HA	HH	K	LA	LB	LL	LE	M	N	S	T	Y	Z
55	225	28	203.5	856.5	18	746.5	269	110	400	350	18.5	5	106.5	8

Table 360: Electrical data

Electrical data		+/-	50 Hz		60 Hz	
			Δ	Y	Δ	Y
Rated voltage	[V]	-	400	690	460	-
Rated current	[A]	-10 %	78.6	45.4	69.5	-
Rated power	[kW]	-	45		45	
Rated speed	[rpm]	Min.	2970		3575	
Rated torque	[Nm]	Min.	145.19		120.66	
Starting current	[IA/IN]	+20 %	9.6		10.5	

Table 361: Mechanical data

Mechanical data		+/-	Values	
			50 Hz	60 Hz
Sound pressure level	[dB]	Max.	77	82
Moment of inertia	[kgm²]	Max.	0.29	0.29
Motor canopy		-	No	

Table 362: Explosion protection

Explosion protection	+/-	Values
Type of protection	-	Without

Table 363: Environment

Environment	+/-	Values
Ambient temperature	[°C]	Max. 40

Table 364: Terminal box

Explosion protection	+/-	Values
Enclosure of terminal box	-	IP55 (TEFC)
Cable entry (Ø)	-	2XM50X1,5

⁷⁹⁾ The name plate data may vary within the tolerances.

⁸⁰⁾ 1800=grey cast iron, 4400=aluminium

V15 type of construction, 2-pole, 250S/M

Material: 2-55,0-250M-BWE7F3NGHDFUWWE⁸¹⁾

Material number: 5443333 (05407861)

Table 367: General data [50 Hz]

General data	+/-	Values
Motor size	Max.	250M
Type of construction	Max.	V15
Starting method	-	Star a. Delta
Motor flange	-	IEC 60072 Type FF
Weight without attachments [kg]	Max.	450
Housing material ⁸²⁾	-	1800
Enclosure	-	IP55
Nominal frequency	[Hz]	50/60
Efficiency class	-	IE4 (Super Premium)
Thermal class F to B	-	Yes
Temperature sensor	-	03
Operation on frequency inverter permitted	-	Yes (acc to motor manufact)
Duty type	-	Continuous running duty (S1)
Direction of rotation	-	Bi-directional
Motor bearings insulated	-	No
Position of fixed motor bearing	-	Drive end
Fixed motor bearing reinforced	-	Without
Radial motor bearing reinforced	-	No
Condensate drain	-	Yes
Bolted feet	-	No

Table 368: Efficiency

Electrical data		+/-	50 Hz			60 Hz		
			4/4	3/4	2/4	4/4	3/4	2/4
Efficiency	[%]	Min.	95.3	94.5	93.5	94.5	94.2	92.7
Power factor		Min.	0.87	0.83	0.75	0.86	0.82	0.73

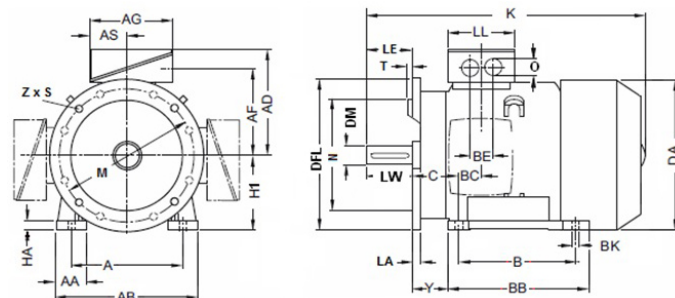

Fig. 41: Envelope dimensions

Table 374: Envelope dimensions A-DFL

A	AA	AB	AD	AF	AG	AS	B	BB	BC	BE	BK	C	DA	DFL
406	100	506	417	334.8	286	165.6	349	500	49	110	24	168	514	550

Table 375: Envelope dimensions DM-Z

DM	H1	HA	HH	K	LA	LB	LL	LE	M	N	S	T	Y	Z
60	250	30	217	965.5	20	825.5	269	140	500	450	18.5	5	120	8

Table 369: Electrical data

Electrical data		+/-	50 Hz		60 Hz	
			Δ	Y	Δ	Y
Rated voltage	[V]	-	400	690	460	-
Rated current	[A]	-10 %	95.7	55.3	84.9	-
Rated power	[kW]	-	55		55	
Rated speed	[rpm]	Min.	2970		3570	
Rated torque	[Nm]	Min.	176.58		147.15	
Starting current	[I _A /I _N]	+20 %	9.3		10.5	

Table 370: Mechanical data

Mechanical data		+/-	Values	
			50 Hz	60 Hz
Sound pressure level	[dB]	Max.	77	82
Moment of inertia	[kgm²]	Max.	0.36	0.36
Motor canopy		-	No	

Table 371: Explosion protection

Explosion protection	+/-	Values
Type of protection	-	Without

Table 372: Environment

Environment	+/-	Values
Ambient temperature [°C]	Max.	40

Table 373: Terminal box

Explosion protection	+/-	Values
Enclosure of terminal box	-	IP55 (TEFC)
Cable entry (O)	-	2XM63X1,5

81 The name plate data may vary within the tolerances.

82 1800=grey cast iron, 4400=aluminium

V15 type of construction, 2-pole, 280S

Material: 2-75,0-280S-BWE7F3NHFDWFWNWKWC⁸³⁾

Material number: 05214352 (5214315)

Table 376: General data [50 Hz]

General data	+/-	Values
Motor size	Max.	280S
Type of construction	Max.	V15
Starting method	-	Star a. Delta
Motor flange	-	IEC 60072 Type FF
Weight without attachments [kg]	Max.	734
Housing material ⁸⁴⁾	-	1800
Enclosure	-	IP55
Nominal frequency	[Hz]	50/60
Efficiency class	-	IE4 (Super Premium)
Thermal class F to B	-	Yes
Temperature sensor	-	03
Operation on frequency inverter permitted	-	No
Duty type	-	Continuous running duty (S1)
Direction of rotation	-	Bi-directional
Motor bearings insulated	-	No
Position of fixed motor bearing	-	Drive end
Fixed motor bearing reinforced	-	Without
Radial motor bearing reinforced	-	No
Condensate drain	-	Yes
Bolted feet	-	Yes

Table 377: Efficiency

Electrical data		+/-	50 Hz			60 Hz		
			4/4	3/4	2/4	4/4	3/4	2/4
Efficiency	[%]	min.	95.6	94.8	93.6	95	94.8	93
Power factor		Min.	0.87	0.84	0.75	0.86	0.83	0.73

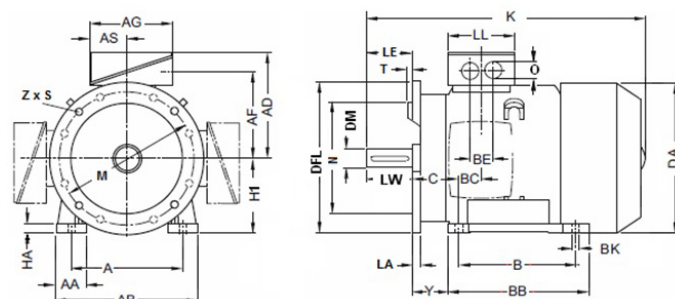

Fig. 42: Envelope dimensions

Table 383: Envelope dimensions A-DFL

A	AA	AB	AD	AF	AG	AS	B	BB	BC	BE	BK	C	DA	DFL
457	100	557	472	391	312	140.4	368	517	76	126	24	190	580	550

Table 384: Envelope dimensions DM-Z

DM	H1	HA	HH	K	LA	LB	LL	LE	M	N	S	T	Y	Z
65	280	42	266	1071	18	931	314	140	500	450	18.5	5	141	8

Table 378: Electrical data

Electrical data		+/-	50 Hz		60 Hz	
			Δ	Y	Δ	Y
Rated voltage	[V]	-	400	690	460	-
Rated current	[A]	-10 %	130	75.5	115	-
Rated power	[kW]	-	75		75	
Rated speed	[rpm]	Min.	2977		3579	
Rated torque	[Nm]	Min.	241		200	
Starting current	[I _A /I _N]	+20 %	8.3		9	

Table 379: Mechanical data

Mechanical data		+/-	Values	
			50 Hz	60 Hz
Sound pressure level	[dB]	Max.	77	81
Moment of inertia	[kgm²]	Max.	0.94	0.94
Motor canopy		-	No	

Table 380: Explosion protection

Explosion protection	+/-	Values
Type of protection	-	Without

Table 381: Environment

Environment	+/-	Values
Ambient temperature [°C]	Max.	40

Table 382: Terminal box

Explosion protection	+/-	Values
Enclosure of terminal box	-	IP55 (TEFC)
Cable entry (O)	-	2XM63X1,5

⁸³ The name plate data may vary within the tolerances.

⁸⁴ 1800=grey cast iron, 4400=aluminium

V15 type of construction, 2-pole, 280M

Material: 2-90,0-280M-BWE7F3NHFDWFNWKWC⁸⁵⁾

Material number: 05214353 (5214316)

Table 385: General data [50 Hz]

General data	+/-	Values
Motor size	Max.	280M
Type of construction	Max.	V15
Starting method	-	Star a. Delta
Motor flange	-	IEC 60072 Type FF
Weight without attachments [kg]	Max.	824
Housing material ⁸⁶⁾	-	1800
Enclosure	-	IP55
Nominal frequency	[Hz]	50/60
Efficiency class	-	IE4 (Super Premium)
Thermal class F to B	-	Yes
Temperature sensor	-	03
Operation on frequency inverter permitted	-	No
Duty type	-	Continuous running duty (S1)
Direction of rotation	-	Bi-directional
Motor bearings insulated	-	No
Position of fixed motor bearing	-	Drive end
Fixed motor bearing reinforced	-	Without
Radial motor bearing reinforced	-	No
Condensate drain	-	Yes
Bolted feet	-	Yes

Table 386: Efficiency

Electrical data		+/-	50 Hz			60 Hz		
			4/4	3/4	2/4	4/4	3/4	2/4
Efficiency	[%]	Min.	95.8	95.2	94.3	95.4	95	93.5
Power factor		min.	0.88	0.85	0.78	0.87	0.84	0.76

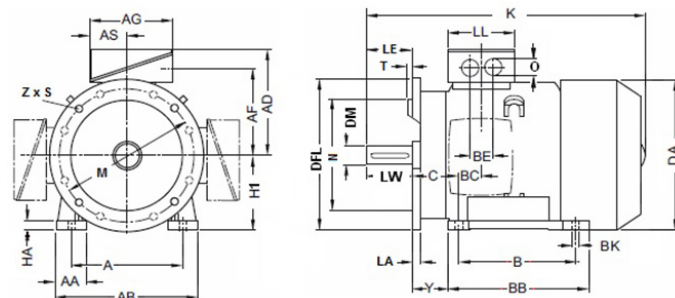

Fig. 43: Envelope dimensions

Table 392: Envelope dimensions A-DFL

A	AA	AB	AD	AF	AG	AS	B	BB	BC	BE	BK	C	DA	DFL
457	100	557	472	391	312	140.4	419	517	76	126	24	190	580	550

Table 393: Envelope dimensions DM-Z

DM	H1	HA	HH	K	LA	LB	LL	LE	M	N	S	T	Y	Z
65	280	42	266	1071	18	931	314	140	500	450	18.5	5	141	8

Table 387: Electrical data

Electrical data		+/-	50 Hz		60 Hz	
			Δ	Y	Δ	Y
Rated voltage	[V]	-	400	690	460	-
Rated current	[A]	-10 %	154	89.3	136	-
Rated power	[kW]	-	90		90	
Rated speed	[rpm]	min.	2975		3578	
Rated torque	[Nm]	Min.	289		240	
Starting current	[IA/IN]	+20 %	7.9		8.7	

Table 388: Mechanical data

Mechanical data		+/-	Values	
			50 Hz	60 Hz
Sound pressure level	[dB]	Max.	77	81
Moment of inertia	[kgm²]	Max.	1	1
Motor canopy		-	No	

Table 389: Explosion protection

Explosion protection	+/-	Values
Type of protection	-	Without

Table 390: Environment

Environment	+/-	Values
Ambient temperature [°C]	Max.	40

Table 391: Terminal box

Explosion protection	+/-	Values
Enclosure of terminal box	-	IP55 (TEFC)
Cable entry (O)	-	2XM63X1,5

⁸⁵ The name plate data may vary within the tolerances.

⁸⁶ 1800=grey cast iron, 4400=aluminium

V15 type of construction, 2-pole, 315S

Material: 2-110 -315S-BWE7F3NHFDUFNWKWC⁸⁷⁾

Material number: 05214354 (5214317)

Table 394: General data [50 Hz]

General data	+/-	Values
Motor size	Max.	315S
Type of construction	Max.	V15
Starting method	-	Star a. Delta
Motor flange	-	IEC 60072 Type FF
Weight without attachments [kg]	Max.	1041
Housing material ⁸⁸⁾	-	1800
Enclosure	-	IP55
Nominal frequency	[Hz]	50/60
Efficiency class	-	IE4 (Super Premium)
Thermal class F to B	-	Yes
Temperature sensor	-	03
Operation on frequency inverter permitted	-	No
Duty type	-	Continuous running duty (S1)
Direction of rotation	-	Bi-directional
Motor bearings insulated	-	No
Position of fixed motor bearing	-	Drive end
Fixed motor bearing reinforced	-	Without
Radial motor bearing reinforced	-	No
Condensate drain	-	Yes
Bolted feet	-	Yes

Table 395: Efficiency

Electrical data		+/-	50 Hz			60 Hz		
			4/4	3/4	2/4	4/4	3/4	2/4
Efficiency	[%]	Min.	96	95.5	94	95.4	95.4	94.5
Power factor		Min.	0.87	0.84	0.76	0.86	0.83	0.74

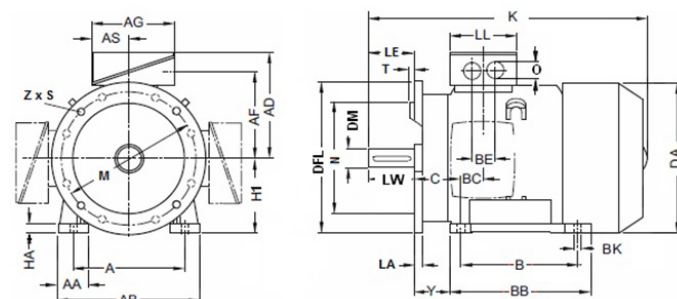

Fig. 44: Envelope dimensions

Table 401: Envelope dimensions A-DFL

A	AA	AB	AD	AF	AG	AS	B	BB	BC	BE	BK	C	DA	DFL
508	120	630	530	432	382	171.9	406	626	48	160	28	216	644	660

Table 402: Envelope dimensions DM-Z

DM	H1	HA	HH	K	LA	LB	LL	LE	M	N	S	T	Y	Z
65	315	48	264	1244	22	1104	379	140	600	550	24	6	146	8

Table 396: Electrical data

Electrical data		+/-	50 Hz		60 Hz	
			Δ	Y	Δ	Y
Rated voltage	[V]	-	400	690	460	-
Rated current	[A]	-10 %	190	110	168	-
Rated power	[kW]	-	110		110	
Rated speed	[rpm]	Min.	2976		3577	
Rated torque	[Nm]	Min.	353		294	
Starting current	[IA/IN]	+20 %	8		8.5	

Table 397: Mechanical data

Mechanical data		+/-	Values	
			50 Hz	60 Hz
Sound pressure level	[dB]	Max.	77	81
Moment of inertia	[kgm²]	Max.	1.4	1.4
Motor canopy		-	No	

Table 398: Explosion protection

Explosion protection	+/-	Values
Type of protection	-	Without

Table 399: Environment

Environment	+/-	Values
Ambient temperature [°C]	Max.	40

Table 400: Terminal box

Explosion protection	+/-	Values
Enclosure of terminal box	-	IP55 (TEFC)
Cable entry (O)	-	2XM63X1,5

⁸⁷ The name plate data may vary within the tolerances.

⁸⁸ 1800=grey cast iron, 4400=aluminium

V15 type of construction, 2-pole, 315M

Material: 2-132 -315M-BWE7F3NGFDWFWNWKWC⁸⁹⁾

Material number: 05214355 (5214318)

Table 403: General data [50 Hz]

General data	+/-	Values
Motor size	Max.	315M
Type of construction	Max.	V15
Starting method	-	Star a. Delta
Motor flange	-	IEC 60072 Type FF
Weight without attachments [kg]	Max.	980
Housing material ⁹⁰⁾	-	1800
Enclosure	-	IP55
Nominal frequency	[Hz]	50/60
Efficiency class	-	IE4 (Super Premium)
Thermal class F to B	-	Yes
Temperature sensor	-	03
Operation on frequency inverter permitted	-	No
Duty type	-	Continuous running duty (S1)
Direction of rotation	-	Bi-directional
Motor bearings insulated	-	No
Position of fixed motor bearing	-	Drive end
Fixed motor bearing reinforced	-	Without
Radial motor bearing reinforced	-	No
Condensate drain	-	Yes
Bolted feet	-	Yes

Table 404: Efficiency

Electrical data		+/-	50 Hz			60 Hz		
			4/4	3/4	2/4	4/4	3/4	2/4
Efficiency	[%]	Min.	96.2	95.9	95.1	95.8	95	93
Power factor		min.	0.88	0.85	0.79	0.88	0.85	0.77

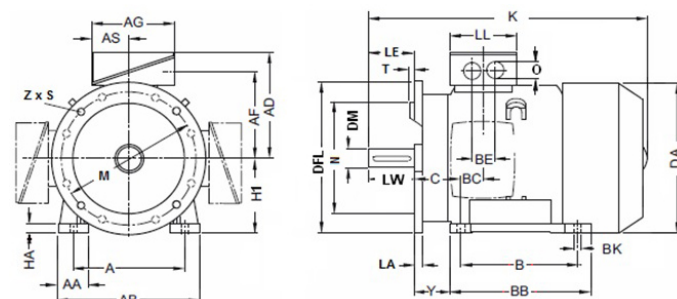


Fig. 45: Envelope dimensions

Table 410: Envelope dimensions A-DFL

A	AA	AB	AD	AF	AG	AS	B	BB	BC	BE	BK	C	DA	DFL
508	120	630	530	432	382	171.9	457	626	48	160	28	216	644	660

Table 411: Envelope dimensions DM-Z

DM	H1	HA	HH	K	LA	LB	LL	LE	M	N	S	T	Y	Z
65	315	48	264	1244	22	1104	379	140	600	550	24	6	146	8

Table 405: Electrical data

Electrical data		+/-	50 Hz		60 Hz	
			Δ	Y	Δ	Y
Rated voltage	[V]	-	400	690	460	-
Rated current	[A]	-10 %	225	130	197	-
Rated power	[kW]	-	132		132	
Rated speed	[rpm]	Min.	2983		3585	
Rated torque	[Nm]	Min.	423		352	
Starting current	[I _A /I _N]	+20 %	8.3		9.2	

Table 406: Mechanical data

Mechanical data		+/-	Values	
			50 Hz	60 Hz
Sound pressure level	[dB]	Max.	77	81
Moment of inertia	[kgm²]	Max.	1.9	1.9
Motor canopy		-	No	

Table 407: Explosion protection

Explosion protection	+/-	Values
Type of protection	-	Without

Table 408: Environment

Environment	+/-	Values
Ambient temperature [°C]	Max.	40

Table 409: Terminal box

Explosion protection	+/-	Values
Enclosure of terminal box	-	IP55 (TEFC)
Cable entry (O)	-	2XM63X1,5

⁸⁹⁾ The name plate data may vary within the tolerances.

⁹⁰⁾ 1800=grey cast iron, 4400=aluminium

V15 type of construction, 4-pole, 180M/L

Material: 4-22,0-180L-BWE7F3NGFDWFUWWE1⁹¹⁾

Material number: 5443334 (05407862)

Table 412: General data [50 Hz]

General data	+/-	Values
Motor size	Max.	180L
Type of construction	Max.	V15
Starting method	-	Star a. Delta
Motor flange	-	IEC 60072 Type FF
Weight without attachments [kg]	Max.	220
Housing material ⁹²⁾	-	1800
Enclosure	-	IP55
Nominal frequency	[Hz]	50/60
Efficiency class	-	IE4 (Super Premium)
Thermal class F to B	-	Yes
Temperature sensor	-	03
Operation on frequency inverter permitted	-	Yes (acc to motor manufact)
Duty type	-	Continuous running duty (S1)
Direction of rotation	-	Bi-directional
Motor bearings insulated	-	No
Position of fixed motor bearing	-	Drive end
Fixed motor bearing reinforced	-	Without
Radial motor bearing reinforced	-	No
Condensate drain	-	Yes
Bolted feet	-	Yes

Table 413: Efficiency

Electrical data		+/-	50 Hz			60 Hz		
			4/4	3/4	2/4	4/4	3/4	2/4
Efficiency	[%]	Min.	94.5	94.1	93.2	94.5	94.1	92.9
Power factor		Min.	0.81	0.74	0.62	0.79	0.72	0.6

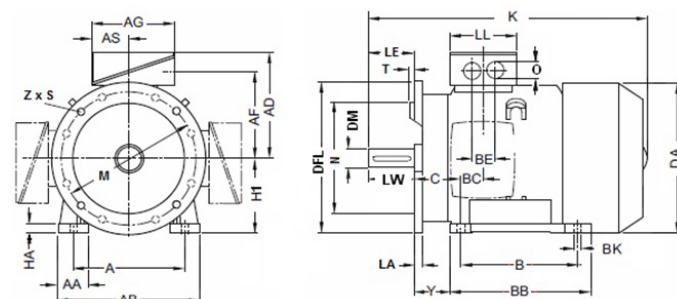

Fig. 46: Envelope dimensions

Table 419: Envelope dimensions A-DFL

A	AA	AB	AD	AF	AG	AS	B	BB	BC	BE	BK	C	DA	DFL
279	51	329	294	239.7	191.5	102.5	279	325	91.6	84	14.5	121	371	350

Table 420: Envelope dimensions DM-Z

DM	H1	HA	HH	K	LA	LB	LL	LE	M	N	S	T	Y	Z
48	180	19	212.5	722.5	16	612.5	198.5	110	300	250	18.5	5	98	4

Table 414: Electrical data

Electrical data		+/-	50 Hz		60 Hz	
			Δ	Y	Δ	Y
Rated voltage	[V]	-	400	690	460	-
Rated current	[A]	-10 %	41.5	24	37	-
Rated power	[kW]	-	22		22	
Rated speed	[rpm]	Min.	1476		1778	
Rated torque	[Nm]	Min.	142.25		118.7	
Starting current	[I _A /I _N]	+20 %	9.1		10.4	

Table 415: Mechanical data

Mechanical data		+/-	Values	
			50 Hz	60 Hz
Sound pressure level	[dB]	Max.	65	68
Moment of inertia	[kgm²]	Max.	0.21	0.21
Motor canopy		-	No	

Table 416: Explosion protection

Explosion protection	+/-	Values
Type of protection	-	Without

Table 417: Environment

Environment	+/-	Values
Ambient temperature [°C]	Max.	40

Table 418: Terminal box

Explosion protection	+/-	Values
Enclosure of terminal box	-	IP55 (TEFC)
Cable entry (O)	-	2XM40X1,5

⁹¹ The name plate data may vary within the tolerances.

⁹² 1800=grey cast iron, 4400=aluminium

V15 type of construction, 4-pole, 200M/L

Material: 4-30,0-200L-BWE7F3NGFDWFUWWE1⁹³⁾

Material number: 5443335 (05407863)

Table 421: General data [50 Hz]

General data	+/-	Values
Motor size	Max.	200L
Type of construction	Max.	V15
Starting method	-	Star a. Delta
Motor flange	-	IEC 60072 Type FF
Weight without attachments [kg]	Max.	255
Housing material ⁹⁴⁾	-	1800
Enclosure	-	IP55
Nominal frequency	[Hz]	50/60
Efficiency class	-	IE4 (Super Premium)
Thermal class F to B	-	Yes
Temperature sensor	-	03
Operation on frequency inverter permitted	-	Yes (acc to motor manufact)
Duty type	-	Continuous running duty (S1)
Direction of rotation	-	Bi-directional
Motor bearings insulated	-	No
Position of fixed motor bearing	-	Drive end
Fixed motor bearing reinforced	-	Without
Radial motor bearing reinforced	-	No
Condensate drain	-	Yes
Bolted feet	-	Yes

Table 422: Efficiency

Electrical data		+/-	50 Hz			60 Hz		
			4/4	3/4	2/4	4/4	3/4	2/4
Efficiency	[%]	Min.	94.9	94.7	94.2	95	94.8	93.7
Power factor		Min.	0.81	0.75	0.64	0.8	0.74	0.62

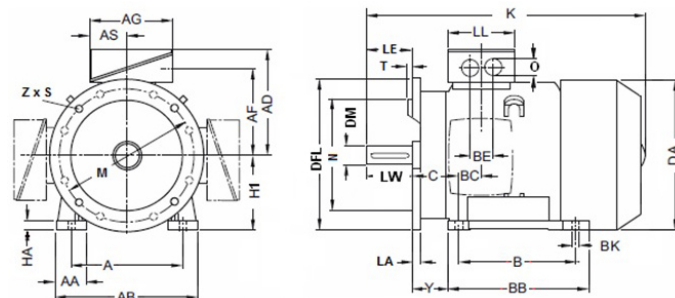

Fig. 47: Envelope dimensions

Table 428: Envelope dimensions A-DFL

A	AA	AB	AD	AF	AG	AS	B	BB	BC	BE	BK	C	DA	DFL
318	65	385	334	272.7	218	119.1	305	369	76	94	18.5	133	414	400

Table 429: Envelope dimensions DM-Z

DM	H1	HA	HH	K	LA	LB	LL	LE	M	N	S	T	Y	Z
55	200	30	209	767.5	18	657.5	228.5	110	350	300	18.5	5	101	4

Table 423: Electrical data

Electrical data		+/-	50 Hz		60 Hz	
			Δ	Y	Δ	Y
Rated voltage	[V]	-	400	690	460	-
Rated current	[A]	-10 %	56.3	32.6	49.5	-
Rated power	[kW]	-	30		30	
Rated speed	[rpm]	Min.	1480		1783	
Rated torque	[Nm]	Min.	193.26		160.88	
Starting current	[I _A /I _N]	+20 %	8.8		9.8	

Table 424: Mechanical data

Mechanical data		+/-	Values	
			50 Hz	60 Hz
Sound pressure level	[dB]	Max.	66	68
Moment of inertia	[kgm²]	Max.	0.28	0.28
Motor canopy		-	No	

Table 425: Explosion protection

Explosion protection	+/-	Values
Type of protection	-	Without

Table 426: Environment

Environment	+/-	Values
Ambient temperature [°C]	Max.	40

Table 427: Terminal box

Explosion protection	+/-	Values
Enclosure of terminal box	-	IP55 (TEFC)
Cable entry (Ø)	-	2XM50X1,5

⁹³ The name plate data may vary within the tolerances.

⁹⁴ 1800=grey cast iron, 4400=aluminium

V15 type of construction, 4-pole, 225S/M

Material: 4-37,0-225S-BWE7F3NGHDWFUWWE1⁹⁵⁾

Material number: 5443336 (05407864)

Table 430: General data [50 Hz]

General data	+/-	Values
Motor size	Max.	225S
Type of construction	Max.	V15
Starting method	-	Star a. Delta
Motor flange	-	IEC 60072 Type FF
Weight without attachments [kg]	Max.	345
Housing material ⁹⁶⁾	-	1800
Enclosure	-	IP55
Nominal frequency	[Hz]	50/60
Efficiency class	-	IE4 (Super Premium)
Thermal class F to B	-	Yes
Temperature sensor	-	03
Operation on frequency inverter permitted	-	Yes (acc to motor manufact)
Duty type	-	Continuous running duty (S1)
Direction of rotation	-	Bi-directional
Motor bearings insulated	-	No
Position of fixed motor bearing	-	Drive end
Fixed motor bearing reinforced	-	Without
Radial motor bearing reinforced	-	No
Condensate drain	-	Yes
Bolted feet	-	No

Table 431: Efficiency

Electrical data		+/-	50 Hz			60 Hz		
			4/4	3/4	2/4	4/4	3/4	2/4
Efficiency	[%]	Min.	95.2	94.8	94	95.4	95.1	94
Power factor		Min.	0.82	0.77	0.66	0.81	0.75	0.63

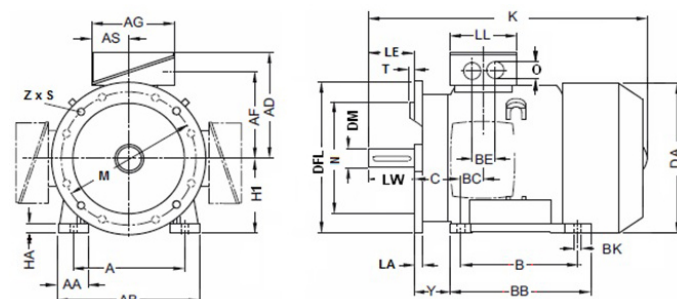

Fig. 48: Envelope dimensions

Table 437: Envelope dimensions A-DFL

A	AA	AB	AD	AF	AG	AS	B	BB	BC	BE	BK	C	DA	DFL
356	80	436	397	314.8	286	165.6	286	439.3	54.6	110	18.5	149	469	450

Table 438: Envelope dimensions DM-Z

DM	H1	HA	HH	K	LA	LB	LL	LE	M	N	S	T	Y	Z
60	225	28	203.5	886.6	18	746.6	269	140	400	350	18.5	5	106.5	8

Table 432: Electrical data

Electrical data		+/-	50 Hz		60 Hz	
			Δ	Y	Δ	Y
Rated voltage	[V]	-	400	690	460	-
Rated current	[A]	-10 %	68.4	39.5	60.1	-
Rated power	[kW]	-	37		37	
Rated speed	[rpm]	Min.	1483		1785	
Rated torque	[Nm]	Min.	238.38		198.16	
Starting current	[IA/IN]	+20 %	8.5		10	

Table 433: Mechanical data

Mechanical data		+/-	Values	
			50 Hz	60 Hz
Sound pressure level	[dB]	Max.	67	70
Moment of inertia	[kgm²]	Max.	0.48	0.48
Motor canopy		-	No	

Table 434: Explosion protection

Explosion protection	+/-	Values
Type of protection	-	Without

Table 435: Environment

Environment	+/-	Values
Ambient temperature [°C]	Max.	40

Table 436: Terminal box

Explosion protection	+/-	Values
Enclosure of terminal box	-	IP55 (TEFC)
Cable entry (Ø)	-	2XM50X1,5

⁹⁵ The name plate data may vary within the tolerances.

⁹⁶ 1800=grey cast iron, 4400=aluminium

V15 type of construction, 4-pole, 225S/M

Material: 4-45,0-225M-BWE7F3NGHDFUWWE1⁹⁷⁾

Material number: 5443337 (05407865)

Table 439: General data [50 Hz]

General data		+/-	Values
Motor size		Max.	225M
Type of construction		Max.	V15
Starting method		-	Star a. Delta
Motor flange		-	IEC 60072 Type FF
Weight without attachments	[kg]	Max.	415
Housing material ⁹⁸⁾		-	1800
Enclosure		-	IP55
Nominal frequency	[Hz]		50/60
Efficiency class		-	IE4 (Super Premium)
Thermal class F to B		-	Yes
Temperature sensor		-	03
Operation on frequency inverter permitted		-	Yes (acc to motor manufact)
Duty type		-	Continuous running duty (S1)
Direction of rotation		-	Bi-directional
Motor bearings insulated		-	No
Position of fixed motor bearing		-	Drive end
Fixed motor bearing reinforced		-	Without
Radial motor bearing reinforced		-	No
Condensate drain		-	Yes
Bolted feet		-	No

Table 440: Efficiency

Electrical data		+/-	50 Hz			60 Hz		
			4/4	3/4	2/4	4/4	3/4	2/4
Efficiency	[%]	Min.	95.4	95.3	94.7	95.4	95.4	94.6
Power factor		Min.	0.84	0.8	0.7	0.83	0.78	0.67

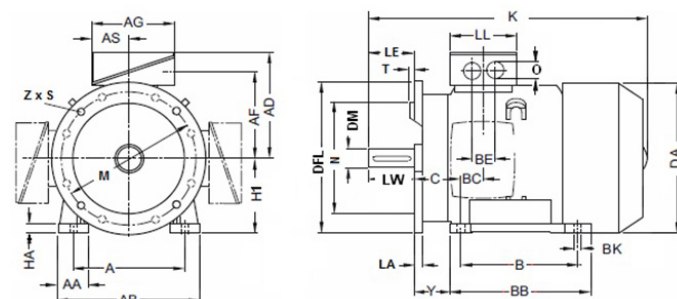

Fig. 49: Envelope dimensions

Table 446: Envelope dimensions A-DFL

A	AA	AB	AD	AF	AG	AS	B	BB	BC	BE	BK	C	DA	DFL
356	80	436	397	314.8	286	165.6	311	439.3	54.6	110	18.5	149	469	450

Table 447: Envelope dimensions DM-Z

DM	H1	HA	HH	K	LA	LB	LL	LE	M	N	S	T	Y	Z
60	225	28	203.5	886.6	18	746.6	269	140	400	350	18.5	5	106.5	8

Table 441: Electrical data

Electrical data		+/-	50 Hz		60 Hz	
			Δ	Y	Δ	Y
Rated voltage	[V]	-	400	690	460	-
Rated current	[A]	-10 %	81.1	46.9	71.3	-
Rated power	[kW]	-	45		45	
Rated speed	[rpm]	Min.	1483		1785	
Rated torque	[Nm]	Min.	290.38		241.33	
Starting current	[IA/IN]	+20 %	8.7		10.2	

Table 442: Mechanical data

Mechanical data		+/-	Values	
			50 Hz	60 Hz
Sound pressure level	[dB]	Max.	67	70
Moment of inertia	[kgm²]	Max.	0.64	0.64
Motor canopy		-	No	

Table 443: Explosion protection

Explosion protection	+/-	Values
Type of protection	-	Without

Table 444: Environment

Environment	+/-	Values
Ambient temperature	[°C]	Max. 40

Table 445: Terminal box

Explosion protection	+/-	Values
Enclosure of terminal box	-	IP55 (TEFC)
Cable entry (Ø)	-	2XM50X1,5

97 The name plate data may vary within the tolerances.

98 1800=grey cast iron, 4400=aluminium

V15 type of construction, 4-pole, 250S/M

Material: 4-55,0-250M-BWE7F3NGHDFUWWE1⁹⁹⁾

Material number: 5443338 (05407866)

Table 448: General data [50 Hz]

General data		+/-	Values
Motor size		Max.	250M
Type of construction		Max.	V15
Starting method		-	Star a. Delta
Motor flange		-	IEC 60072 Type FF
Weight without attachments	[kg]	Max.	470
Housing material ¹⁰⁰⁾		-	1800
Enclosure		-	IP55
Nominal frequency	[Hz]		50/60
Efficiency class		-	IE4 (Super Premium)
Thermal class F to B		-	Yes
Temperature sensor		-	03
Operation on frequency inverter permitted		-	Yes (acc to motor manufact)
Duty type		-	Continuous running duty (S1)
Direction of rotation		-	Bi-directional
Motor bearings insulated		-	No
Position of fixed motor bearing		-	Drive end
Fixed motor bearing reinforced		-	Without
Radial motor bearing reinforced		-	No
Condensate drain		-	Yes
Bolted feet		-	No

Table 449: Efficiency

Electrical data		+/-	50 Hz			60 Hz		
			4/4	3/4	2/4	4/4	3/4	2/4
Efficiency	[%]	Min.	95.7	95.4	94.8	95.8	95.6	94.7
Power factor		Min.	0.83	0.78	0.67	0.82	0.76	0.65

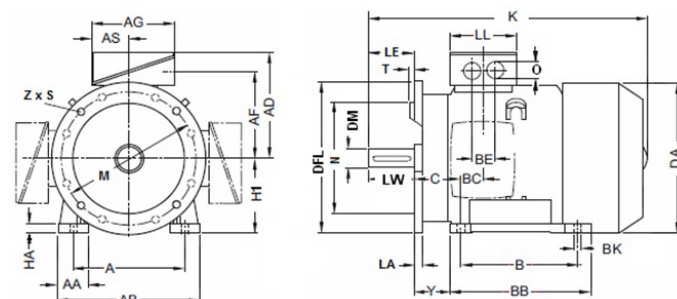

Fig. 50: Envelope dimensions

Table 455: Envelope dimensions A-DFL

A	AA	AB	AD	AF	AG	AS	B	BB	BC	BE	BK	C	DA	DFL
406	100	506	417	334.8	286	165.6	349	500	49	110	24	168	514	550

Table 456: Envelope dimensions DM-Z

DM	H1	HA	HH	K	LA	LB	LL	LE	M	N	S	T	Y	Z
65	250	30	217	965.5	20	825.5	269	140	500	450	18.5	5	120	8

Table 450: Electrical data

Electrical data		+/-	50 Hz		60 Hz	
			Δ	Y	Δ	Y
Rated voltage	[V]	-	400	690	460	-
Rated current	[A]	-10 %	100	57.8	87.9	-
Rated power	[kW]	-	55		55	
Rated speed	[rpm]	Min.	1485		1787	
Rated torque	[Nm]	Min.	354.14		294.3	
Starting current	[IA/IN]	+20 %	8.9		10.2	

Table 451: Mechanical data

Mechanical data		+/-	Values	
			50 Hz	60 Hz
Sound pressure level	[dB]	Max.	67	71
Moment of inertia	[kgm ²]	Max.	0.78	0.78
Motor canopy		-	No	

Table 452: Explosion protection

Explosion protection	+/-	Values
Type of protection	-	Without

Table 453: Environment

Environment	+/-	Values
Ambient temperature	[°C]	Max. 40

Table 454: Terminal box

Explosion protection	+/-	Values
Enclosure of terminal box	-	IP55 (TEFC)
Cable entry (Ø)	-	2XM63X1,5

⁹⁹ The name plate data may vary within the tolerances.

¹⁰⁰ 1800=grey cast iron, 4400=aluminium

V15 type of construction, 4-pole, 280S

Material: 4-75,0-280S-BWE7F3NGFDWFWNWKWC¹⁰¹⁾

Material number: 05214356 (5214319)

Table 457: General data [50 Hz]

General data		+/-	Values
Motor size		Max.	280S
Type of construction		Max.	V15
Starting method		-	Star a. Delta
Motor flange		-	IEC 60072 Type FF
Weight without attachments	[kg]	Max.	802
Housing material ¹⁰²⁾		-	1800
Enclosure		-	IP55
Nominal frequency	[Hz]		50/60
Efficiency class		-	IE4 (Super Premium)
Thermal class F to B		-	Yes
Temperature sensor		-	03
Operation on frequency inverter permitted		-	No
Duty type		-	Continuous running duty (S1)
Direction of rotation		-	Bi-directional
Motor bearings insulated		-	No
Position of fixed motor bearing		-	Drive end
Fixed motor bearing reinforced		-	Without
Radial motor bearing reinforced		-	No
Condensate drain		-	Yes
Bolted feet		-	Yes

Table 458: Efficiency

Electrical data		+/-	50 Hz			60 Hz		
			4/4	3/4	2/4	4/4	3/4	2/4
Efficiency	[%]	Min.	96	95.8	95.3	96.2	95.7	94.8
Power factor		Min.	0.83	0.8	0.7	0.82	0.77	0.66

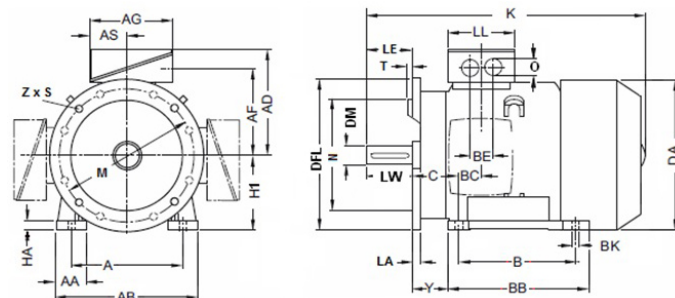

Fig. 51: Envelope dimensions

Table 464: Envelope dimensions A-DFL

A	AA	AB	AD	AF	AG	AS	B	BB	BC	BE	BK	C	DA	DFL
457	100	557	472	391	312	140.4	368	517	76	126	24	190	580	550

Table 465: Envelope dimensions DM-Z

DM	H1	HA	HH	K	LA	LB	LL	LE	M	N	S	T	Y	Z
75	280	42	266	1071	18	931	314	140	500	450	18.5	5	141	8

Table 459: Electrical data

Electrical data		+/-	50 Hz		60 Hz	
			Δ	Y	Δ	Y
Rated voltage	[V]	-	400	690	460	-
Rated current	[A]	-10 %	136	78.8	119	-
Rated power	[kW]	-	75		75	
Rated speed	[rpm]	Min.	1486		1788	
Rated torque	[Nm]	Min.	482		401	
Starting current	[I _A /I _N]	+20 %	7.7		8.7	

Table 460: Mechanical data

Mechanical data		+/-	Values	
			50 Hz	60 Hz
Sound pressure level	[dB]	Max.	69	73
Moment of inertia	[kgm ²]	Max.	1.7	1.7
Motor canopy		-	No	

Table 461: Explosion protection

Explosion protection	+/-	Values
Type of protection	-	Without

Table 462: Environment

Environment	+/-	Values
Ambient temperature	[°C]	Max. 40

Table 463: Terminal box

Explosion protection	+/-	Values
Enclosure of terminal box	-	IP55 (TEFC)
Cable entry (Ø)	-	2XM63X1,5

101 The name plate data may vary within the tolerances.

102 1800=grey cast iron, 4400=aluminium

V15 type of construction, 4-pole, 280M

Material: 4-90,0-280M-BWE7F3NGFDWFNWKWC¹⁰³⁾

Material number: 05214357 (5214360)

Table 466: General data [50 Hz]

General data	+/-	Values
Motor size	Max.	280M
Type of construction	Max.	V15
Starting method	-	Star a. Delta
Motor flange	-	IEC 60072 Type FF
Weight without attachments [kg]	Max.	790
Housing material ¹⁰⁴⁾	-	1800
Enclosure	-	IP55
Nominal frequency	[Hz]	50/60
Efficiency class	-	IE4 (Super Premium)
Thermal class F to B	-	Yes
Temperature sensor	-	03
Operation on frequency inverter permitted	-	No
Duty type	-	Continuous running duty (S1)
Direction of rotation	-	Bi-directional
Motor bearings insulated	-	No
Position of fixed motor bearing	-	Drive end
Fixed motor bearing reinforced	-	Without
Radial motor bearing reinforced	-	No
Condensate drain	-	Yes
Bolted feet	-	Yes

Table 467: Efficiency

Electrical data		+/-	50 Hz			60 Hz		
			4/4	3/4	2/4	4/4	3/4	2/4
Efficiency	[%]	Min.	96.1	96	95.5	96.2	96.1	95.5
Power factor		Min.	0.84	0.81	0.72	0.83	0.79	0.69

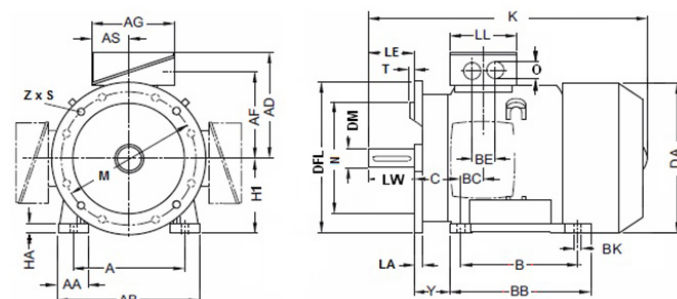

Fig. 52: Envelope dimensions

Table 473: Envelope dimensions A-DFL

A	AA	AB	AD	AF	AG	AS	B	BB	BC	BE	BK	C	DA	DFL
457	100	557	472	391	312	140.4	419	517	76	126	24	190	580	550

Table 474: Envelope dimensions DM-Z

DM	H1	HA	HH	K	LA	LB	LL	LE	M	N	S	T	Y	Z
75	280	42	266	1071	18	931	314	140	500	450	18.5	5	146	8

Table 468: Electrical data

Electrical data		+/-	50 Hz		60 Hz	
			Δ	Y	Δ	Y
Rated voltage	[V]	-	400	690	460	-
Rated current	[A]	-10 %	161	93.3	141	-
Rated power	[kW]	-	90		90	
Rated speed	[rpm]	Min.	1486		1788	
Rated torque	[Nm]	Min.	579		481	
Starting current	[IA/IN]	+20 %	8		8.9	

Table 469: Mechanical data

Mechanical data		+/-	Values	
			50 Hz	60 Hz
Sound pressure level	[dB]	Max.	69	73
Moment of inertia	[kgm²]	Max.	2	2
Motor canopy		-	No	

Table 470: Explosion protection

Explosion protection	+/-	Values
Type of protection	-	Without

Table 471: Environment

Environment	+/-	Values
Ambient temperature [°C]	Max.	40

Table 472: Terminal box

Explosion protection	+/-	Values
Enclosure of terminal box	-	IP55 (TEFC)
Cable entry (O)	-	2XM63X1,5

¹⁰³ The name plate data may vary within the tolerances.

¹⁰⁴ 1800=grey cast iron, 4400=aluminium

V15 type of construction, 4-pole, 315M

Material: 4-110 -315M-BWE7F3NGFDWFNWKWC¹⁰⁵⁾

Material number: 05214358 (5214361)

Table 475: General data [50 Hz]

General data	+/-	Values
Motor size	Max.	315M
Type of construction	Max.	V15
Starting method	-	Star a. Delta
Motor flange	-	IEC 60072 Type FF
Weight without attachments [kg]	Max.	935
Housing material ¹⁰⁶⁾	-	1800
Enclosure	-	IP55
Nominal frequency	[Hz]	50/60
Efficiency class	-	IE4 (Super Premium)
Thermal class F to B	-	Yes
Temperature sensor	-	03
Operation on frequency inverter permitted	-	No
Duty type	-	Continuous running duty (S1)
Direction of rotation	-	Bi-directional
Motor bearings insulated	-	No
Position of fixed motor bearing	-	Drive end
Fixed motor bearing reinforced	-	Without
Radial motor bearing reinforced	-	No
Condensate drain	-	Yes
Bolted feet	-	Yes

Table 476: Efficiency

Electrical data		+/-	50 Hz			60 Hz		
			4/4	3/4	2/4	4/4	3/4	2/4
Efficiency	[%]	Min.	96.3	95.4	96	96.2	96.1	95.3
Power factor		Min.	0.83	0.79	0.69	0.82	0.77	0.67

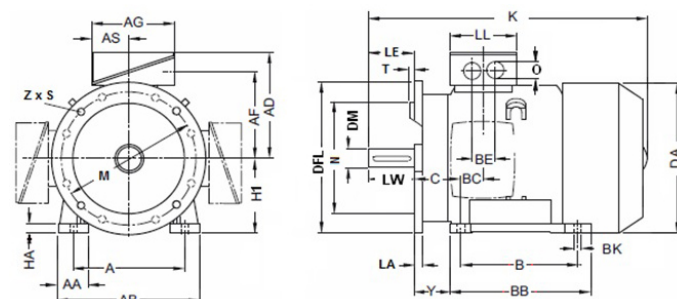

Fig. 53: Envelope dimensions

Table 482: Envelope dimensions A-DFL

A	AA	AB	AD	AF	AG	AS	B	BB	BC	BE	BK	C	DA	DFL
508	120	630	530	432	382	171.9	457	626	48	160	28	216	644	660

Table 483: Envelope dimensions DM-Z

DM	H1	HA	HH	K	LA	LB	LL	LE	M	N	S	T	Y	Z
80	315	48	264	1274	22	1104	379	170	600	550	24	6	146	8

Table 477: Electrical data

Electrical data		+/-	50 Hz		60 Hz	
			Δ	Y	Δ	Y
Rated voltage	[V]	-	400	690	460	-
Rated current	[A]	-10 %	199	115	175	-
Rated power	[kW]	-	110		110	
Rated speed	[rpm]	Min.	1491		1792	
Rated torque	[Nm]	Min.	706		586	
Starting current	[IA/IN]	+20 %	7		7.9	

Table 478: Mechanical data

Mechanical data		+/-	Values	
			50 Hz	60 Hz
Sound pressure level	[dB]	Max.	71	75
Moment of inertia	[kgm²]	Max.	2.7	2.7
Motor canopy		-	No	

Table 479: Explosion protection

Explosion protection	+/-	Values
Type of protection	-	Without

Table 480: Environment

Environment	+/-	Values
Ambient temperature [°C]	Max.	40

Table 481: Terminal box

Explosion protection	+/-	Values
Enclosure of terminal box	-	IP55 (TEFC)
Cable entry (O)	-	2XM63X1,5

105 The name plate data may vary within the tolerances.

106 1800=grey cast iron, 4400=aluminium

V15 type of construction, 4-pole, 315M

Material: 4-132 -315M-BWE7F3NGFDWFNWKWC¹⁰⁷⁾

Material number: 05214359 (5214362)

Table 484: General data [50 Hz]

General data	+/-	Values
Motor size	Max.	315M
Type of construction	Max.	V15
Starting method	-	Star a. Delta
Motor flange	-	IEC 60072 Type FF
Weight without attachments [kg]	Max.	980
Housing material ¹⁰⁸⁾	-	1800
Enclosure	-	IP55
Nominal frequency	[Hz]	50/60
Efficiency class	-	IE4 (Super Premium)
Thermal class F to B	-	Yes
Temperature sensor	-	03
Operation on frequency inverter permitted	-	No
Duty type	-	Continuous running duty (S1)
Direction of rotation	-	Bi-directional
Motor bearings insulated	-	No
Position of fixed motor bearing	-	Drive end
Fixed motor bearing reinforced	-	Without
Radial motor bearing reinforced	-	No
Condensate drain	-	Yes
Bolted feet	-	Yes

Table 485: Efficiency

Electrical data		+/-	50 Hz			60 Hz		
			4/4	3/4	2/4	4/4	3/4	2/4
Efficiency	[%]	Min.	96.4	96.1	95.4	96.5	96.2	95.4
Power factor		Min.	0.83	0.78	0.68	0.82	0.77	0.66

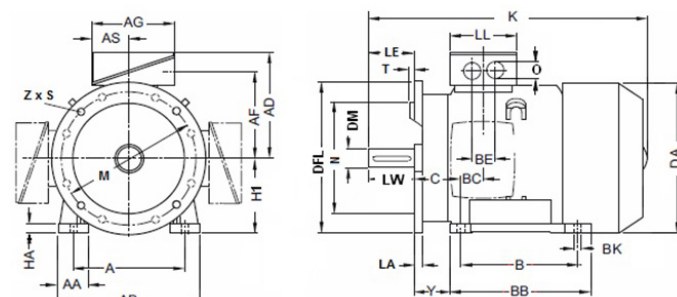

Fig. 54: Envelope dimensions

Table 491: Envelope dimensions A-DFL

A	AA	AB	AD	AF	AG	AS	B	BB	BC	BE	BK	C	DA	DFL
508	120	630	530	432	382	171.9	457	626	48	160	28	216	644	660

Table 492: Envelope dimensions DM-Z

DM	H1	HA	HH	K	LA	LB	LL	LE	M	N	S	T	Y	Z
80	315	48	264	1274	22	1104	379	170	600	550	24	6	146	8

Table 486: Electrical data

Electrical data		+/-	50 Hz		60 Hz	
			Δ	Y	Δ	Y
Rated voltage	[V]	-	400	690	460	-
Rated current	[A]	-10 %	238	138	209	-
Rated power	[kW]	-	132		132	
Rated speed	[rpm]	Min.	1491		1792	
Rated torque	[Nm]	Min.	847		704	
Starting current	[I _A /I _N]	+20 %	7.3		8.3	

Table 487: Mechanical data

Mechanical data		+/-	Values	
			50 Hz	60 Hz
Sound pressure level	[dB]	Max.	71	75
Moment of inertia	[kgm²]	Max.	3.1	3.1
Motor canopy		-	No	

Table 488: Explosion protection

Explosion protection	+/-	Values
Type of protection	-	Without

Table 489: Environment

Environment	+/-	Values
Ambient temperature [°C]	Max.	40

Table 490: Terminal box

Explosion protection	+/-	Values
Enclosure of terminal box	-	IP55 (TEFC)
Cable entry (O)	-	2XM63X1,5

¹⁰⁷ The name plate data may vary within the tolerances.

¹⁰⁸ 1800=grey cast iron, 4400=aluminium

Product information as per Regulation (EU) 2019/1781

Table 493: Efficiency

Model ID						Efficiency at 50 Hz			Efficiency at 60 Hz		
Factory-dependent motor material number			KSB material number			η_N	$\eta_{75\%}^{109)}$	$\eta_{50\%}^{109)}$	η_N	$\eta_{75\%}^{109)}$	$\eta_{50\%}^{109)}$
						[%]	[%]	[%]	[%]	[%]	[%]
5407377	5407841	5407857	5443310	5443313	5443329	94,0	93,6	92,5	93,0	93,0	92,2
5407852	5407846	5407862	5443324	5443318	5443334	94,5	94,1	93,2	94,5	94,1	92,9
5407378	5407842	5407858	5443311	5443314	5443330	94,5	94,0	92,8	93,6	93,4	91,7
5407853	5407847	5407863	5443325	5443319	5443335	94,9	94,7	94,2	95,0	94,8	93,7
5407379	5407843	5407859	5443312	5443315	5443331	94,8	94,3	93,3	94,1	93,8	92,4
5407854	5407848	5407864	5443326	5443320	5443336	95,2	94,8	94,0	95,4	95,1	94,0
5407850	5407844	5407860	5443322	5443316	5443332	95,0	94,3	93,2	94,5	94,1	92,6
5407855	5407849	5407865	5443327	5443321	5443337	95,4	95,3	94,7	95,4	95,4	94,6
5407851	5407845	5407861	5443323	5443317	5443333	95,3	94,5	93,5	94,5	94,2	92,7
5407856	5408280	5407866	5443328	5443339	5443338	95,7	95,4	94,8	95,8	95,6	94,7
5214341	5214248	5214315	5214325	5214272	5214352	95,6	94,8	93,6	95,0	94,8	93,0
5214338	5214333	5214319	5214322	5214277	5214356	96,0	95,8	95,3	96,2	95,7	94,8
5214342	5214249	5214316	5214326	5214273	5214353	95,8	95,2	94,3	95,4	95,0	93,5
5214339	5214334	5214360	5214323	5214278	5214357	96,1	96,0	95,5	96,2	96,1	95,5
5214343	5214330	5214317	5214327	5214274	5214354	96,0	95,5	94,0	95,4	95,4	94,5
5214340	5214335	5214361	5214324	5214279	5214358	96,3	96,0	95,4	96,2	96,1	95,3
5214337	5214331	5214318	5214321	5214275	5214355	96,2	95,9	95,1	95,8	95,0	93,0
5214332	5214336	5214362	5214276	5214320	5214359	96,4	96,1	95,4	96,5	96,2	95,4

Efficiency class: IE4
 Manufacturer: KSB SE & Co. KGaA
 Johann-Klein-Straße 9
 67227 Frankenthal
 HRB 65657 Ludwigshafen

Table 494: Model-dependent electrical data

Model ID						Number of poles	Rated output power	Rated input frequency	Nominal voltage	Rated speed	Rated input frequency	Nominal voltage	Rated speed
							P _N	f ₁	U ₁		f ₁	U ₁	
Factory-dependent motor material number			KSB material number					[kW]	[Hz]	[V]	[rpm]	[Hz]	[V]
5407377	5407841	5407857	5443310	5443313	5443329	2	22	50	400	2960	60	460	3565
5407852	5407846	5407862	5443324	5443318	5443334	4	30	50	400	1476	60	460	1778
5407378	5407842	5407858	5443311	5443314	5443330	2				2965			3570
5407853	5407847	5407863	5443325	5443319	5443335	4				1480			1783
5407379	5407843	5407859	5443312	5443315	5443331	2	37			2965			3570
5407854	5407848	5407864	5443326	5443320	5443336	4	1483			1785			
5407850	5407844	5407860	5443322	5443316	5443332	2	45			2970			3575
5407855	5407849	5407865	5443327	5443321	5443337	4	1483			1785			
5407851	5407845	5407861	5443323	5443317	5443333	2	55			2970			3570
5407856	5408280	5407866	5443328	5443339	5443338	4	1485			1787			
5214341	5214248	5214315	5214325	5214272	5214352	2	75			2977			3579
5214338	5214333	5214319	5214322	5214277	5214356	4	90			1486			1788
5214342	5214249	5214316	5214326	5214273	5214353	2				2975			3578
5214339	5214334	5214360	5214323	5214278	5214357	4				1486			1788
5214343	5214330	5214317	5214327	5214274	5214354	2	110			2976			3577
5214340	5214335	5214361	5214324	5214279	5214358	4	132			1488			1792
5214337	5214331	5214318	5214321	5214275	5214355	2				2979			3585
5214332	5214336	5214362	5214276	5214320	5214359	4				1488			1792

Number of phases: 3
 Installation altitude [m]: 1000
 Ambient air temperature range [°C]: -20 to +40
 Maximum in-service temperature [°C]: 130

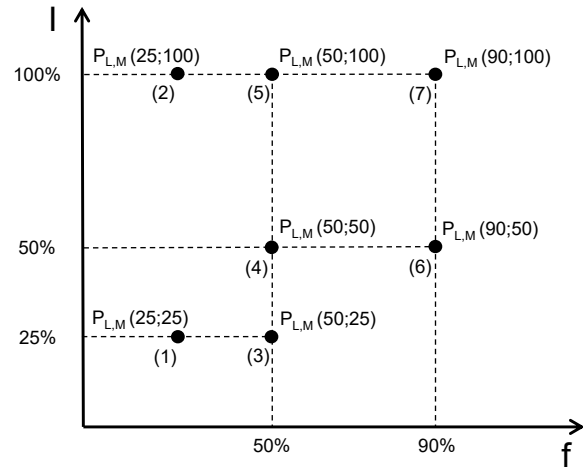


Fig. 55: Power losses (%) of rated output power (speed versus torque)

Table 495: Power losses

Model ID						Power losses						
						P1 (25;25)	P2 (25;100)	P3 (50;25)	P4 (50;50)	P5 (50;100)	P6 (90;50)	P7 (90;100)
Factory-dependent motor material number		KSB material number				[%]	[%]	[%]	[%]	[%]	[%]	[%]
5407377	5407841	5407857	5443310	5443313	5443329	0,7 %	4,3 %	1,2 %	1,9 %	4,7 %	3,1 %	6,0 %
5407852	5407846	5407862	5443324	5443318	5443334	0,9 %	4,2 %	1,3 %	1,9 %	4,5 %	2,9 %	5,6 %
5407378	5407842	5407858	5443311	5443314	5443330	0,7 %	3,6 %	1,2 %	1,7 %	4,0 %	3,0 %	5,4 %
5407853	5407847	5407863	5443325	5443319	5443335	0,8 %	3,9 %	1,1 %	1,7 %	4,2 %	2,5 %	5,1 %
5407379	5407843	5407859	5443312	5443315	5443331	0,6 %	3,5 %	1,1 %	1,6 %	3,9 %	2,8 %	5,2 %
5407854	5407848	5407864	5443326	5443320	5443336	0,7 %	3,3 %	1,1 %	1,6 %	3,7 %	2,5 %	4,8 %
5407850	5407844	5407860	5443322	5443316	5443332	0,6 %	3,1 %	1,0 %	1,5 %	3,6 %	2,8 %	4,9 %
5407855	5407849	5407865	5443327	5443321	5443337	0,6 %	3,2 %	0,9 %	1,4 %	3,5 %	2,3 %	4,6 %
5407851	5407845	5407861	5443323	5443317	5443333	0,5 %	2,9 %	0,9 %	1,4 %	3,3 %	2,5 %	4,6 %
5407856	5408280	5407866	5443328	5443339	5443338	0,6 %	3,1 %	1,0 %	1,4 %	3,5 %	2,2 %	4,4 %
5214341	5214248	5214315	5214325	5214272	5214352	0,5 %	2,4 %	1,0 %	1,3 %	2,9 %	2,6 %	4,2 %
5214338	5214333	5214319	5214322	5214277	5214356	0,5 %	2,6 %	0,8 %	1,2 %	2,9 %	1,9 %	3,8 %
5214342	5214249	5214316	5214326	5214273	5214353	0,5 %	2,4 %	0,9 %	1,2 %	2,8 %	2,3 %	4,0 %
5214339	5214334	5214360	5214323	5214278	5214357	0,5 %	2,8 %	0,8 %	1,3 %	3,1 %	2,0 %	3,9 %
5214343	5214330	5214317	5214327	5214274	5214354	0,5 %	2,4 %	0,9 %	1,3 %	2,8 %	2,4 %	4,0 %
5214340	5214335	5214361	5214324	5214279	5214358	0,6 %	2,7 %	1,0 %	1,4 %	3,0 %	2,2 %	3,9 %
5214337	5214331	5214318	5214321	5214275	5214355	0,5 %	2,3 %	1,0 %	1,3 %	2,7 %	2,4 %	3,9 %
5214332	5214336	5214362	5214276	5214320	5214359	0,6 %	2,6 %	1,0 %	1,4 %	3,0 %	2,2 %	3,9 %



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